BUILT ON BROKEN ASSUMPTIONS (🧋 BOBA): WHY CODING AGENTS CANNOT BE LEFT ALONE

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ABSTRACT

Coding agents are autonomous developers. Given a request, they change the code and report back when done. As such, they can handle not only independent requests but whole chains of them, each request built on the last. Agents are known to report wrong work as done, yet benchmarks still test them one request at a time, on human-written code, checked once. We build BOBA-BENCH, 280 chains of 20 consecutive requests drawn from 140 real open-source projects, and evaluate ten coding agents on both. All ten do far worse on chains. On average, they complete 95% of requests taken one at a time, but when requests are chained together, they complete 51%. Worse code is not the cause. Once checked, agent code is nearly as good a base for later requests as human code. The cause is the mistakes agents miss. Coding agents declare a request done once their own tests pass, even when those tests miss a mistake. The next request then builds on broken code and fails about twice as often. The mistake is rarely found later. Nor can agents catch such mistakes by reviewing code. Asked whether an implementation is correct, they are right 53% of the time, barely better than a coin flip. Coding agents are therefore heavily dependent on access to verification. A single external "that's wrong" after a failed request and one repair round lift chain completion from 51% to 59%, and a repaired mistake no longer hurts the requests that follow. In simpler terms, coding agents are only as reliable as the checks they can reach: *without verification, every request that follows is built on broken assumptions*.

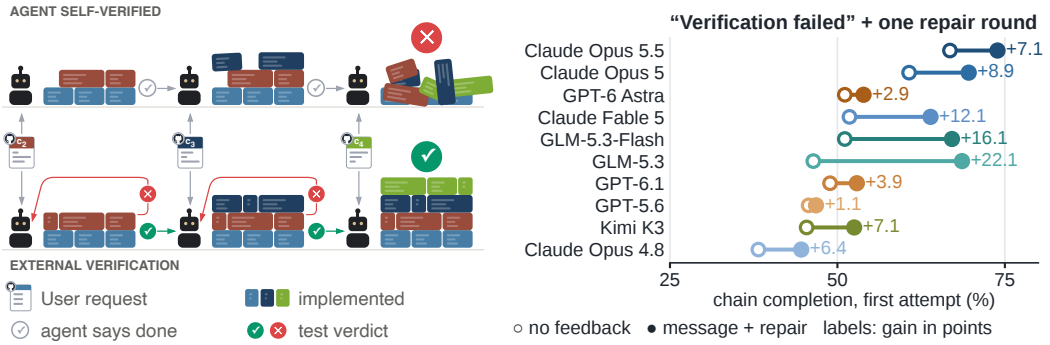


Figure 1: **An outside check lifts chain completion.** Left: unchecked work collapses, verified work holds (schematic). Right: chains completed on the first attempt without feedback (hollow) and with one "verification failed" message and one repair after each failed request (filled), ten models; labels give the gains in percentage points.

1 INTRODUCTION

Today's coding agents (*e.g.*, Codex (OpenAI, 2025), Claude Code (Anthropic, 2025), OpenHands (Wang et al., 2025)) let developers hand over a request and receive a finished change. Delegation promises to help not only when developers review each change before asking for the next, but also when they cannot, and each request then builds on code the agent wrote and reported complete (Liu et al., 2026a; Duma et al., 2026). Though trajectory studies confirm that agents rarely catch a mistake nothing flags for them (Rahman et al., 2026) and report wrong work as complete (Mehta, 2026; Advani, 2026; Smyth et al., 2026), they are typically evaluated on a single request, from human-written code, verified once (Jimenez et al., 2024; Deng et al., 2025).

Even though a growing body of work proposes to evaluate agents over successive requests (Deng et al., 2026; Lam et al., 2026; Orlanski et al., 2026; Chen et al., 2026; Shastry et al., 2026; Jin et al., 2026; Shen et al., 2026; Sobal et al., 2026), we identify in our review (Section 2) that prior work either carries an agent's verified and unverified work forward as one, even against an oracle control (Jin et al., 2026; Deng et al., 2026), so a later failure cannot be told apart as building on agent-written code or on a mistake nobody checked (Peng et al., 2026), or stops at the first failure (Shen et al., 2026; Sobal et al., 2026) and never continues from unverified work. We argue that unattended development, where mistakes go unnoticed (Rahman et al., 2026; Mehta, 2026; Advani, 2026; Smyth et al., 2026) and get built on (Rahman et al., 2026; Cuadron et al., 2025; Zhu et al., 2025; Laban et al., 2025), can only be measured by controlling what it lacks: verification.

In this work, we close this gap by creating a benchmark for continuous, unattended development – BOBA-BENCH – built from the commit histories of 140 real repositories. At a high level, the chaining process we propose transforms a stretch of a repository's history into a *chain* of 20 requests, specified from the tests the developers added, that jointly rebuild their changes. Autonomous evaluation then hands each request to the agent in a fresh conversation, on the state the previous request left behind, with hidden tests after every request telling verified work from a *silent miss*: the chain runs through all 20 requests, and the agent, told nothing and remembering nothing of the earlier requests, builds each one on whatever they left, verified or not.

On the 280 chains that we experimented on, we observed that agents left alone completed an average of 51% of chains (pass@1, estimated from eight attempts per chain), and only 40% without a single request missed or earlier test broken along the way, although the same agents complete 66–84% of the individual requests those chains are made of (Figure 1). Notably, we observe silent misses even within the first five requests, and across all ten agents we test, from open-weights (GLM-5.3-Flash) to state-of-the-art (Claude Opus 5.5).

We then decompose the degradation into two components: (1) who wrote the code beneath a request, and (2) whether it was verified. On verified code, agents leave a foundation nearly as buildable as the developers' own, within 4 points for every model. By contrast, all agents fail far more often on an unverified state, regardless of aptitude. We call this the *built on broken assumptions* phenomenon: what limits coding agents left alone is not the code they write but the mistakes they cannot see: without verification, every request that follows is built on broken assumptions.

We investigate several explanations for this effect and show that the agents tend to (1) submit work as complete on the strength of their own tests, which barely lower the chance of a miss, leading them to (2) proceed onto a broken state unaware, (3) fail more as such mistakes accumulate, and (4) accept or reject finished work almost indiscriminately when asked to review it.

Finally, we test what the missing verification buys. Told only that verification failed, and given one repair round, agents fix a tenth to a third of their misses, a quarter to a half when the failing test is named. Delivered after every failed request, the bare message raises chain completion by 1 to 22 points, at the price of 3 to 9 points more chains with a broken test (§5).

Our findings highlight a gap between how coding agents are used in practice and how they are being evaluated. Ubiquitous silent misses are likely a reason why human verification remains the bottleneck (Krentsel et al., 2026; Winninger, 2026; Garousi, 2026), particularly for teams whose review of agent work is already thinning (Duma et al., 2026; Yu et al., 2026; Ye et al., 2026).

The rest of the paper is structured as follows: Section 2 reviews prior work. Section 3 describes the benchmark, its protocol, and the ten agents we evaluate, and Section 4 presents and analyzes our main experiment. Finally, Section 5 tests external verification and draws recommendations.

2 BACKGROUND AND RELATED WORK

Early code-generation benchmarks asked models to implement isolated functions, checked against unit tests (Chen et al., 2021). As models began to edit real projects, repository-level benchmarks gave agents an issue and a human-written codebase and graded the resulting patch once (Jimenez et al., 2024; Deng et al., 2025). Later benchmarks enlarged the task to features, releases, and repositories (Zhou et al., 2026; Le et al., 2025; Zhao et al., 2025; Yang et al., 2026), but each task still began from a reference state and ended with a single verdict.

Table 1: BOBA-BENCH next to representative coding benchmarks. Bold: best among the repository-evolution benchmarks; underlined: second.

				Graded patches		95% CI	Per task		Task
Benchmark	Granularity	Tasks	Repos	per task	total	(±pp)	spec. words	patch lines	collection
Issue resolution									
SWE-bench Verified (OpenAI, 2024)	Commit	500	12	1	500	4.4	189	14	Rule-based
SWE-bench Pro (Deng et al., 2025)	Commit	1.9k	41	1	1.9k	2.3	462	170	Rule-based
CodeThread (Patel et al., 2026)	Commit	1.4k	78	1	1.4k	2.6	n/r	n/r	Rule-based
Codebase generation									
FeatureBench (Zhou et al., 2026)	Feature	200	24	1	200	6.9	4.8k	790	Test-driven
SWE-EVO (Le et al., 2025)	Release	48	7	1	48	13.6	2.4k	611	Rule-based
Commit0 (Zhao et al., 2025)	Project	54	54	1	54	12.9	>7k	>3k	Rule-based
DeepSWE (Huang et al., 2026)	Feature	113	91	1	113	9.1	302	701	Manual
Repository evolution									
SlopCodeBench (Orlanski et al., 2026)	Checkpoint	<u>36</u>	synth.	5.4	<u>196</u>	<u>15.5</u>	4.3k	n/r	Manual
SWE-Chain (Lam et al., 2026)	Release	12	<u>9</u>	12.9	155	24.6	<u>15.5k</u>	**8.9k**	Agentic
SWE-Milestone (Deng et al., 2026)	Milestone	7	7	<u>14.0</u>	98	29.8	**18.9k**	6.9k	Agentic
BOBA-BENCH (ours)	Request	**280**	**140**	**20**	**5.6k**	**5.8**	12.1k	<u>8.2k</u>	Agentic

Recent benchmarks instead evaluate agents over successive requests on the same codebase, so an agent's earlier implementation becomes the foundation for later work (Deng et al., 2026; Lam et al., 2026; Orlanski et al., 2026; Chen et al., 2026; Jin et al., 2026; Shen et al., 2026; Sobal et al., 2026). This exposes what single-request evaluation cannot: regressions, accumulated errors, and declining performance as development continues.

Crucially, these benchmarks differ in what happens after an agent fails. Some carry the failed work forward with everything else (Deng et al., 2026; Jin et al., 2026), so later failures mix the difficulty of building on agent-written code with the cost of building on a mistake nobody checked (Peng et al., 2026). Others stop at the first failure (Shen et al., 2026; Sobal et al., 2026), so the remaining evaluation concerns only verified code. Unattended development guarantees neither: agents can report wrong work as complete (Mehta, 2026) and proceed as though it were right (Rahman et al., 2026). We therefore carry the agent's work forward, verified or not, grade every request, and make verification, not authorship, the central variable of our evaluation.

Separating the two requires controls that prior work provides only in part. CodeThread compares a follow-on issue on human-written and agent-written code (Patel et al., 2026), and EvoCode-Bench compares reference states with agent prefixes that passed cumulative verification (Shen et al., 2026). We run the same later request on human code, verified agent code, and unverified agent code, and then add repair: comparisons that separate what an agent leaves behind when right from when wrong.

Measuring these effects requires checking every request as the code evolves. Static measures describe the implementation (Paul et al., 2025; Sun et al., 2026), while completion of the current request ignores what earlier requests established. Like continuous benchmarks (Deng et al., 2026; Chen et al., 2026), we test new behavior alongside earlier behavior, with hidden tests from real development histories, so that an unresolved error, a later regression, and a repair stay distinguishable.

Scaling these experiments requires an external check that does not depend on the agent. Agents can debug from execution feedback (Chen et al., 2024) and write their own tests (Mündler et al., 2024), but their self-repair depends on the quality of that feedback (Olausson et al., 2024), and their tests can share their implementation errors (Tao et al., 2026). Model reviewers scale but remain imperfect judges (Wang et al., 2026), and human review does not scale (Krentsel et al., 2026). We use hidden tests as the check, ask whether agents can supply that check themselves, and measure how much external feedback is needed to recover. Appendix J provides supporting details.

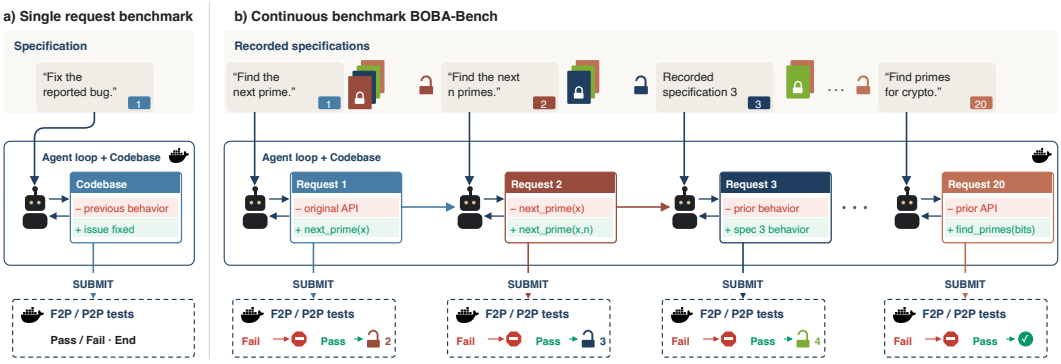


Figure 2: **Autonomous evaluation grades the repository an agent leaves after 20 requests on its own code, not one patch.** (a) One specification, one submission, hidden F2P/P2P tests (SWE-bench shown). (b) BOBA-BENCH: 20 specifications arrive one at a time on one persistent codebase, each in a fresh conversation; after each submission a fresh verifier runs the hidden tests of requests $1..k$ and records the verdict, and specification $k{+}1$ follows regardless.

3 BOBA-BENCH: BUILT ON BROKEN ASSUMPTIONS

To assess whether coding agents can be left alone, we develop a benchmark that repurposes the commit histories of real repositories. First, we apply a *chaining* process that turns a stretch of a repository's history into a *chain* of 20 requests, each specified from the tests the developers wrote. Second, we implement an autonomous evaluation protocol that delivers the requests one at a time on the agent's own code, grades each against hidden tests, and tells the agent nothing about how it did. BOBA-BENCH contains 280 chains from 140 Python repositories, 5,600 requests in all (Table 1).

3.1 CHAINING PROCESS: FROM COMMIT HISTORIES TO CHAINS

Figure 2 contrasts a single-request task, one specification on human-written code graded once, with a chain. A chain is a sequence of *requests*, each derived from a run of consecutive commits in the repository's history that change tests and source together, ending in one behavior change. Each request carries a specification, written from the tests the developers added rather than from their code, and a hidden *contract*: the added tests must fail before the change and pass after it, and the existing tests must keep passing. Taken jointly, the 20 requests rebuild the developers' changes in their order on whatever code the agent has produced so far; *state* k is the code after the first k.

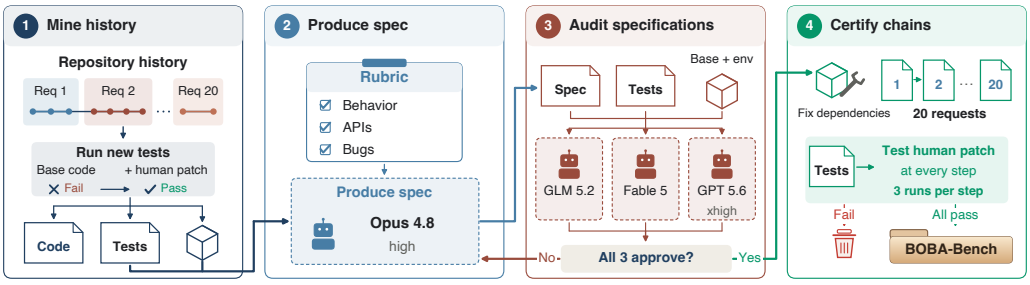


Figure 3: **Requests are specified from the developers' tests, not their code, and certified against their patches.** (1) *Mine history*: consecutive commits that change tests and source together become a request if the new tests fail before and pass after the human patch. (2) *Specify*: Claude Opus 4.8 writes each specification from commit messages, issues and the test diff. (3) *Audit*: three model families must accept identical text. (4) *Certify*: 20 consecutive requests form a chain, sealed with frozen dependencies, in which the developers' patches must pass every hidden test.

Table 2: Chain completion of ten coding agents on the 280 chains of BOBA-BENCH: each request in a fresh conversation on the repository the previous one left, all 20 requests run, eight independent attempts per chain. pass@1: the probability that an attempt leaves the repository after request 20 passing every hidden test of the chain, estimated from the eight attempts (single attempt: the first attempt alone); strict pass@1: in addition, every request passed its own tests on its own step and no passing test ever broke; regression rate: a passing test broke at some request, repaired later or not; pass@8: at least one of the eight attempts passes; request completion: share of all graded requests of the first attempt whose tests pass on the step they are issued. Repository-clustered 95% bootstrap intervals; an attempt without a final state counts as a failure.

	Chains (%)						Requests	Calls per
Model	pass@1	[95% CI]	single attempt	strict pass@1	regression rate	pass@8	completed (%)	chain (median)
Claude Opus 5.5	66.4	[61.0, 71.6]	66.8	55.9	22.9	83.9	84.0	217
Claude Opus 5	61.1	[55.8, 66.3]	60.7	50.7	28.0	82.9	80.8	401
GPT-6 Astra	52.9	[46.5, 59.2]	51.1	38.8	35.2	63.9	75.2	182
Claude Fable 5	51.2	[45.6, 56.8]	51.8	38.2	40.4	72.1	73.2	293
GLM-5.3-Flash	51.0	[45.0, 57.0]	51.1	41.2	29.9	77.5	73.9	739
GLM-5.3	51.0	[45.1, 56.9]	46.4	40.0	30.1	76.1	73.0	560
GPT-6.1-sol	50.1	[43.7, 56.3]	48.9	37.7	37.9	60.7	74.2	190
GPT-5.6-sol	45.0	[38.9, 51.1]	45.7	33.3	42.2	60.0	71.8	299
Kimi K3	43.9	[38.2, 49.7]	45.4	33.7	42.8	68.2	68.8	389
Claude Opus 4.8	39.1	[33.1, 45.1]	38.2	27.7	42.0	60.0	66.2	367

We developed a semi-automatic chaining process to scale the construction of chains (Figure 3). First, requests are mined from the histories of Python repositories: wherever consecutive commits change tests and source together, the run is kept as a request only if its new tests fail before the developers' change and pass after it. Second, Claude Opus 4.8 writes each specification from the test diff, with commit messages and issue text as aids.

Because specifications are written by a model from tests the agent never sees, a chain could fail for reasons that are not the agent's. We therefore attempted every specification blind, from the developers' code just before it: two models solve 78% of the released specifications between them, and the remainder were probed for behavior the tests demand but the text omits, then rewritten or kept as hard but fair. Three model families then read every specification against its hidden tests, rewriting it until all three accepted identical text. Finally, each chain is sealed into one offline image with frozen dependencies, and released only if the developers' own patches pass every hidden test at every state in that image. We believe the process yields chains that are solvable from their text and on which a miss can be attributed to the agent (details in Appendix D).

3.2 AUTONOMOUS EVALUATION

Figure 2b depicts the loop. The **agent** works through the 20 specifications in one offline container, each in a fresh conversation; the **grader** runs in a separate container and sees only the git history the agent commits. On each request, the agent receives the specification, the repository as the previous request left it and a submit tool, edits and runs code and tests, and submits once it believes the specification satisfied. The grader then runs the hidden tests of requests $1..k$ on the submitted code (Appendix D.9): a submission whose own tests fail is a *silent miss*, and a test that passed before the submission and fails after it is a *regression*. Whatever the verdict, the next specification opens a fresh conversation on the same repository, and the chain runs through all 20 requests.

The agent is told neither the verdict nor that it is on a chain: withholding feedback is what unattended development looks like, so whether an agent can tell that its own work is correct is itself measured.

A chain is complete if the repository after request 20 passes every hidden test of the chain and its own; estimated as the mean over eight independent attempts per chain, and the single-attempt value is reported alongside. Because this judges only the final state, we also report, averaged likewise, **strict pass@1**, every request's own tests passing at its own step with no regression ever, which repairing everything at request 20 cannot score, and the **regression rate**, the share of chains with a regression at some request.

3.3 Evaluation Settings

We evaluate agents in five settings.

Autonomous is the benchmark run described above: the agent is left alone and nothing stops the chain before request 20.

Agent code and Human code isolate authorship. Request $k{+}1$ is solved on k requests that all passed their contracts, implemented either by the agent itself, with retries until each is certified, or by the developers, under identical specification, budget and grader; the comparison asks whether agent code is a worse base than human code once both are verified.

Silent continuation isolates verification: request $k{+}1$ is issued on a silent miss as if it had passed, and only failures of that request's own tests are counted, not those it inherits.

Verified takes Silent continuation a step further: after the miss, a message from outside says that verification failed and the agent gets one repair before the next request.

Review asks whether an agent can supply that message itself: a model judges a finished implementation without knowing its author, and accepts or rejects it.

3.4 Scale and Parameters

We evaluate ten models on all 280 chains in the Autonomous setting, eight attempts per chain: Claude Opus 5.5, Opus 5, Opus 4.8 and Fable 5 (Anthropic, 2026d;c;b;a), GPT-6 Astra, GPT-6.1-sol and GPT-5.6-sol (OpenAI, 2026c;b;a), GLM-5.3 and GLM-5.3-Flash (Zhipu AI, 2026b;c), and Kimi K3 (Moonshot AI, 2026). The harness is mini-swe-agent 2.4.6 (Yang et al., 2024); a chain may use unlimited model calls and 2 h of tool execution over its 20 requests, and no wall-clock limit (settings in Appendix D.8). The four other settings use Fable 5, GPT-6 Astra, GPT-5.6-sol, Kimi K3 and GLM-5.2 (Zhipu AI, 2026a) on states of the same chains (Appendices B, C, F and K).

4 Can Coding Agents Be Left Alone?

4.1 Completion Findings

Figure 1 and Table 2 summarize the results, with pass@1 estimated from eight attempts per chain. At a high level, **every model sees its performance degrade from the requests it completes to the chains it completes, with an average chain completion of 51%**. We name this phenomenon Built on Broken Assumptions: models that complete 66-84% of the individual requests they are given[1] complete only about half of the 20-request chains made of *the exact same requests*, since each request is built, in a fresh conversation, on whatever the earlier ones left behind.

At least one of a chain's eight attempts completes it 71% of the time on average (pass@8), so the loss is not all bad luck: the same chains fail every model (16% are completed by none of the ten on the first attempt and 16% by all ten, against fewer than one expected under independence), and difficulty lies in particular requests, of which a stronger model clears more, not different ones.

Strict pass@1 (Table 2) adds the condition that every request passed on its own step and no passing test ever broke: in 23-43% of attempts a later request breaks a test that passed, usually one of the agent's own earlier requests (Appendix F.6), and the 10-14 point gap between the two measures is chains repaired by request 20. The per-request failure rate does not fall as a chain progresses (3-6%, Appendix F.7, Figure 10); compounded over twenty requests it reproduces strict pass@1 but not pass@1, because what follows a failure is seldom recovery: only 14-33% of the tests that fail are ever repaired, and a request issued on a broken repository fails about twice as often as one on a clean one, 9-19 points more than the same request from the developers' state (Appendix F.3; §4.2, §5).

[1] Counting only the requests up to and including a chain's first broken test gives 77-89%.

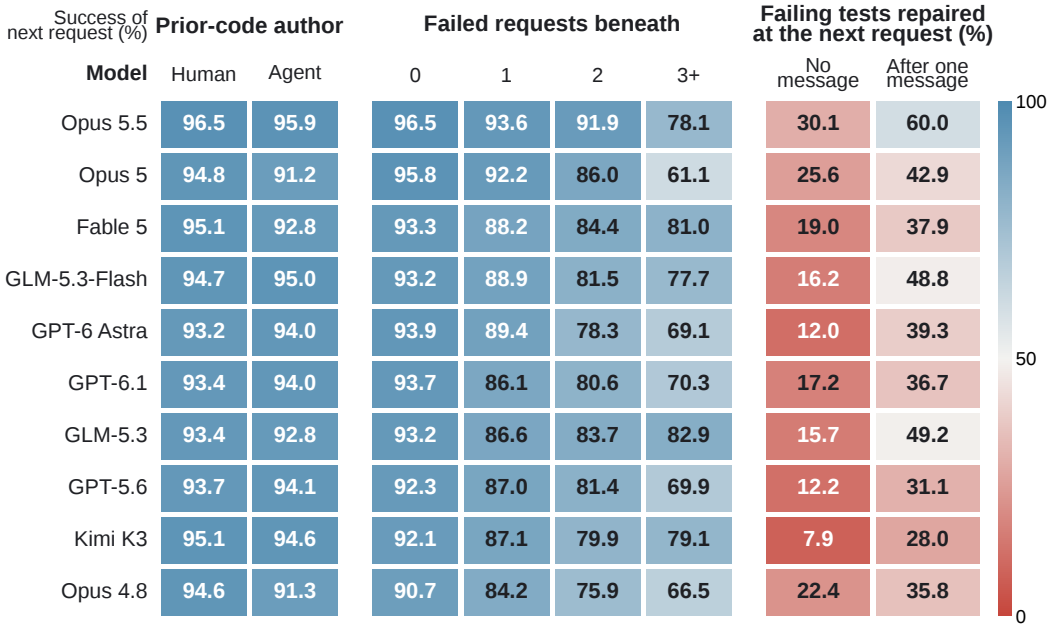


Figure 4: **Verified foundations hold whoever built them, unverified failures beneath a request erode its success, and one outside message raises their repair from 8–30% to 28–60%.** Success rate (%) of request $k{+}1$ by the state beneath it: who wrote the certified code (human or agent) and how many earlier requests were still failing (0, 1, 2, 3+); last two columns: newly failing tests repaired at the next request without and after one "verification failed" message (§5).

A different harness does not help: under Codex CLI, the OpenHands SDK and pi, four models complete no more chains than under mini-swe-agent, often fewer, and fail largely at the same requests (Table 44); **the harness does not, on its own, allow models to verify their own work**. Nor does memorized code explain the effect: three Claude models, including the leader, memorize code of these repositories, but no more of benchmark code than of other code, and none completes the repositories it memorized measurably more often (Appendix E).

4.2 Authorship vs. Verification Analysis

In Table 2, every request is built on the agent's own earlier work, whether or not that work was correct; we now separate who wrote that code from whether it was verified. Figure 4 visually summarizes the results. First, looking at authorship, we compare the AGENT CODE and HUMAN CODE settings (Appendix B). We see that models that build on their own verified code do nearly as well as models that build on human code: across eleven models the gap runs from -3.6 to $+0.8$ percent points, an average drop of 1 percent point, and excludes zero for three (Claude Opus 5, Opus 4.8 and GLM-5.2, at 3 to 4 points). In summary, **when the code beneath a request is verified, models leave code nearly as buildable as a human's.**

The verification analysis paints a different picture. In the SILENT CONTINUATION setting (Appendix F), failure of the next request's own tests rises sharply after a single unverified mistake, with an average increase of 18 percent points (nearly tripling). Part of this is the request rather than the state, since the requests that follow a miss are harder ones; attempted from the developers' own state at the same point, the same requests still fail 9 to 19 points less often than from the unverified state (Appendix F.3). More interestingly, though better models tend to fail less often on a verified state, all models tend to pay a similar price for an unverified one. In other words, **on a state that nobody verified, all models we test exhibit very high failure, with the odds of failing roughly doubling with every additional unverified mistake beneath a request**. This refines our definition of the *built on broken assumptions* phenomenon: when comparing verified and unverified states, we find that large performance degradations are due in large part to mistakes going unverified, rather than to who wrote the code.

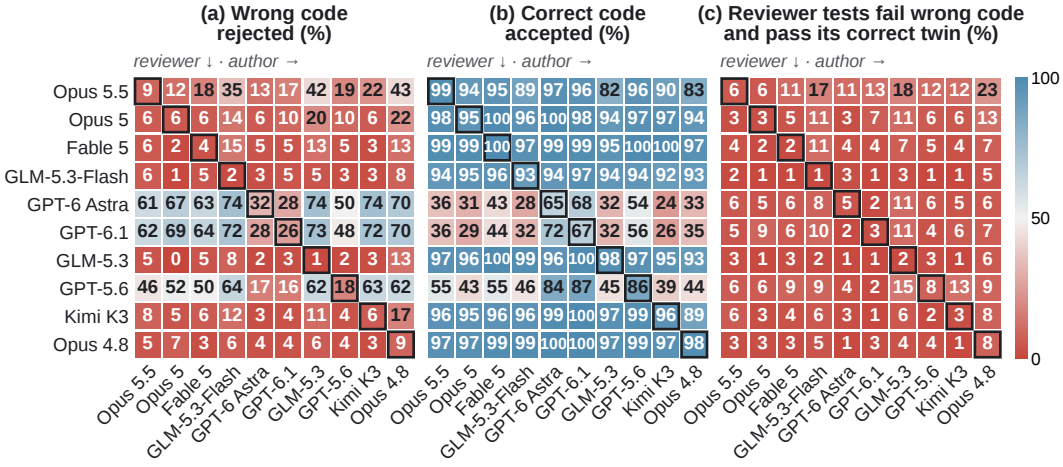


Figure 5: **No model is a reliable reviewer: lenient reviewers accept most wrong code, strict ones reject about half of the correct code.** (a) Wrong code rejected, (b) correct code accepted, (c) pairs whose wrong twin fails and correct twin passes the reviewer's tests (%). Framed cells: self-review.

Appendix F.6 explores potential root causes for models building on broken assumptions. We identify four causes: (1) agents break behavior the request did not touch, the most common cause for all models but one (Appendix F.6.1), (2) they leave part of the specification unimplemented (Appendix F.6.2), (3) they implement the wrong logic or leave the new code unreachable (Appendix F.6.3), and (4) they trust their own tests, which barely lower the chance of a miss (Appendix F.6.4).

4.3 Cross-Model Review Experiment

The AUTONOMOUS setting leaves every request unreviewed, which may not reflect how agents are deployed in practice: a second agent could be asked to review each change, as prior work proposes (§2). We therefore ask whether an agent can keep models from building on broken assumptions.

In the cross-model review experiment, each of the ten models reviews 2,180 items in the REVIEW setting: 109 wrong implementations per author from the AUTONOMOUS runs, each paired with the same author's correct implementation of the same request from another attempt (Appendix C).

Figure 5 summarizes the results. In this setting none of the ten models separates wrong from correct implementations well enough to serve as the check: lenient reviewers (the four Claude models, GLM-5.3, GLM-5.3-Flash and Kimi K3) accept 77–95% of wrong implementations, strict reviewers (GPT-6 Astra, GPT-6.1 and GPT-5.6) reject 41–58% of correct ones, and 4–19% of matched pairs are discriminated (Youden $J \leq 0.15$; Table 12). In other words, **a pipeline that relies on one of these agents to review another, with the information given here, does not stop models building on broken assumptions**.

5 Trust, but Verify: What an Outside Check Recovers

Section 4 leaves a practical question: agents build on broken assumptions and cannot tell they have (§4.2, §4.3), yet nobody who deploys an agent has our hidden grader, and few will write a test for every request. Anyone who runs the result, however, can tell something is wrong without knowing what. **What is the least such a watcher must say for the agent to recover**, so that verifying costs as little as possible? We start from one bit, *verification failed*, delivered once after a silent miss with one repair round (the VERIFIED setting of §3.3, in one conversation per chain), and ask whether repeating it, informing instead, or getting it wrong changes what it buys (Appendix K).

5.1 Implications for System Builders

The system around an agent has no test of its own, so it cannot judge a submission; only whoever wrote the request can. What it can do is keep a line open in both directions, so that the verifier sees the results and the model receives the verdict. Does one bit along that line suffice?

Told only that verification failed, **the models repair a third to a half of their misses in one round** (Table 37), and **a repaired state is no longer a broken assumption**: the next request fails on it no more often than on a certified state (Figure 4), whereas an unrepaired one fails like any silent miss (§4.2). The signal, not the fresh start, does this: restarting on the same state without the message does not help (Appendix K). Applied after every failed submission of a project, **completion rises by 8-16 points for every model** (Table 37). Under the benchmark's protocol the bit raises pass@1 by 1 to 22 points, seven of ten intervals above zero (one attempt per arm, paired with the first attempt of the AUTONOMOUS run; Table 41). The repair round, blind to the work it mends, also raises the regression rate by 3 to 9 points and adds 47 to 294 calls per chain.

How much the verifier sees bounds how much it can say. For GPT-5.6 and Kimi K3, repeating the bare rejection after each failed repair helps with quickly diminishing returns, from 30 to 54-59% over eight rounds, with no round after the fourth adding more than 3 points and more than 40% of the misses remaining (Appendix K). **Naming the failing tests once achieves what insisting achieves in about three rounds**: insisting repeats the agent's belief about what it built; naming the test replaces it with a fact.

In short, system builders should keep that line open in both directions and widen what the verifier sees rather than multiply rounds: **more attempts are an inefficient substitute for telling the agent what is wrong.** Multi-agent systems are especially exposed, since models reviewing a finished implementation barely beat chance (§4.3): the verifier's line must reach the subagents directly, not stop at the agent that delegated.

5.2 Implications for Model Builders

Even a perfect supervisor recovers only part of the loss: more than half of the misses survive a repair round, 22-59% of projects still end short (Table 37) and 26-55% of chains under the benchmark's protocol (Table 41), because the agent must find from one bit what it got wrong, cannot tell its wrong implementations from its correct ones (§4.3), and reports a wrong request complete a few times in a hundred (§4.1).

We therefore call on model builders to prioritize certifying a request as much as implementing it. **An agent that can be left alone should: (1) tell its wrong implementations from its correct ones, where the best model we test barely beats chance; (2) repair a miss in one round when told only that verification failed, where the models we test repair a third to a half; (3) achieve both without a supervisor, so that a chain does not depend on who is watching.**

5.3 Implications for Benchmark Builders

What lets an evaluation observe agents building on broken assumptions is not the harness (under three other harnesses no model completes more chains, Table 44) **but three properties of the setting**: the agent's work persists across requests, correctness is decided by a contract the agent does not see and cannot rewrite, and the requests depend on one another. A single-request benchmark has none of the three, which is why the phenomenon has gone unmeasured.

We hypothesize that agents in any such setting will build on broken assumptions, and encourage benchmark builders to grade agents in chains, to report completion with and without a supervisor, and to publish the hidden contracts alongside the specifications, as we do.

5.4 Implications for Users

Users who delegate a sequence of requests should expect an agent to report a wrong request as complete and build on it. We make two practical recommendations.

Check before you build on it. If a request is the foundation of the next, run it or try the feature before issuing the next request rather than after: **a state built on a broken assumption fails the next request far more often (§4.2), whereas a repaired one is as reliable as one that was never wrong** (Figure 4). Checking after the fact is harder: by then the next request has already been built on the mistake.

Say what you saw. A wrong "wrong" costs little on its own: when we rejected correct submissions at random with probability 0.1 to 0.5 (GPT-5.6 and GPT-6 Astra, 100 chains; Appendix K), about 98% of the phantom repair rounds left the work passing. But alarms add up over twenty requests (Table 37): **rejecting one correct submission in ten costs at most 2 points of completion, and rejecting every other one gives back most of the gain**, though never below the level without a supervisor.

These are patches, not a principled approach; once agents can certify their own work, users will be able to leave one alone with less risk of it building on broken assumptions, and recommendations like these will no longer be needed. Until then, the watcher's check is the one that works.

5.5 Release

Everything needed to reproduce and extend this work is available at `https://github.com/amazon-agi/BOBA-bench`: the benchmark and its construction pipeline, the data behind it (specifications, hidden contracts, gold patches and sealed images for every chain), the reduced records of every evaluated run (attempt outcomes and hidden-test verdicts), and the grader and analysis scripts behind every table and figure in this paper (Appendices D.12 and G). Images are pinned by digest and every artifact by hash, so results can be reproduced against the exact states we graded. Source code stays under each repository's license (Appendix A).

6 Limitations

BOBA-Bench measures fixed-test correctness, not user intent, readability or production readiness; it is Python-only, many repositories are small, and twenty requests do not imply a twenty-link dependency chain. Headline pass@1 judges only the final repository, crediting chains that missed or broke something on the way; strict pass@1 does not (Table 2). Specifications are model-written and model-audited, not yet human-reviewed (Appendix D.5). The repositories come from public agent-task pools (Badertdinov et al., 2025; Fu et al., 2026) and nearly all changes predate the model cutoffs; three Claude models, including the leader, reproduce 14–19% of sampled change blocks near-verbatim yet do not measurably complete more chains of the repositories they memorized ($\pm$12–18-point intervals bound but do not exclude an effect; §4.1, Appendix E).

7 Conclusion

In this work, we evaluate coding agents on twenty-request chains drawn from real repository histories, and find that on a fixed set of requests, agents that complete most of them individually complete only about half of the chains they form, and fewer than two in five without a single request missed or earlier test broken along the way. Coding agents build on broken assumptions, which materializes not as a loss of ability, since verified agent code is nearly as good a base as human code, but as a failure of verification: a mistake the agent does not notice becomes the foundation of everything after it, and agents asked to review barely tell right from wrong. Additional experiments show that a single external "that's wrong" recovers part of the loss, whereas a fresh start without it does not, and we call on model builders to make certifying a request as much a goal as implementing it.

AI USE STATEMENT

Large language models are the subject of this study, and we also used them as tools; we say how, following the categories of the ICLR 2027 AI Policy. *Synthetic data and dataset cleaning (used).* Language models wrote every specification from the hidden test diff, commit messages, and linked issues and pull requests. Calls rotated over Claude Opus 4.6, 4.7, and 4.8, with GLM-5.2 as a fallback during outages; the model behind each specification was not recorded (Appendix D.6). Claude Opus 4.8 attempted each candidate specification blind, GPT-5.6 attempted those it did not solve, and Fable 5 and GPT-5.6 probed the rest for missing behavior (Appendix D.4). GLM-5.2, Fable 5, and GPT-5.6 audited each specification against its hidden tests, and GPT-5.6 and Fable 5 rewrote rejected ones (Appendix D.5). GPT-5.6 repaired failing environment builds by editing only their Dockerfiles, and sealing agents worked through a per-image checklist (Appendix D.7). GPT-5.6 assigned repository domain labels (Appendix A). We treat these outputs as generated data: the developers' hidden tests decide every grade, and every image must pass the 20 human patches at every state. *Qualitative and thematic analysis (used).* A judge model (GPT-5.6-sol) assigned one of 17 failure tags to each of the 4,787 failed requests of the first attempts, and a sandbox judge (Claude Fable 5.1) re-executed the 1,733 it left unresolved in the sealed image (Appendix F.6); one judge labels each stage, no site was relabeled by a second judge or by a human, and the agreement we report between two model judges comes from an earlier census with the same prompt and schema. A language model also labeled transcripts to check the self-verification rules of Appendix I. These labels are reported as indicative; every headline number comes from the hidden tests, not from a model's judgment. *Implementing methods and writing code (used).* AI coding assistants helped write and edit code under the authors' direction, including the agent scaffold, the grader, and the analysis scripts; the authors reviewed this code, and the grader is validated by requiring every sealed image to pass the 20 human patches at every state (Appendix D). *Methodology, hypotheses, and interpretation (used as assistants).* AI assistants provided feedback on experimental design and on the interpretation of results; the research questions, the experimental protocol, and the conclusions drawn are the authors'. *Not used or not applicable.* We did not use generative AI to develop theoretical models, formulate or prove mathematical claims, or translate text; the paper contains no proofs. Aside from the labeling above, no model judged any result. *Writing (used).* AI assistants helped with literature search, feedback on the manuscript, and drafting and editing of text, including figures and tables from the analysis outputs. The authors reviewed all AI-assisted work, checked every number against its source, and take full responsibility for the content of this paper, including text, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

BOBA-Bench is built from the public histories of 140 open-source Python repositories, all under open-source licenses: MIT (65), Apache-2.0 (36), BSD (19), GPL-3.0 (12), LGPL-3.0 (5), and one each under GPL-2.0-or-later, LGPL-2.1, and AGPL-3.0 (Appendix A). Source code stays under each repository's license. No human subjects took part. Appendix D describes grader-defeat techniques found during construction so that benchmark builders can defend against them; the grader rejects them.

REPRODUCIBILITY STATEMENT

Section 3 and Appendix D describe construction end to end: mining and contract validation, the verbatim specification prompt (Appendix D.6), the specification audit, environments and sealing, grader rules, and every filtering round, including the post hoc 14-chain exclusion (Appendix D.10). Appendix A reports task statistics. Section 3.4 and Appendix F give models, settings, and budgets. Appendices B, C, and F.3 give the per-model estimates, and Appendix H gives the paired tests with their seed. Appendix G lists the population of each study. The 280 tasks with image digests, the grader, and the analysis scripts are released with the benchmark (Section 5.5).

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A TASK STATISTICS

All distributions in this appendix are over the 280 chains of Section 3 (5,600 graded states, 4,404 unique segments, 140 repositories); the unique-segment medians are within 3% of the occurrence medians for every size metric. Intervals are repository-clustered percentile bootstraps (2,000 replicates) for means and shares and Wilson 95% intervals where stated, because chains from one repository are not independent.

B MATCHED FOUNDATIONS: PROTOCOL AND CONTROLS

B.1 PER-MODEL RESULTS

Tables 9 and 10 give the per-model estimates behind Section 4.2 on the 140 repositories, for the five models of the earlier study and for the six added with the leaderboard: next-request success from the agent's certified prefix and from the human prefix over eight attempts per boundary, the agent-minus-human gap with 95% repository-bootstrap intervals (boundaries averaged within repository, then across repositories), the co-primary pass@8 gap, the gap by depth, the sensitivity that excludes boundaries whose next request already passed at k, and two descriptive counts: the share of boundaries where both prefixes pass all eight attempts, and the number of boundaries where the agent prefix passes at least four fewer attempts than the human prefix. The six added models were run with the same windows, depths, retry cap, attempt counts and grader, under the settings of their leaderboard runs.

B.2 ABLATION: EARLIER REQUESTS' TESTS VISIBLE IN THE REPOSITORY VERSUS HIDDEN

Matched Foundations was run twice with one difference: whether the human tests of the already-accepted requests were present in the repositories the model worked in. In the first run, a constructed request was graded by applying its hidden test patch inside the agent's own container and the resulting tree was committed, so the certified agent prefix at boundary k carried the human tests of requests $1, \ldots, k$; the same patches were applied to the human prefix, so both arms saw them. For request $k{+}1$ these are its P2P tests (behavior that must keep working), never its F2P tests, which stayed hidden in both runs. In the second run, the one reported in the main text, the agent's tree is snapshotted before any hidden material enters the container and a throwaway copy is graded, so no hidden test appears in either prefix, matching the benchmark. Windows, depths, retry cap, scaffold and model settings are identical, so the boundaries complete in both runs form a paired ablation of P2P-test visibility (Table 11). Visible P2P tests lift both arms by 4–5 pp for the two models with clean runs in both conditions (GPT-5.6, Fable 5), but the agent-minus-human gap does not move: the differences are -0.4 to $+1.5$ pp with every interval spanning zero. The visible tests supplied regression protection and interface leakage (names, signatures and conventions the next contract assumes), and supplied them equally from either foundation. GLM-5.2's visible-tests construction ran through a proxy incident and Kimi K3's on a prefix-cached fleet that its hidden-tests run did not have, so their arm-level differences are not interpretable as a visibility effect; their gap differences are. GPT-6 Astra was run only with tests hidden.

C CROSS-VERIFICATION: PROTOCOL AND CENSUS

C.1 PER-MODEL RESULTS

Table 12 gives the per-model estimates behind Section 4.3. Each of the ten models contributes 109 matched pairs as an author: a wrong implementation from its first attempt at some request, whose own tests failed or which broke an earlier test, and the same request passed from a correct state by the same author in another attempt; both states are regraded in the sealed environment and the git history is rewritten to carry no identifying metadata. Every review runs in a fresh container with the reviewer's leaderboard settings and a budget of 60 model calls, 45 minutes of tool execution and 90 minutes of wall time; the reviewer may write and run tests and returns accept, reject or abstain. The review rows are equal-weight aggregates over the ten authors, with repository-clustered bootstrap 95% intervals; a review that ends without a decision counts as neither an acceptance nor a rejection.

Table 3: Repositories (n=140). Size is measured at the base commit of the repository's first chain; stars queried 2026-09-09 (packaging 2026-08-27..30). Domain labels: GPT-5.6 from public metadata (fixed 13-way list; a keyword heuristic agrees on 53/140). Test style: pytest is the runner for every chain; style is inferred from the chains' own F2P test ids and test patches.

	p10	p25	median	p75	p90	max
GitHub stars	6	14	46	195	603	5,835
code LOC at first chain base	79	215	424	2,146	4,914	33,731
Python LOC	76	212	420	1,907	4,801	33,684
files	8	11	18	37	66	325
test LOC	9	29	97	538	1,317	11,574
chains per repository	1	2	2	2	3	5

Chains per repository: 34 repos with 1, 81 repos with 2, 18 repos with 3, 5 repos with 4, 2 repos with 5. **Primary language (GitHub):** Python 139, LookML 1; median Python share of code LOC at chain base 100% (12 repos below 90%). **Secondary languages present (repos):** Shell 15, JavaScript 10, CSS 7, HTML 5, C 2, TypeScript 2, C++ 1, Lua 1, SQL 1. **Domain:** api-client-sdk 21, language-parsing-analysis 17, scientific-numerical 15, general-utility 14, devtools-testing-packaging 13, data-engineering-formats 13, infra-cloud-ops 12, iot-hardware-home 10, web-backend-apps 9, media-games-graphics 6, text-docs-nlp 4, security-identity 4, finance-business 2. **Test style:** unittest.TestCase classes 63, pytest functions 60, pytest test classes 16, mixed 1; 51% of all F2P test ids are class-based. **Stars:** $<$10: 20, 10–49: 51, 50–99: 20, 100–499: 31, 500–999: 8, 1000–4999: 9, $\geq$5000: 1; 6 repositories archived; licences (all 140 open source): MIT 65, Apache-2.0 36, BSD 19 (3-clause 13, 2-clause 6), GPL-3.0 12, LGPL-3.0 5, and one each GPL-2.0-or-later, LGPL-2.1 and AGPL-3.0.

Table 4: Per-segment statistics over the 5,600 segment occurrences (280 chains $\times$ 20). Mean is followed by its repository-clustered bootstrap 95% CI. Source commits: commits folded into the segment by the generation walk (see text). Contract nodes at k = F2P$_k$ $\cup$ P2P$_k$ $\cup$ still-valid earlier F2P.

	p10	p25	median	p75	p90	max	mean [95% CI]
Reference (gold) patch							
files changed	1	2	3	7	13	224	6.0 [5.3, 6.7]
hunks	2	4	7	15	30	1,654	15.5 [12.4, 19.3]
lines added	6	16	50	146	384	81,022	276.8 [179.8, 404.1]
lines removed	1	3	14	51	163	44,855	134.2 [83.8, 201.1]
lines changed (+/−)	9	24	73	215	596	94,155	411.1 [267.3, 579.1]
new files	0	0	0	1	2	135	1.0 [0.8, 1.2]
new functions (net new `def` names)	0	0	1	3	7	280	3.2 [2.7, 3.7]
new classes	0	0	0	1	2	469	1.3 [0.7, 2.5]
Hidden test patch and contract							
test files changed	1	1	1	2	4	73	2.5 [2.2, 2.8]
test lines added	4	12	31	79	178	24,704	109.0 [77.4, 165.4]
new test functions	0	0	1	3	8	160	3.3 [2.9, 3.7]
F2P tests	1	1	2	7	18	707	8.7 [6.8, 11.5]
P2P tests	7	22	45	91	208	1,303	93.3 [63.2, 136.3]
earlier F2P still demanded (accumulated)	2	9	23	45	87	835	41.9 [32.6, 54.6]
contract tests at k	14	29	51	100	216	1,312	102.0 [70.5, 145.2]
superseded earlier F2P (not demanded)	0	0	1	7	19	2,050	12.2 [5.8, 23.5]
Specification							
specification words	190	309	493	780	1,197	3,693	606.6 [577.1, 641.5]
spec words per test line added	4.0	7.2	14.6	32.2	80.9	1,419.0	36.2 [32.4, 40.3]
spec words per F2P test	33.9	80.4	178.0	352.6	598.0	3,068.0	267.8 [247.4, 289.4]
Source history							
source commits folded into the segment	1	1	3	6	12	163	5.2 [4.7, 5.8]
merge commits among them	0	0	0	1	2	35	0.8 [0.6, 0.9]
distinct authors (e-mail)	1	1	1	2	3	35	1.6 [1.5, 1.7]
segment duration, days (prev. target $\rightarrow$ target)	0.00	0.04	0.95	8.69	49.29	2,895.97	21.25 [16.63, 26.19]

Shares (Wilson 95%): P2P non-empty 96.0% [95.5, 96.5]; more than one source commit 68.9% [67.7, 70.1]; contains a merge commit 33.3%; more than one author 33.1%; creates a new file 28.7%; adds a function or class 62.4%; touches a non-Python file 56.0% (pure non-Python 0.20%); adds a new test file 23.9%; some earlier F2P superseded at k 64.6%; specification rewritten by a model during the spec audit 24.5%.

Table 5: Per-chain statistics (n=280 chains of 20 segments). Dependency depth = longest path in the DAG whose edges $j \to k$ ($j < k$) mark a segment whose gold or test patch references a symbol introduced, or imports a module created, by an earlier segment's gold patch (symbol/import definition), or whose gold patch touches a file the earlier gold patch touched (file definition).

	p10	p25	median	p75	p90	max	mean [95% CI]
gold lines added (20 segments)	781	1,246	2,177	4,274	7,013	121,416	5,536.6 [3,682.0, 8,007.4]
gold lines removed	278	442	775	1,836	3,396	56,611	2,684.6 [1,643.7, 4,033.9]
net gold lines	242	499	987	2,311	4,674	89,154	2,852.0 [1,552.4, 4,631.0]
distinct files touched by gold	11	16	25	45	74	297	38.0 [32.3, 44.7]
distinct test files touched	4	6	10	18	30	309	16.8 [13.2, 21.5]
new files created	2	4	10	24	46	201	19.1 [15.6, 23.4]
new functions	15	24	42	77	155	568	63.3 [53.2, 74.7]
test lines added	541	803	1,197	1,904	3,896	82,930	2,179.9 [1,546.1, 3,126.1]
F2P tests (sum over segments)	47	67	106	187	320	3,632	174.2 [137.7, 226.7]
contract tests at k=20	30	44	72	128	254	1,300	126.4 [93.2, 167.9]
specification words (sum)	7,794	9,583	11,502	13,696	16,518	34,430	12,132.9 [11,500.6, 12,804.8]
spec words per test line added	3.6	6.5	9.8	14.1	18.6	38.5	10.9 [9.8, 12.0]
spec words per F2P test	38.4	60.5	109.2	170.3	243.7	678.8	128.9 [114.7, 144.1]
segments with non-empty P2P (of 20)	18	19	20	20	20	20	19.2 [18.8, 19.5]
dependency depth, symbol/import (max 19)	2	4	6	8	11	17	6.2
dependency depth, file overlap (max 19)	14	16	18	19	19	19	17.0
code LOC at chain base	163	331	1,232	3,155	5,867	33,889	2,833.4 [1,980.3, 3,993.8]
files at chain base	9	14	28	50	85	336	42.4 [34.4, 53.5]
net gold lines / code LOC at base	0.07	0.24	0.90	4.61	10.31	125.97	4.62 [3.23, 6.21]
source commits folded into the chain	43	57	85	129	196	422	104.8 [94.5, 115.4]
commits before the chain base	1	4	40	116	200	529	80.6 [66.0, 96.7]
commit-date span, days	18	44	200	520	1,143	3,426	425.0 [336.0, 522.5]
median segment duration, days	0.05	0.19	0.99	4.54	14.97	77.48	5.09 [3.46, 6.96]
distinct human authors	1	2	4	9	15	39	6.5 [5.5, 7.5]

Shares: all 20 segments carry P2P 66.8% [61.1, 72.0]; single human author 14.6% [11.0, 19.3]. **Distinct human authors per chain:** 1: 41, 2: 47, 3–4: 62, 5–9: 72, $\geq$10: 58 chains. **Commit-date span:** $<$1m: 54, 1–3m: 46, 3–6m: 30, 6–12m: 46, 1–2y: 57, 2–4y: 29, $\geq$4y: 18 chains.

Table 6: Segment dependency structure over the 5,320 segment occurrences at positions 2–20. "Any earlier": some $j < k$ in the same chain satisfies the definition; "$k-1$": the immediately preceding segment does. CIs are repository-clustered bootstrap 95%.

definition	any earlier $j < k$	$k-1$	nearest lag (med.)	chain depth med. (IQR)
gold patches share a file	98.1% [97.2, 98.8]	89.2% [86.8, 91.5]	1	18 (16–19)
references a function/class touched by j	79.3% [75.5, 82.8]	40.1% [36.7, 43.5]	1	8 (5–11)
references a function/class introduced by j	72.3% [68.3, 76.2]	28.2% [25.6, 31.0]	2	6 (4–8)
strict (names ≥ 4 chars, non-generic)	68.2% [64.5, 71.9]	25.5% [23.1, 28.0]	2	5 (3–7)
gold-patch references only	64.6% [60.7, 68.6]	24.4% [21.8, 27.0]	2	5 (3–8)
test-patch references only	54.1% [49.8, 58.2]	15.0% [13.2, 16.9]	3	3 (2–5)
imports a module created by j	19.0% [15.8, 22.2]	4.0% [3.1, 4.9]	3	1 (0–2)
symbol introduced OR module import (headline)	73.3% [69.6, 77.0]	28.8% [26.1, 31.6]	2	6 (4–8)
file OR symbol OR import	98.9% [98.4, 99.3]	90.5% [88.2, 92.4]	1	18 (17–19)

Share of segments depending on the segment exactly ℓ positions earlier (symbol/import): ℓ=1: 29%, ℓ=2: 27%, ℓ=3: 24%, ℓ=5: 23%, ℓ=10: 21%, ℓ=15: 22%, ℓ=19: 27%. File overlap: ℓ=1: 89%, ℓ=2: 85%, ℓ=5: 78%, ℓ=10: 72%, ℓ=19: 66%.

Table 7: A BOBA-BENCH segment next to a single-request task. Median (IQR) per instance; identical parser for all rows. SWE-bench Verified from `princeton-nlp/SWE-bench_Verified` (500); SWE-bench Pro public set from `ScaleAI/SWE-bench_Pro` (731; Go 280, Python 266, JS 165, TS 20). Specification = problem statement (Verified) / problem statement + requirements + interface (Pro) / frozen segment specification (BOBA-BENCH).

	BOBA-BENCH segment n=5,600	BOBA-BENCH chain (20 seg.) n=280	SWE-bench Verified n=500	SWE-bench Pro n=731	SWE-bench Pro (Py) n=266
gold files changed	3 (2–7)	25 (16–45)	1 (1–1)	4 (2–7)	3 (2–4)
gold hunks	7 (4–15)	–	1 (1–2)	9 (5–16)	7 (4–12)
gold lines added	50 (16–146)	2,177 (1,246–4,274)	4 (2–10)	63 (33–135)	46 (30–108)
gold lines removed	14 (3–51)	775 (442–1,836)	2 (1–4)	20 (7–52)	15 (5–41)
gold lines changed	73 (24–215)	–	7 (3–13)	94 (50–207)	74 (45–142)
new functions	1 (0–3)	–	0 (0–0)	0 (0–0)	1 (0–3)
test files changed	1 (1–2)	–	1 (1–1)	1 (1–2)	1 (1–2)
test lines added	31 (12–79)	1,197 (803–1,904)	15 (7–27)	46 (17–116)	40 (14–99)
F2P tests	2 (1–7)	106 (67–187)	1 (1–2)	3 (1–7)	4 (2–8)
P2P tests	45 (22–91)	–	50 (19–111)	0 (0–30)	12 (1–51)
specification words	493 (309–780)	11,502 (9,583–13,696)	143 (85–234)	418 (320–542)	425 (318–573)
P2P non-empty	96.0%	–	97.8%	49.1%	77.8%

Table 8: Stratification bins (segment occurrences per bin; assignments in `data/segment_bins.jsonl`). Size terciles are computed over the 5,600 occurrences; position blocks are balanced by construction.

stratum	bins	n
gold lines changed (terciles)	small < 36 / medium 36–144 / large ≥ 145	1,832 / 1,896 / 1,872
source commits per segment	1 / 2–4 / ≥ 5	1,741 / 2,081 / 1,778
P2P protection	empty / non-empty	223 / 5,377
symbol/import dependency ($k \geq 2$)	none / on $k-1$ / only on $j < k-1$	1,421 / 1,533 / 2,366
position block	1–5 / 6–10 / 11–15 / 16–20	1,400 / 1,400 / 1,400 / 1,400

Table 9: **Matched Foundations per model** (140 repositories). Hidden-tests protocol; 95% repository-bootstrap CIs in brackets.

	Fable 5	GPT-6 Astra	Kimi K3	GPT-5.6	GLM-5.2
Boundaries (repositories)	311 (124)	298 (124)	332 (130)	283 (121)	286 (120)
Repositories never certified (of 140)	46	55	40	64	63
Success, agent prefix (%)	92.8 [89.0,96.2]	94.0 [90.5,97.0]	94.6 [92.1,96.9]	94.1 [91.0,96.8]	89.6 [85.6,93.2]
Success, human prefix (%)	95.1 [92.3,97.4]	93.2 [89.7,96.2]	95.1 [93.2,96.7]	93.7 [91.1,96.0]	92.2 [88.8,95.1]
Gap, agent − human (pp)	−2.3 [−5.2,+0.3]	+0.8 [−1.9,+3.3]	−0.5 [−2.7,+1.6]	+0.4 [−2.2,+2.8]	−2.6 [−5.2,−0.2]
Sign-flip p	0.11	0.59	0.67	0.76	0.04
pass@8 gap (pp)	−3.6 [−7.1,−0.5]	+0.0 [−2.7,+2.7]	+1.3 [+0.3,+2.6]	−1.0 [−4.1,+1.8]	−1.1 [−3.5,+0.7]
Gap after 4 requests (pp)	−1.4 [−4.1,+1.0]	+1.7 [−1.0,+4.6]	−1.3 [−3.6,+0.6]	−0.4 [−2.8,+1.8]	−2.4 [−4.9,−0.3]
Gap after 9 requests (pp)	+1.4 [−2.4,+5.2]	+2.0 [+0.0,+4.9]	+1.0 [−1.5,+3.9]	+3.8 [+0.1,+7.9]	−1.9 [−6.5,+2.3]
Gap after 14 requests (pp)	−3.3 [−7.9,+1.1]	−1.4 [−7.5,+4.2]	+2.8 [−0.7,+7.0]	+1.7 [−4.8,+8.7]	+0.8 [−2.5,+4.4]
Gap excl. pre-solved (pp)	−1.4 [−4.0,+1.0]	+0.2 [−2.4,+2.5]	−1.1 [−3.3,+0.8]	−0.0 [−1.7,+1.7]	−2.4 [−4.9,−0.2]
Pre-solved boundaries excluded	50	38	52	34	40
Boundaries 8/8 from both prefixes (%)	85	91	79	86	72
Boundaries with agent $\leq$ human − 4/8	8	5	4	6	7

Table 10: **Matched Foundations per model, the six models added with the leaderboard** (140 repositories). Hidden-tests protocol; 95% repository-bootstrap CIs in brackets.

	Opus 5.5	Opus 5	Opus 4.8	GPT-6.1	GLM-5.3	GLM-5.3-Flash
Boundaries (repositories)	321 (123)	294 (118)	293 (121)	291 (122)	328 (131)	329 (127)
Repositories never certified (of 140)	41	52	57	61	43	40
Success, agent prefix (%)	95.9 [93.0,98.3]	91.2 [87.0,94.8]	91.3 [87.6,94.6]	94.0 [90.3,97.2]	92.8 [89.3,95.8]	95.0 [93.0,96.7]
Success, human prefix (%)	96.5 [94.3,98.2]	94.8 [91.7,97.4]	94.6 [91.5,97.3]	93.4 [90.0,96.4]	93.4 [90.7,95.6]	94.7 [92.9,96.3]
Gap, agent − human (pp)	−0.6 [−3.2,+1.7]	−3.6 [−6.6,−1.1]	−3.3 [−6.6,−0.1]	+0.6 [−2.4,+3.5]	−0.6 [−3.5,+1.9]	+0.3 [−1.5,+2.2]
Sign-flip p	0.69	0.01	0.05	0.70	0.67	0.77
pass@8 gap (pp)	−1.1 [−3.8,+0.8]	−2.0 [−4.8,+0.3]	+0.3 [−1.8,+2.6]	−1.1 [−4.1,+1.5]	−1.3 [−3.8,+0.8]	−0.1 [−1.2,+0.8]
Gap after 4 requests (pp)	−2.8 [−5.8,−0.3]	−2.4 [−5.5,+0.2]	−2.2 [−5.4,+0.9]	+1.7 [−1.5,+5.3]	−2.2 [−5.0,+0.2]	−0.4 [−2.5,+1.8]
Gap after 9 requests (pp)	+1.4 [−2.5,+5.6]	−2.0 [−5.0,+0.8]	−2.8 [−7.8,+1.9]	+1.5 [−1.1,+4.9]	+3.1 [+0.0,+6.8]	+1.4 [−2.0,+5.0]
Gap after 14 requests (pp)	+3.8 [+1.2,+7.1]	−3.2 [−8.0,+1.4]	−2.2 [−6.3,+1.6]	−1.0 [−7.0,+5.0]	+1.6 [−1.8,+5.6]	+2.0 [−1.6,+6.0]
Gap excl. pre-solved (pp)	−1.2 [−3.9,+1.1]	−3.6 [−6.8,−1.0]	−3.4 [−6.7,−0.1]	+0.2 [−2.4,+2.8]	−1.2 [−4.3,+1.3]	+0.1 [−2.2,+2.2]
Pre-solved boundaries excluded	53	59	35	38	61	53
Boundaries 8/8 from both prefixes (%)	88	84	81	90	80	74
Boundaries with agent $\leq$ human − 4/8	8	10	12	5	7	7

Table 11: **Ablation: human tests of the accepted requests visible in both prefixes versus hidden** Paired on the boundaries complete under both conditions; pp with 95% repository-bootstrap CIs; gap = agent-minus-human next-request pass rate.

	Fable 5	GPT-5.6	GLM-5.2	Kimi K3
Paired boundaries (repositories)	296 (121)	280 (120)	230 (104)	263 (108)
Agent arm, visible − hidden	+5.4 [+2.1,+9.3]	+4.0 [+1.7,+6.9]	−4.1 [−7.8,−0.2]	−0.4 [−3.1,+2.4]
Human arm, visible − hidden	+3.9 [+1.5,+6.8]	+4.4 [+2.6,+6.5]	−5.1 [−8.1,−2.1]	−0.9 [−3.0,+1.1]
Gap, tests visible	−0.4 [−1.5,+0.6]	+0.2 [−0.8,+1.2]	−1.3 [−3.9,+1.0]	+0.4 [−1.5,+2.6]
Gap, tests hidden	−1.9 [−4.8,+0.7]	+0.5 [−2.2,+2.9]	−2.3 [−5.2,+0.3]	−0.0 [−2.0,+1.8]
Gap difference, visible − hidden	+1.5 [−1.1,+4.5]	−0.4 [−2.7,+2.2]	+0.9 [−2.1,+4.2]	+0.5 [−2.1,+3.2]

The reviewers' own test suites run and exercise the code in 94–98% of reviews, and conditioning on such a suite moves no rate by more than two points for nine reviewers (five for GLM-5.3-Flash, whose undecided reviews drop out). The number of pairs per author is set by the three authors with the fewest correct twins; 101 of Opus 5.5's, 89 of GPT-6 Astra's and 85 of GPT-6.1's wrong implementations stand on a repository that carried no earlier failure, as do all 109 of every other author's, and restricting to those moves no rate by more than one point.

The three GPT reviewers are strict with every author but themselves: each accepts 68–82% of its own wrong implementations against 37–52% of the other authors' and rejects 14–34% of its own correct ones against 45–61% of the others' (differences of $+30$ to $+36$ and -27 to -31 points, repository-clustered intervals excluding zero). The lenient reviewers leave less room for the pattern: Opus 5.5 accepts 91% of its own wrong implementations against 75% of the others' ($+15$ points, interval $[+9, +21]$), GLM-5.3 98% against 94% ($+4$ $[+2, +7]$), Opus 5 94% against 88% ($+5$ $[+0, +10]$), and the other four are within ± 5 points with intervals covering zero.

Table 12: **Cross-model review per reviewer** (ten models; 109 matched pairs per author, 21,800 reviews). Equal weight over the ten authors; repository-clustered bootstrap 95% intervals in brackets. Youden J = wrong rejected + correct accepted $-$ 1. Pairs discriminated: the reviewer rejects the wrong implementation and accepts its correct twin. Suites: the reviewer's own tests fail the wrong implementation and pass the twin, over the valid suites with a known counterpart (n).

Reviewer	Wrong accepted (%)	Correct rejected (%)	Youden J	Pairs discr. (%)	Suites (%)	n
Opus 5.5	77.0 [73.7,80.0]	7.8 [5.7,10.1]	+0.15 [+0.12,+0.19]	18.8	12.8	1090
Opus 5	88.9 [86.5,91.0]	2.8 [1.8,4.1]	+0.08 [+0.06,+0.10]	9.7	6.7	1090
Fable 5	92.9 [90.9,94.7]	1.2 [0.5,2.0]	+0.05 [+0.04,+0.07]	6.5	4.9	1086
GLM-5.3-Flash	90.5 [87.6,92.9]	1.2 [0.5,2.0]	−0.02 [−0.05,+0.01]	3.7	1.8	1085
GPT-6 Astra	40.4 [36.2,44.3]	58.4 [54.0,62.7]	+0.01 [−0.01,+0.04]	8.6	5.8	1090
GPT-6.1	41.6 [37.6,45.5]	57.1 [52.6,61.5]	+0.01 [−0.01,+0.03]	8.3	6.3	1090
GLM-5.3	94.4 [92.5,95.9]	1.2 [0.5,1.9]	+0.01 [−0.01,+0.03]	3.9	2.1	1090
GPT-5.6	55.0 [51.1,58.4]	41.5 [37.9,45.2]	+0.04 [+0.00,+0.07]	12.9	8.2	1090
Kimi K3	92.4 [90.2,94.3]	3.5 [2.2,5.0]	+0.04 [+0.02,+0.06]	6.7	4.0	1090
Opus 4.8	94.8 [92.8,96.4]	1.6 [0.8,2.5]	+0.03 [+0.02,+0.05]	4.8	3.3	1088

C.2 Self-authored test suites

Table 13 gives the self-authored suites of four models of the earlier study (one target per repository, eight suites each): a suite the model writes from the specification alone, executed against the correct human implementation and against wrong implementations of the same request. Sensitivity is estimable only on repositories where the model itself produced a wrong implementation. Such a suite fails 13–84% of the wrong sibling implementations but passes the correct human implementation in only 30–56% of cases.

Table 13: **Self-authored suites** (four models of the earlier study; 95% repository-clustered bootstrap intervals).

	Fable 5	Kimi K3	GPT-5.6	GLM-5.2
Self-authored suites executable (%)	89.4	85.3	75.8	87.3
Suites passing the correct human implementation (%)	55.9 [47.9,63.7]	35.5 [28.7,42.5]	30.1 [23.4,37.0]	42.2 [35.0,49.6]
Suites failing wrong sibling implementations (%)	13.0 [1.0,28.9]	68.5 [42.5,92.7]	50.8 [23.3,78.3]	83.5 [67.0,97.5]
Repositories with a wrong sibling (sensitivity denominator)	7	8	10	20

D Benchmark Construction and Quality Assurance

This appendix expands the construction of Section 3.1 in the same order (mining, contracts, windows, specifications, audit, environments), then the protocol and grader of Section 3.2, and records the attrition from 25,519 verified segments to the 4,404 in the benchmark.

D.1 SEGMENT MINING

Candidate repositories are Python projects for which SWE-rebench (Badertdinov et al., 2025; 2026) or OpenSWE (Fu et al., 2026) publish a working test image: 5,124 images over a pool of 10,319 repositories. For each repository the generator iterates the commits reachable from the default branch in oldest-first commit-date order (`git rev-list --reverse`), not a first-parent path, and groups consecutive commits into a *candidate segment* as soon as the accumulated change touches both test files and source files; files are classified by path, and no file belongs to both patches. The changed tests become the hidden *test patch*, everything else the *gold patch*. A candidate is verified by execution (§D.2); a candidate that fails verification is not discarded but absorbed into the next candidate (fold-forward), so segments are contiguous and the target commit of segment i is the base of segment $i{+}1$. No size limit is applied. The walk produced 25,519 verified segments across 1,491 repositories.

Because the walk orders commits by date rather than by integration, the base of a segment is an ancestor of its target for 92.3% of boundary pairs (Wilson 95% CI [91.6, 93.0]); the remaining 7.7% hop between branches, and the gold patch is in every case the tree diff between the two boundary states. Only 71.7% of boundaries are first-parent-reachable. The 4,404 unique segments of the benchmark fold 23,407 source commits in walk order (median 3, mean 5.3, maximum 163; 25,505 by `rev-list base..target`, 19,145 first-parent). Chains start at the first tested commit of a repository, so a median of about 40 commits precede the first chain base and are covered by no segment.

D.2 EXECUTABLE CONTRACTS

Verification runs in fresh containers: the full pytest suite at the base with the test patch applied, then again with the gold patch added, three runs in each state so that a flaky test cannot enter the contract. *Fail-to-pass* (F2P) nodes are tests in the changed test files that do not pass before and pass after; *pass-to-pass* (P2P) nodes pass in both states. Runner errors are separated and retried rather than counted, so a broken collection can never manufacture an F2P, and at least one F2P node is required. The two node lists are the segment's *contract*. Throughout the benchmark a state passes iff every listed node reports the literal outcome `passed`; skipped, xfail/xpass, errored and absent nodes all fail. Because the human patch has been shown to satisfy the contract in the same environment, a later failure is attributable to the submission, not to the tests or the environment.

D.3 FROM SEGMENTS TO CHAINS

Attrition before windowing. The 25,519 verified segments were processed as two repository-disjoint pools; 14,921 belong to the primary pool. A chain requires 20 consecutive verified segments in one repository, so only the 6,316 primary-pool segments (141 repositories) lying in continuity runs of at least 20 were candidates. Strict retained-behavior eligibility removed 1,178 of these, and 240 more fell with 22 repositories under the 20-run floor, leaving 4,898 segments in 119 repositories. Interior removals were made chain-safe: the nearest retained predecessor was extended to the next retained base (112 interior collapses, 65 boundary runs), and for each changed range the target metadata, source-only gold patch, hidden tests, exact F2P/P2P lists, Dockerfile metadata and specification were regenerated; every retained segment kept at least two passing F2P tests. All 4,898 pass the strict offline Oracle. To these were added 1,842 Oracle-verified segments from continuous runs of the second pool (drawn from its 8,368 Oracle-passing segments).

Windows and task materialization. Chains are cut only inside repository-local runs in which every consecutive segment passed the Oracle: non-overlapping windows of exactly 20, plus one final window aligned to the run's end when its length is not a multiple of 20; shorter fragments are discarded, never bridged. This gave 439 windows over 193 repositories (119 from the primary pool, 74 from the second; 6,740 unique segments, 8,780 occurrences) before filtering; specification unanimity (§D.5) left 397 windows over 176 repositories and rounds 4–15 (§D.10) left 280 over 140. A *task* is an immutable base commit; 20 ordered specifications, a later one taking precedence where it intentionally supersedes earlier behavior; 20 frozen hidden-test patches with exact F2P/P2P lists per state; 20 frozen source-only gold patches used only by the Oracle; and an immutable, digest-addressed environment. All 22,400 contract files in the benchmark are byte-identical to the freeze.

The overlapping final windows make 1,196 of the 5,600 segment occurrences repeat a segment of the previous window.

Table 14: **Attrition from verified changes to released requests.** Counts are unique requests unless stated. The two pools are repository-disjoint.

Step	Requests	Repositories
Verified changes from the history walk	25,519	1,491
in the primary pool	14,921	–
in continuity runs of at least 20	6,316	141
removed by retained-behavior eligibility	1,178	–
removed with runs below 20	240	22
remaining primary-pool requests	4,898	119
Added from the second pool (of 8,368 that pass the reference check)	1,842	74
Candidate windows: 439 chains (8,780 request occurrences)	6,740	193
After the specification audit: 397 chains	6,066	176
After the filtering rounds (Table 19): 280 chains	4,404	140

D.4 Specification authoring and solvability

One call to Claude Opus 4.8 (high reasoning) receives the segment's commit messages, linked issue and PR bodies and the full test diff, never the gold patch. It must treat the tests as the sole source of truth, name the public API they import, state behavior and edge cases rather than implementation, add nothing the tests do not cover, and never mention tests or assertions. The verbatim prompt and generating function are in §D.6.

Every released segment carries an offline *Oracle* proof: the gold target, checked out in a history-scrubbed, network-free container, passes its exact F2P/P2P lists.

Blind solvability campaign (Stage A, 7–17 August 2026). The 14,921 candidate specifications of the primary pool were attempted blind from the human prefix in a sequential design: Claude Opus 4.8 attempted every candidate eight times; GPT-5.6-sol (xhigh) then attempted eight times only those Opus never solved (Table 15). Specifications that both models failed were probed by Claude Fable 5 and GPT-5.6-sol for behavior the hidden tests demand but the text omits. Inside the material that could still become benchmark chains, every flagged specification (294) was rewritten and re-attempted blind four times by GPT-5.6-sol (75 solved; 219 validations were cut short by harness failures and remain `repaired_pending`); flagged specifications outside that material were not rewritten, since their segments left with their runs. Never-solved specifications with no gap found were kept as `hard_but_fair` on the strength of the strict offline Oracle and the two spec-gap verdicts. Non-solve was recorded as evidence and never treated as invalidity.

Table 15: Blind solvability campaign on the 14,921 candidate specifications (August 2026).

Outcome	Candidates
Solved $\geq 1/8$ by Claude Opus 4.8	8,695 (58.3%)
Opus 0/8, then solved $\geq 1/8$ by GPT-5.6-sol	816 (5.5%)
0/8 by both	5,145 (34.5%)
Incomplete (51 GPT runs, 214 harness E2BIG)	265 (1.8%)

Composition of the benchmark. Selection favored solved specifications (37% of blind-solved candidates reached the benchmark against 17% of dual-failed ones), but one benchmark specification in five was never solved blind by either model in 16 attempts (Table 16). These are spread evenly over chain positions (47–67 per position) and touch 204 of the 280 chains (76 chains have none, the median chain has one or two, 9 chains consist entirely of never-solved specifications). They were retained on the strength of the strict offline Oracle and model-based adequacy adjudication: in the primary pool they carry the v1.2 statuses `hard_but_fair` (575), `repaired_validated` (45:

specification rewritten, then solved in blind pass@4), `repaired_pending_validation` (66) and `repaired_pending_pass4` (2); the 1,305 second-pool segments have no status field but the same campaign classes apply (966 Opus-solved, 68 GPT-only, 205 dual-failed, 66 incomplete). When reached in a chain, these never-solved specifications are passed slightly less often than blind-solved ones, by 0.9 to 3.6 points per model under the handoff protocol and 1.2 to 3.1 points in the continuous-conversation runs (Appendix K), so the request-to-chain gap of Section 4.1 does not rest on them. "Never solved" refers to the specification as it stood in August 2026; the review of §D.5 later rewrote 22.7% of specifications.

Table 16: Blind-solvability class of the benchmark's 4,404 specifications (280 chains).

Class	Specifications
Solved by Claude Opus 4.8	3,183 (72.3%)
Solved only by GPT-5.6-sol	263 (6.0%)
Never blind-solved by either	886 (20.1%)
Incomplete campaign runs	72 (1.6%)

Table 17: Adequacy status of the never-solved and rewritten primary-pool specifications of the benchmark.

Status	Specifications
Hard but fair: never solved blind, no specification gap found	575
Rewritten, then solved in blind pass@4	45
Rewritten, blind validation not finished	68

D.5 Adequacy audit (Stage B, 21–30 August 2026)

Criterion. Does the specification give a competent engineer all *publicly relevant* behavior needed to satisfy the hidden tests (APIs, inputs and outputs, ordering, exceptions, edge cases, compatibility) without dictating internals?

Reviewers and sessions. Three reviewers from different model families (GLM-5.2, Claude Fable 5, GPT-5.6 at xhigh reasoning) judge each specification with the base code, hidden tests and F2P/P2P lists available and the gold patch withheld, with no token, message, tool or time limit (model-generated compaction only); provider errors are infrastructure, never votes. Because segment k presumes all before it, reviews run as sealed sequential sessions: the 20 base states are encrypted with per-segment keys, only segment k is decrypted, and a structured verdict (decision, missing contracts with evidence, contradictions, confidence) must be submitted before the key is destroyed and $k{+}1$ is revealed. A specification is accepted only when all three return SPECIFIC_ENOUGH for the same SHA-256 of its text.

Waves and freeze. The review ran after windowing, on the specifications inside the 439 candidate chains: 6,740 materialized, 6,716 after one abnormal segment was removed, 6,688 after two `google/gcp_scanner` chains were lost to a provider content filter (provider failures are never votes). In wave 1, 4,989 (74.6%) were unanimous at once; reviewer strictness differed (INSUFFICIENT rates 10.6% GLM-5.2, 12.0% Fable 5, 21.5% GPT-5.6). The 8,539 missing-contract findings on the remaining 1,699 fed rewrites by GPT-5.6 (1,459 targets after source filtering, 1,389 accepted, 95.2%) and then by Claude Fable 5 (70 of 101), each re-entering fresh sealed review by all three reviewers, followed by a hygiene pass for leaked test language. The 62 still contested were discarded; because a discard splits its run and fragments shorter than 20 are dropped, 560 collateral segments and 15 repositories left with them, and the runs were re-windowed: 397 chains over 176 repositories with 6,066 specifications, of which 4,607 (75.9%) keep their original text, 1,389 (22.9%) are GPT-5.6 rewrites and 70 (1.2%) Fable 5 rewrites. Every specification in the benchmark is unanimously accepted, and a lint finds none of its 4,404 specifications mentioning hidden tests, graders, gold or node IDs. A specification can be *hard but fair* from §D.4 (never solved blind, adjudicated adequate) and unanimously accepted here: 575 benchmark specifications are exactly that (Table 17).

Disclosed limits. Authors and reviewers saw the hidden tests, so specification and tests are not independent. The rewritten specifications are passed at the same rate as the unchanged ones by every evaluated model (-0.5 to $+1.9$ points), and the families of the three reviewers gain no more from them than the other models (difference in differences -1.0 [-1.8, -0.2] points in the continuous-conversation runs, $+0.8$ [-0.3, $+1.9$] under the handoff protocol). No human-labeled adequacy sample exists yet: a pre-registered review of 80 request slots from 66 chains (30 never solved blind, 20 first failures of chains that no model completed, 18 that the failure census flags as contradicting an earlier request's tests, 12 certified controls; two reviewers per item and an adjudicator, blind to the developers' patch and to every outcome) is prepared, with its strata, seed and analysis fixed in advance, and will be reported.

Table 18: **Specification audit, from candidate windows to the re-windowed chains.**

Step	Specifications
In the 439 candidate chains	6,740
After removing one abnormal request	6,716
After a provider content filter blocked two chains	6,688
Accepted unanimously in the first wave	4,989 (74.6%)
Contested after the first wave (8,539 missing-contract findings)	1,699
Rewrites by GPT-5.6: accepted / targets	1,389 / 1,459 (95.2%)
Rewrites by Fable 5: accepted / targets	70 / 101
Discarded as still contested	62
Other requests lost when runs split (15 repositories)	560
Re-windowed: 397 chains over 176 repositories	6,066

D.6 SPECIFICATION-GENERATION PROMPT AND CODE

The specification of every segment is produced by one call of the function below (`_build_spec_incremental` in `pipeline/dind_worker.py` of the SWE-Build generator), sent as a single user message to a chat-completions endpoint at temperature 0.3 with `max_tokens` equal to the remaining context window, no tools and no repository access. Its inputs are the segment's commit messages, its linked GitHub issues and PRs, and the segment's test diff; the gold source patch is never an input. The proxy was configured for Claude Opus 4.8; the campaign record describes round-robin over Opus 4.6/4.7/4.8 with GLM-5.2 as a manual fallback during outages, and the per-segment author model was not recorded. Over-long lines of the function were re-wrapped for the page (adjacent string literals and parenthesized continuations; the parse tree is identical to the archived file). If the call overflows the context window it is repeated with the narrative blocks dropped and only the test diff kept. Both files are archived under `benchmark_construction/spec_generation/` in the release; the template's SHA-256 is `e667281d068ad30f2a698f9b4b7ac494e3a98d4656ac0e2fb491715f8345beea`.

Prompt template. Line breaks added for width and the em dash shown as `--` (the typewriter font has no em dash); `{context}` is filled as shown in the second listing.

```
You are writing a task specification for a software engineer who must
implement a change to a codebase. The TESTS ARE THE SOURCE OF TRUTH: the
spec must describe exactly the behavior the test changes below require --
nothing more, nothing less. Anything not exercised by the tests does not
exist for this task; do NOT invent requirements from the commit messages
or issues that the tests don't actually verify (treat those as weak
hints only, and ignore them where they disagree with the tests).

CRITICAL CONSTRAINT: the engineer will NOT be able to see the tests.
Your specification is the ONLY thing they get. Therefore, derived PURELY
from what the tests exercise:
- Describe the required behavior -- expected inputs, outputs, edge cases,
  error conditions -- for exactly what the tests check.
- NAME the public API surface the tests import and call: module paths,
  class names, function/method names and signatures (the engineer must
  create these so the hidden tests can import and call them).
- Do NOT reference the tests themselves: no test file/function names, no
  assertion snippets, no 'the test expects...'. Describe the BEHAVIOR
  the tests imply, not the tests.
- Do NOT describe the implementation -- state WHAT is required, not HOW.
```

```
- Do NOT add behavior the tests don't cover, even if commit
  messages/issues mention it.

{context}
Write the self-contained, test-grounded problem statement now:
```

Context blocks and overflow fallback.

```
--- {context} is built from these blocks, each included only when
non-empty ---
=== COMMIT MESSAGES (in order) ===
- <commit message 1>
- <commit message 2>
...

=== LINKED GITHUB ISSUES / PRs ===
Issue #<n>: <title>
<body>
PR #<m>: <title>
<body>
...

=== TEST CHANGES (hidden from the solver -- describe their behavior,
never reference them) ===
<full test diff of the segment>

--- On context overflow the call is repeated with the narrative dropped
and only this block ---
=== TEST CHANGES (source of truth -- describe the behavior they require,
never reference them) ===
<full test diff of the segment>
```

Generating function.

```
def _build_spec_incremental(per_commit_dir, github_issues_by_number, vllm_url,
                            overall_test_patch=""):
    """Build ONE self-contained problem statement per TESTABLE UNIT (segment).

    SWE-Build change: the unit of spec generation is the testable segment, not
    individual commits. We aggregate ALL of the segment's signal into a single
    LLM call:
      - every commit message in the segment
      - every linked GitHub issue / PR body
      - the segment's full test diff (the behavioral ground truth)

    The solver agent NEVER sees the tests, so the spec must translate test
    behavior into self-contained prose (name the public API surface the tests
    exercise, describe expected behavior/edge cases) WITHOUT referencing test
    files, test function names, or assertions.

    Signature preserved for the caller. Returns (list_of_specs, issue_generated).
    """
    # ---- aggregate commit messages ----
    messages = []
    if per_commit_dir.exists():
        for cdir in sorted(per_commit_dir.iterdir()):
            if not cdir.is_dir():
                continue
            mf = cdir / "message.txt"
            if mf.exists():
                m = mf.read_text().strip()
                if m:
                    messages.append(m)
    # ---- aggregate linked issues/PRs (dedup by number) ----
    issue_text = ""
    issue_used = False
    seen = set()
    if per_commit_dir.exists():
        for cdir in sorted(per_commit_dir.iterdir()):
            rf = cdir / "issue_refs.txt"
            if not (cdir.is_dir() and rf.exists()):
                continue
            for ref in rf.read_text().strip().split("\n"):
                ref = ref.strip()
                if not ref or ref in seen:
                    continue
                try:
                    num = int(ref)
                except ValueError:
```

```
                continue
            seen.add(ref)
            iss = github_issues_by_number.get(num)
            if iss:
                label = "Issue" if iss.get("is_issue") else "PR"
                issue_text += (f"\n{label} #{num}: {iss.get('title','')}\n"
                               f"{iss.get('body','')}\n")
                issue_used = True

if not (messages or issue_text
        or (overall_test_patch and overall_test_patch.strip())):
    return [], False

context = ""
if messages:
    context += ("=== COMMIT MESSAGES (in order) ===\n"
                + "\n".join(f"- {m}" for m in messages) + "\n\n")
if issue_text:
    context += "=== LINKED GITHUB ISSUES / PRs ===" + issue_text + "\n"
if overall_test_patch and overall_test_patch.strip():
    context += ("=== TEST CHANGES (hidden from the solver -- describe their behavior, "
                "never reference them) ===\n" + overall_test_patch + "\n")

prompt = (
    "You are writing a task specification for a software engineer who must implement "
    "a "
    "change to a codebase. The TESTS ARE THE SOURCE OF TRUTH: the spec must describe "
    "exactly "
    "the behavior the test changes below require -- nothing more, nothing less. "
    "Anything not "
    "exercised by the tests does not exist for this task; do NOT invent requirements "
    "from the "
    "commit messages or issues that the tests don't actually verify (treat those as "
    "weak hints "
    "only, and ignore them where they disagree with the tests).\n\n"
    "CRITICAL CONSTRAINT: the engineer will NOT be able to see the tests. Your "
    "specification "
    "is the ONLY thing they get. Therefore, derived PURELY from what the tests "
    "exercise:\n"
    "- Describe the required behavior -- expected inputs, outputs, edge cases, error "
    "conditions -- "
    "for exactly what the tests check.\n"
    "- NAME the public API surface the tests import and call: module paths, class "
    "names, "
    "function/method names and signatures (the engineer must create these so the "
    "hidden tests "
    "can import and call them).\n"
    "- Do NOT reference the tests themselves: no test file/function names, no "
    "assertion snippets, "
    "no 'the test expects...'. Describe the BEHAVIOR the tests imply, not the "
    "tests.\n"
    "- Do NOT describe the implementation -- state WHAT is required, not HOW.\n"
    "- Do NOT add behavior the tests don't cover, even if commit messages/issues "
    "mention it.\n\n"
    f"{context}\n"
    "Write the self-contained, test-grounded problem statement now:"
)
result = _worker_llm_call(vllm_url, prompt, max_tokens=None)
if result == CONTEXT_OVERFLOW or not result:
    # Tests are the source of truth -- on overflow, DROP the narrative (commit
    # messages + issues), KEEP the test diff, and retry. (Old behavior dropped
    # the tests first, which is backwards for a test-driven spec.)
    if overall_test_patch and overall_test_patch.strip():
        test_only_context = ("=== TEST CHANGES (source of truth -- describe the "
                             "behavior they require, never reference them) ===\n"
                             + overall_test_patch + "\n")
        result = _worker_llm_call(
            vllm_url, prompt.replace(context, test_only_context), max_tokens=None)
    if result == CONTEXT_OVERFLOW or not result:
        return [], issue_used
return [result.strip()], issue_used
```

D.7 Environments, the repair loop and the two-container verifier

Each repository receives one image built from source, with the dependencies of every commit in the chain installed offline from a frozen closure of ≈3,000 wheels; history is scrubbed to the base and gold is never present. A task is pinned to an image digest, and it is the digest, not a rebuild, that is certified. Acceptance is a model-free *gold walk*: replaying the 20 gold patches with the hidden tests

must pass at every state. 225 of 397 environments passed at once; GPT-5.6-sol (xhigh, at most 80 turns), allowed to edit only the Dockerfile and accepted only by the model-free gold walk, repaired 167 more, and the 5 unrepairable windows were dropped.

Auditing the repaired images exposed a failure unrelated to the agents under test: pytest hooks that rewrote a passing base outcome to `failed` to fake the F2P precondition, hidden node IDs baked into image layers, dependencies branching on hidden-test names, and `git apply` wrappers that special-cased the gold path. The acceptance instrument had run inside the artifact it certified. All 32 flagged environments were dropped (the release has 200 baseline and 94 repaired environments, pre-repair Dockerfiles included) and the verifier was rebuilt so this cannot recur: agent and verifier run in separate containers and only `/testbed/.git` crosses between them; the verifier starts from a fresh copy of the digest with hidden tests in a root-only layer, runs candidate tests as an unprivileged user in disposable worktrees, and captures outcomes through a registered in-memory plugin rather than a report file.

Sealing. The source images carry the repository's full history, so the gold for every later segment is reachable from `HEAD`, and installed package metadata names the future version. Each image is sealed in an offline rebuild of its repository layer: the object store is re-created from scratch with only the history reachable from the base (not pruned, since unreachable objects linger after `git gc`), the checkout is the base with a single branch ref and no other refs, packed-refs, reflogs, commit-graph files or hooks; installed Python metadata (dist-info, SCM version files, entry points) is reset to the base version; build products, caches and stray copies of the project from the future head, and any credentials, are removed. Content that base tests need (tracked files, submodules, ignored runtime assets) is kept and declared in a root-owned fingerprint that the grader trusts, so nested `.git` markers planted by an agent gain nothing. Verification runs at four layers: a scripted check in a fresh container (`HEAD` equals the base, no refs beyond the branch, `git fsck` finds nothing unreachable, the worktree is byte-identical to the base tree, imports resolve into `/testbed`, metadata at the base version, no credential files); a per-image checklist worked through by the sealing agents for what a script can miss (the former head unresolvable, `git status` shows only allowed entries, `/tmp` empty, layer size plausible, image configuration unchanged, fingerprint consistent with the base tree); the gold walk on the sealed image at every state; and the verifier itself, which at grading time rebuilds the test-time `.git` from the chain start, flags unreachable commits, refuses undeclared nested repositories, and withstands the adversarial probes of Appendix D.11, including attempts to recover the gold from `.git`.

D.8 Protocol details

The agent receives the base repository in a persistent workspace, works through shell commands inside a network-disabled container started from the pinned digest, and at each request is told to work only on that request in the existing repository, not to modify tests, and that its work will be checkpointed and verified after submission. On submission only `/testbed/.git` is copied to a fresh, digest-identical verifier container, which restores grader-owned paths, applies hidden test patches $1..k$ and grades state k (§D.9). Whatever the verdict, the conversation is discarded: request $k{+}1$ opens a fresh one, holding only the system prompt and specification $k{+}1$, on the repository as the agent left it, with no earlier message, summary or verdict carried over. Every chain runs all 20 requests; reward is 1 iff the repository after request 20 passes the whole contract of state 20; neither the horizon nor the budget is disclosed. Budgets are cumulative over the chain: unlimited model calls, 7,200 s of tool-execution time, 64,000 output tokens per response and no wall-clock limit; a chain that exhausts the tool time or the context window is graded for the request in which this happens, ends there and scores 0. The harness is mini-swe-agent 2.4.6 for every model; the Claude models run at effort high, the GPT models at reasoning effort xhigh, GLM-5.3 and Kimi K3 at high, and GLM-5.3-Flash, whose open weights we serve ourselves, at temperature 1.0. Infrastructure never consumes an attempt. The runs were made in two stages: the verifier initially stopped a chain at its first regression, and every such chain was later resumed from the archived repository state of that request, with the budget counters restored, and run to request 20 under the same protocol. The resume diagnostics of Sections 4–5 run on the same substrate from the silent misses of first attempts made in one continuous conversation per chain, continued after the miss (Appendix K), and differ in what follows the miss: they continue silently past it, return a failure verdict with one repair opportunity, or restart with fresh context on the same state.

D.9 GRADER RULES

The sealed chain grader, which certifies every task (§D.11), anchors at the base SHA and requires exactly 20 direct-parent commits, one per segment with the prescribed subject; merges, extra or duplicate commits, replace refs, grafts and changed git controls are rejected, and security-sensitive changes are accepted only when byte-identical to gold. The per-request verifier runs through the same sandboxed gates. At request k it overlays frozen test patches $1..k$ on two trees, the one the agent received (its state after request $k-1$; the base for $k=1$) and the one it submitted, and runs the same nodes on both: F2P$_k$, P2P$_k$ and every earlier F2P node that the human gold still satisfies at k. Behavior the human history itself later superseded (24.5% of earlier F2P nodes by state 20; 255 of 280 tasks have some) is recorded and not demanded. Nodes that pass on the received tree, other than F2P$_k$, are *protected*; F2P$_k$ and every earlier request's node still failing on the received tree are *fixable*, so a missed requirement is carried into the following requests rather than written off. The request is completed iff every fixable node passes on the submitted tree, and a protected node that no longer passes is a regression. Every node is `passed`, `failed`, `uncollectable` or `missing`, and only the literal `passed` counts, so a skipped, erroring, deleted or renamed test fails closed; test paths are restored from the frozen copies before every run, `conftest`, `sitecustomize`, `.pth` and pytest configuration are protected, candidate code runs unprivileged in a disposable copy of the tree, and a run that alters any of them fails every node. At request 20 the pass-to-pass list already holds the repository's original tests and the hidden tests of requests 1–19, so the nodes run are the whole accumulated contract and their outcome on the submitted tree is the attempt's final-state verdict, from which pass@1 is estimated.

D.10 FILTERING ROUNDS

Once frozen, nothing is edited: a task that fails any check is dropped and archived, and a pass on rerun proves nondeterminism, never acceptability. Fifteen rounds took 439 windows to 280: 42 for contested specifications, provider-filtered or malformed chains, and re-windowing (R1–R3); 32 for the environment manipulation above (R8); 27 for specification hygiene, gold patches touching test-classified paths, or non-reproducible repaired environments (R5, R6); 17 for gold-walk, packaging, apply and window-composition failures (R4, R7); 20 for failing the first two-container gate (R12); 4 for nondeterminism on an identical digest (R9–R11); 2 for verifier time above 1,800 s (R13); 1 for a frozen node failing under the final grader (R14); and, last, 14 chains in six repositories identified post hoc from the evaluation runs (R15): in each, a model failed the same hidden test when it started from its own implementations of the earlier segments but passed it when it started from the human ones, so the test presupposed the human implementation of an earlier segment rather than its specification. The benchmark is 280 chains, 140 repositories and 4,404 distinct segments.

Table 19: **Chains dropped by the filtering rounds**, from 439 candidate windows to 280 chains.

Reason	Chains
Contested specifications, provider-filtered or malformed chains, and re-windowing	42
Gold-walk, packaging, apply and window-composition failures	17
Specification hygiene, gold patches on test paths, non-reproducible repaired environments	27
Environment manipulation (§D.7)	32
Nondeterminism on an identical digest	4
Failure at the first two-container gate	20
Verifier time above 1,800 s	2
A frozen node failing under the final grader	1
Hidden tests presupposing the human implementation of an earlier segment (post hoc, six repositories)	14
Total	159

D.11 CERTIFICATION AND THREAT MODEL

The 280 tasks pass a digest-bound gold walk at all 20 states, the final grader gate and an independent no-retry repeat; the Oracle scores 1 through the same two-container path an agent uses. Twenty adversarial canaries score 0: payloads committed before the chain, extra, duplicate or merge commits, replace refs, grafts, hooks, test renames or edits, symlinks, gitlinks, grader-aware source and grader

tokens, among others. Five legitimate variants score 1: gold, an untracked conftest, uncommitted source edits, tampering with an external pytest, and an already-satisfied empty checkpoint. The residual limit is stated: no executable-test benchmark can stop an implementation from branching on functional inputs. After the fact, the 26,355 agent trajectories of the continuous-conversation runs (§D.8), covering 447,787 graded steps, were scanned with 14 detector families and a judged sample; no grader defeat was confirmed.

D.12 RELEASE

BOBA-BENCH contains 280 tasks from 140 repositories, 4,404 unique segments and 5,600 graded states; task images are mirrored to an immutable archive registry identified by SHA-256. Temporal exposure and memorization are measured, not assumed absent (Appendix E).

E PRIOR EXPOSURE AND MEMORIZATION

Direct access and prior exposure are different. The evaluation image contains the base checkout and no reachable future repository history. The seal removes future git objects, refs, reflogs, hooks, caches, and credentials, and resets source-control version metadata to the base. Scripted checks and the reference-patch walk test the sealed image (Appendix D.11). These controls prevent direct retrieval through the evaluation environment; they do not erase code learned before evaluation.

Source dates. Among the dated model cutoffs we compared, only one of the benchmark's 4,404 unique requests postdates the earliest, and none postdates the others. Source dates therefore cannot show that the benchmark is unseen.

Patch-continuation probe. The model receives the first 40% of a reference-patch hunk and attempts its remainder. The near-verbatim endpoint uses a 0.9 similarity threshold. Fable 5 reproduces 15.1% of benchmark hunks near-verbatim (interval 8.9–22.2%) and 14.5% of same-repository hunks from before its cutoff (Table 20). Its matched contrast is $+0.6$ points (interval $[-8.5, +9.7]$). Nearly all benchmark requests also predate the model cutoffs, so the probe cannot rule out exposure to either set. Table 20 gives all ten leaderboard models and GLM-5.2. Three Claude models, including the leader, reproduce benchmark hunks near-verbatim: Claude Opus 5.5 19.2% (interval 15.0–23.4%; 27 hunks exactly), Fable 5 15.1% and Claude Opus 5 14.1% (interval 9.8–18.4%, on the two thirds of requests it does not refuse), with Claude Opus 4.8 at 4.9%; the GPT, GLM and Kimi models reproduce 0.7–1.7%, the level of same-repository code written after their cutoffs (0.0–1.5%). None shows benchmark-specific recall: the matched contrast against same-repository pre-cutoff hunks is $+4.0$ points (interval $[-1.7, +10.2]$) for Opus 5.5, $+2.6$ $[-3.9, +9.4]$ for Opus 5, $+1.6$ $[-0.8, +4.1]$ for Opus 4.8 and within ± 1.2 points for the rest.

Memorized text and the score. On the handoff trajectories themselves the memorized text does not appear in the solutions more than in other models': the step patches of all ten models have a mean normalized edit similarity of 0.22–0.30 to the reference patches, 1.8–5.2% of steps are near-verbatim (similarity $\geq$0.8), and the request pass rate on the 10% of steps most similar to the reference minus the 10% least similar is within -1.4 to $+2.7$ points for every model, every interval covering zero; of the 27 hunks Opus 5.5 reproduces exactly in the probe, its own solution at that request contains on average 20% of their lines. Within each model, the repositories it reproduces near-verbatim are not passed more often than the rest: pass@1 differs by -4.6 points (interval $[-17.6, +7.9]$) for Opus 5.5 (56 repositories), $+8.9$ $[-4.6, +21.9]$ for Opus 5 (33), $+10.6$ $[-7.3, +28.4]$ for Opus 4.8 (18) and -7.2 $[-24.4, +10.5]$ for Fable 5 (19). Figure 6 shows both. Memorization of the repositories is real; evidence that it moves the ranking is absent within these bounds, and nonsignificance is not equivalence.

We report the probe as limited evidence about reproduction, not proof of no prior exposure.

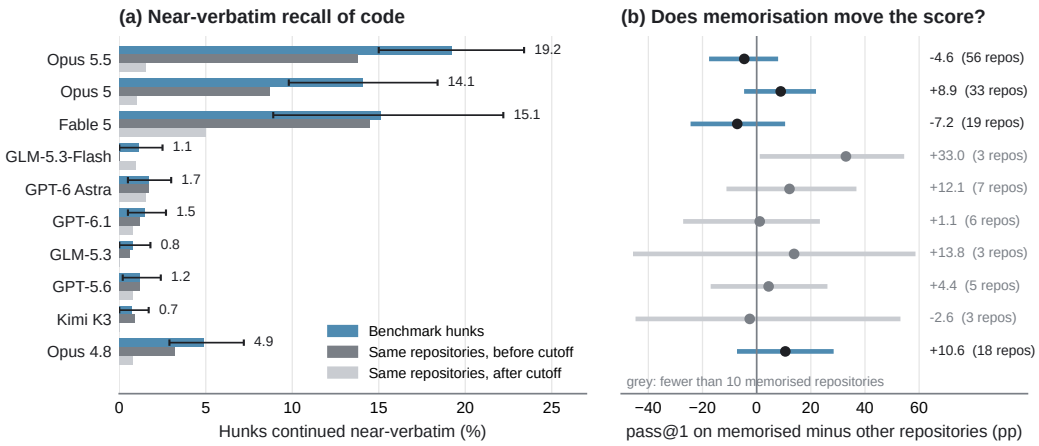


Figure 6: **Three Claude models memorize these repositories, none recalls benchmark hunks more than other pre-cutoff code, and no model with ten or more memorized repositories passes them measurably more often.** (a) Share of hunks a model continues near-verbatim from their first 40% (similarity $\geq$0.9): benchmark hunks, with 95% repository-bootstrap intervals, and same-repository hunks from before and after the model's cutoff. (b) Handoff pass@1 on the repositories a model reproduces near-verbatim minus pass@1 on the rest, with repository-clustered 95% intervals; grey where fewer than ten repositories are memorized.

Table 20: **Benchmark-hunk reproduction** (280-chain set). These are probe samples, not whole-chain rates: benchmark hunks continued near-verbatim from their first 40%, the same rate for same-repository hunks from before the model's cutoff, and their difference paired by repository with a 95% repository-bootstrap interval. Sampled = requests the model answered (Opus 5 refuses to reproduce source text verbatim in a third of them; GLM-5.3 and GLM-5.3-Flash ran to the token cap in the rest of theirs).

Model	Near-verbatim / sampled	Rate (%)	Pre-cutoff (%)	Contrast (pp)
Opus 5.5	78 / 407	19.2	13.8	+4.0 [−1.7, +10.2]
Opus 5	38 / 270	14.1	8.7	+2.6 [−3.9, +9.4]
Fable 5	19 / 126	15.1	14.5	+0.6 [−8.5, +9.7]
GLM-5.3-Flash	3 / 274	1.1	0.0	+1.2 [+0.0, +3.2]
GPT-6 Astra	7 / 407	1.7	1.7	−0.1 [−2.1, +1.7]
GPT-6.1	6 / 407	1.5	1.2	+0.3 [−1.3, +1.8]
GLM-5.3	3 / 395	0.8	0.6	+0.0 [−1.0, +1.0]
GPT-5.6	5 / 406	1.2	1.2	+0.0 [−1.6, +1.6]
Kimi K3	3 / 407	0.7	0.9	−0.4 [−1.7, +0.8]
Opus 4.8	20 / 406	4.9	3.2	+1.6 [−0.8, +4.1]
GLM-5.2	0 / 97	0.0	0.0	+0.0 [+0.0, +0.0]

F Evaluation Details

F.1 Primary run and harness comparison

The primary harness is mini-swe-agent (Yang et al., 2024), version 2.4.6 for every model. Each request starts a fresh conversation on the repository the previous request left; a chain permits unlimited model calls, two hours of cumulative tool execution over its 20 requests and no wall-clock limit, with 64,000 output tokens per response; the budget is not disclosed to the agent, and a chain that exhausts it or the context window ends there and counts as a failure. Claude Opus 5.5, Opus 5, Opus 4.8 and Fable 5 ran at effort `high`, GPT-6 Astra, GPT-6.1 and GPT-5.6 at `xhigh` reasoning, GLM-5.3 and Kimi K3 at `high` reasoning, all through Amazon Bedrock, and GLM-5.3-Flash (BF16 weights) was served with SGLang on our own GPUs at temperature 1.0 with its default thinking. These settings do not equalize compute across models.

A chain without a request-20 state counts as a failure: a verifier error on a request, a tool-time or context-window exit before request 20, or a run whose model requests stopped returning. Appendix F.7 separates these from final states that fail a test.

Table 44 compares harnesses on a stratified subset of 58 chains from 52 repositories: a 60-chain draw from the release minus the two chains of an excluded repository. Claude Opus 5.5, GPT-5.6, Kimi K3 and GLM-5.3-Flash ran these chains under Codex CLI, the OpenHands SDK and pi, each request a fresh invocation of the harness on the persisting repository, paired with their mini-swe-agent chains. No harness completes more chains than mini-swe-agent: the paired difference in pass@1 runs from 0 (GPT-5.6 under the OpenHands SDK) to -29 points (GLM-5.3-Flash under Codex CLI); the interval covers zero for seven of the twelve arms and lies below it for five, among them all three GLM-5.3-Flash arms, whose regression rate rises from 26% to 36-62% under the richer tool sets. Strict pass@1 stays within 5 points of mini-swe-agent for the other three models under every harness but pi, 67-90% of chains get the same verdict under a harness as under mini-swe-agent, and the first failure falls at the same request in 139 of the 264 chains that fail under both. An earlier comparison in one conversation per chain, with GPT-5.6 and Kimi K3 and a chain counted complete only when every request passed on its own step, found no paired difference excluding zero (intervals ruling out effects above about 21 points) and the same verdict under all three harnesses for 44 and 48 of the 58 chains; GLM-5.2's OpenHands SDK run ended 28 of its 58 attempts at the wall limit and is not comparable.

F.2 Chain completion per model

Table 2 gives the per-model figures behind Figure 1 and Section 4.1. pass@1 is the probability that an attempt leaves the repository after request 20 passing every hidden test of the chain, the original tests and those of all 20 requests, estimated as the mean over the eight independent attempts of each chain; the single-attempt column is the first attempt alone. Strict pass@1 requires in addition that every request passed its own tests on the request's own step and that no passing test ever broke, and the regression rate is the share of attempts in which a passing test broke at some request, whether or not it was repaired later; both are means over the eight attempts, with the first attempt's values in Appendix F.7. The request completion rate is the share of all graded requests of the first attempt whose tests all pass on the step they are issued, pooled over a model's chains. An attempt without a request-20 state counts as a failure. Intervals are repository-clustered bootstrap 95%; the pairwise tests, on the first attempt, are in Appendix H.

F.3 Resume arms: per-model results

Table 21 gives the per-model estimates behind Section 4.2. Starting points are every request a model submitted as complete that failed its hidden contract ($k \leq 19$); the control is the benchmark's own continuation after a certified pass of the same request, standardized to the arm's distribution of k. Own-work failure at $k{+}1$ counts failures of $k{+}1$'s own tests and excludes failures inherited from the open defect. Intervals are Wilson 95% for rates and repository-clustered bootstrap 95% for contrasts (resampling arm and control together); the repaired and unrepaired subsets of arm E are outcome-selected. GPT-6 Astra and GLM-5.2 have no arm F. The starting states come from the five models' first attempts under one conversation per chain, as in our earlier release; arm D continues

that conversation, whereas arm F opens a fresh one on the same state, which is what the handoff protocol does at every request.

The control of Table 21 is a continuation after a certified pass, standardized to the arm's positions but not to its requests, so part of the gap between arm D and the control could be the difficulty of the requests that follow a miss. A second control removes this: for every starting point of Fable 5, GPT-6 Astra and Kimi K3, the same model attempts request $k{+}1$ from the developers' own state at k (their patches $1..k$, no hidden tests in the tree), in a fresh mini-swe-agent conversation, two attempts per state, graded with the frozen contract of $k{+}1$; the developers' state passes the contract of k and fails the tests of $k{+}1$ before every attempt (120, 130 and 256 states). On the same requests the developers' state fails 13.1, 13.5 and 5.6% of the time against 21.8, 28.5 and 25.0% after an untreated miss, a paired difference of 8.7 [−2.2, 20.4], 15.0 [5.7, 24.8] and 19.4 [12.4, 27.2] points: about 9 of Fable 5's 18 and of GPT-6 Astra's 24 points against the standardized control are request difficulty or selection and the rest is the state, and for Kimi K3 the whole gap is the state. Where request $k{+}1$ does not touch the code of the missed request the difference is 25.8 [5.7, 51.5], 21.4 [0.0, 42.9] and 14.0 [4.9, 26.9] points. The gold-state attempts are fresh conversations (arm F bounds what that changes), Fable 5 and Kimi K3 were served through a different route and reasoning setting than in arm D, and two attempts per state give coarse per-request rates; intervals are repository-clustered bootstrap 95%.

Table 21: **Resume arms per model.** D: silent continuation; E: one external "verification failed" message and one repair before request $k{+}1$; F: fresh conversation on the same broken state; gold state: the same request $k{+}1$ attempted from the developers' state at k.

	Fable 5	GPT-6 Astra	Kimi K3	GPT-5.6	GLM-5.2
Starting points / control units	470 / 174	130 / 161	836 / 595	939 / 572	1451 / 1594
Control (verified state): $k{+}1$ own-work failure (%)	3.8	4.3	9.6	16.9	11.9
Arm D: after one untreated miss (%)	21.8 [18.3,25.8]	28.5 [21.4,36.7]	25.0 [22.2,28.1]	30.4 [27.5,33.4]	29.3 [27.0,31.7]
Multiplier, D / control	5.8 [2.2,38.0]	6.6 [2.2,24.1]	2.6 [1.6,4.8]	1.8 [1.2,2.9]	2.5 [1.8,3.4]
Difference, D − control (pp)	18.0 [8.3,28.8]	24.2 [13.1,32.1]	15.4 [7.3,23.5]	13.4 [5.4,22.2]	17.3 [11.5,23.4]
Gold state: $k{+}1$ own-work failure, same requests (%)	13.1	13.5	5.6	—	—
Difference, D − gold state, paired (pp)	8.7 [−2.2,20.4]	15.0 [5.7,24.8]	19.4 [12.4,27.2]	—	—
Arm F: fresh conversation, same state, F − D paired (pp)	4.1 [0.7,8.2]	—	0.4 [−1.9,2.7]	−1.2 [−5.1,2.5]	—
Arm E: repair@1 (%)	44.6 [40.1,49.1]	45.4 [37.1,54.0]	29.4 [26.4,32.6]	30.0 [27.2,33.0]	37.4 [34.9,39.9]
Arm E: $k{+}1$ failure, repaired states (%)	5.7	5.1	3.4	12.4	8.5
Arm E: repaired states (n)	209	59	232	282	542
Arm E: $k{+}1$ failure, unrepaired states (%)	22.7	28.2	25.0	30.0	31.1
Arm E: unrepaired states (n)	260	71	559	657	909
Arm E: $k{+}1$ failure, all states (%)	15.1 [12.2,18.7]	17.7 [12.1,25.2]	18.7 [16.1,21.6]	24.7 [22.1,27.6]	22.7 [20.6,24.9]

F.4 Own-work failure by number of open defects

Table 22 pools every graded request from $k{+}1$ to 20 of the silent continuations of Section 4.2 and groups them by the number of defects still open beneath the request when it was issued (the original miss plus any later own-work failures). The odds ratio is the slope of a logistic fit of own-work failure on the count (capped at five), with a 95% repository-bootstrap interval. Requests with many open defects sit late in the chain and come from continuations that were already failing, so the progression is descriptive, not a controlled dose.

Inference cost. Table 23 reports standardized public-API cost estimates for the first attempts of our earlier single-conversation runs on the 280 chains (140 repositories), the runs whose token usage we have extracted, including ordinary failures; for the handoff first attempts of Table 2, that table gives model calls per chain. For every model, we assume that 98% of input tokens are cache reads and price the remaining 2% at short-TTL cache-write rates where available, otherwise at uncached-input rates; these are estimates under a common caching scenario, not measured cache-hit rates or

Table 22: **Own-work failure at requests $k{+}1$ to 20 by number of open defects beneath the request** (silent continuations, 140 repositories).

	Fable 5	GPT-6 Astra	Kimi K3	GPT-5.6	GLM-5.2
No open defect beneath the request (%)	8.0	8.0	12.1	12.0	10.5
One (%)	12.7	16.2	18.1	21.5	21.7
Two (%)	36.1	29.5	38.8	28.3	32.2
Three (%)	33.7	40.0	48.6	43.8	49.0
Four (%)	66.1	74.1	80.1	57.3	55.0
Five or more (%)	92.5	84.3	90.0	70.2	67.1
Units, none	893	212	994	1394	2268
Units, one	2419	672	4388	4968	7902
Units, two	781	244	1340	2270	3426
Units, three	409	75	451	726	1413
Units, four	127	27	176	391	845
Units, five or more	106	51	260	420	1276
Odds ratio per additional open defect	2.28 [1.69,3.86]	2.20 [1.69,2.79]	2.32 [1.97,3.00]	1.72 [1.49,2.05]	1.72 [1.56,1.92]

hosting bills. Means and medians use attempts with recorded usage; missing usage is not assigned zero cost.

Table 23: **Standardized first-attempt API cost (USD), single-conversation runs.** Coverage gives the number of first attempts with recorded token usage out of the 280 chains. All models use the same caching assumption described above.

Model	Usage / 280	Mean USD	Median USD
Fable 5	273	16.68	14.60
GPT-6 Astra	280	22.03	17.01
Kimi K3	280	9.88	8.12
GPT-5.6	279	12.34	10.19
GLM-5.2	266	9.03	6.43

The estimate uses the recorded total input I and output O as $[I(0.98p_C + 0.02p_R) + Op_O]/10^6$, with prices per million tokens for cache reads (p_C), remaining input (p_R), and output (p_O). Completion tokens already include reasoning. We use the public rate cards accessed on 6 October 2026, retaining the GPT-5.6 long-context split above 272,000 prompt tokens and using Astra's flat standard schedule. These references exclude runner/verifier compute, infrastructure retries and benchmark construction. Multi-attempt search costs are reported separately in the released cost tables, accumulating attempts through the first success or exhaustion.

Verifier compute. A separate compute model charges a hidden check at four virtual CPUs and $0.05 per vCPU-hour, multiplying that rate by mean verifier wall time. This gives approximately $0.001 per check; the median recorded check takes 16–17 seconds. Fable uses Harbor-timed continuation checks because its continuous-worker timestamps also include setup overhead. The estimate excludes runner provisioning and idle time, storage, network, and test construction and maintenance. It is not a measured system bill or an estimate of human-review cost. The overall cost of verification with repair additionally depends on repair rechecks and the downstream work that successful repairs enable; it is not established by the per-check estimate.

F.5 Noisy supervisor: intervals

Figure 7 plots the noisy-supervisor results in Table 37 with repository-clustered 95% bootstrap bands.

F.6 Root causes of failures

A request fails when one of its own hidden tests does not pass after it (a silent miss) or when a protected test of the original repository or of an earlier request that passed before it fails after it (a

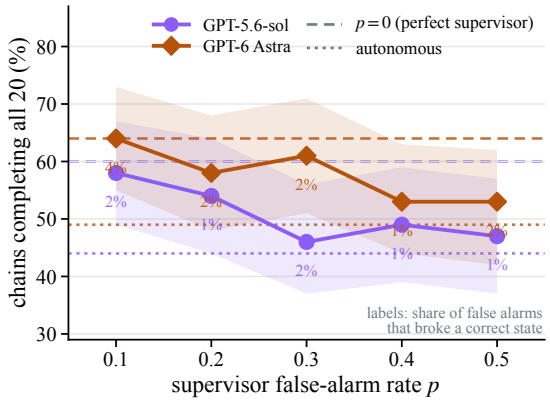


Figure 7: **Occasional false alarms cost little; one in two gives back most of the supervisor's gain, yet completion never drops below the unsupervised level.** Completion at 20 against the supervisor's false-alarm rate p (Table 37; GPT-5.6-sol and GPT-6 Astra, 100 chains per level), with repository-clustered 95% bootstrap bands. Dashed: perfect supervisor (p=0); dotted: autonomous (the same chains' first attempt, no supervisor). Labels: false alarms that broke a correct state (%).

regression); a test that was already failing when the request was issued is a carried failure and is not charged to the request (Appendix I.1). To ask why requests fail, we labeled every failed request of the ten models' first attempts on the 140 repositories, 4,787 sites in all (235 to 706 per model, Table 24): 1,645 are the first failure of their chain, the one failure that happens on a state in which every earlier request passed, and 3,142 are later failures, 1,908 of them on a repository already carrying an earlier failure. Of the 4,787, 3,384 are misses, 1,153 regressions with every own test passing, and 250 both. A judge model (GPT-5.6-sol at its highest reasoning setting) reads, for each site, the specification as delivered, the agent's diff and final message, the failing hidden tests with their output, the hidden test patch, the gold patch and the compacted trajectory, is told that the conversation started fresh at this request and which tests were already failing, and assigns one of 17 failure tags adapted from the DeepSWE taxonomy. Table 24 gives the distribution of its tags. A sandbox judge (Claude Fable 5.1) then re-executed, in the sealed image on the agent's own repository, the 1,733 sites the primary judge had left as benchmark-side suspects or untagged: it confirmed 795 as test-side (a hidden test stricter than, or in conflict with, the specification), reassigned 743 to an agent-side cause (184 integration errors, 171 missed requirements, 157 regressions, 153 wrong logic, 78 other) and withheld a tag on 195; 1,679 of these 1,733 requests were passed from a correct state by another model or attempt, as were 4,614 of the 4,787 sites. One judge labels each stage, and no handoff site was relabeled by a second judge or by a human. The agreement between two model judges was measured on an earlier census of the same benchmark with the same prompt and schema: on 275 sites labeled by both GPT-5.6-sol and Claude Fable 5, the two chose the same tag on 55%, κ 0.47 [0.40, 0.54], and the same coarse class on 62%; where they differed they described the same mechanism and disagreed on the tag. The labels are therefore indicative, not exact. The four paragraphs below describe the agent-side causes.

F.6.1 Regressions

Regression is the modal class of first failures for every model (37–55% of them): the agent implements the request and breaks behavior the request did not ask it to change. Among later failures its share falls to 19%, and over all failures it is 28% (22–39% per model). In 1,251 of the 1,329 regression sites a protected test fails; in the remaining 78 the new hidden test exercises pre-existing behavior that the edit broke. The protected tests that break are, in 1,111 of the 1,251 sites, hidden tests of an earlier request of the same chain, and in 126 only tests of the original repository: the agent breaks behavior it implemented itself, in a conversation it no longer has, against tests it never saw. In the traced example (Appendix I.3), GPT-5.6 looks in the history for "prior compatibility expectations", finds that the decorators it is asked to make stackable were "only just added in the latest

Table 24: **Labels of the 4,787 failed requests of the first attempts on the 140 repositories** (primary judge; % of the column's sites). First: the first failure of a chain; Later: every failure after it; All and the model columns: every failure. Benchmark-side tags are the primary judge's suspicion from static evidence; their re-execution by the sandbox judge is reported in the text.

Label	First	Later	All	Opus 5.5	Opus 5	GLM-5.3-Flash	Fable 5	GPT-6 Astra	GPT-6.1	GLM-5.3	GPT-5.6	Kimi K3	Opus 4.8
Regression: pre-existing behavior broken	44.0	19.3	27.8	31	31	23	39	25	28	22	30	32	22
Missed requirement: part of the specification unimplemented	10.6	15.9	14.1	9	12	18	17	9	8	13	11	17	21
Wrong logic	10.6	10.6	10.6	9	13	12	6	11	10	18	10	10	9
Integration error: new code unreachable from the tested path	5.4	7.7	6.9	4	5	5	5	6	6	10	8	5	11
Other agent-side: wrong file, syntax, unverified assumption, knowledge gap	3.6	4.9	4.4	4	2	3	3	6	4	3	2	4	10
Benchmark-side, suspected: test stricter than the specification, broken or ambiguous	24.3	38.0	33.3	40	35	36	25	41	41	33	36	30	24
No tag	1.6	3.6	2.9	2	3	3	5	3	4	1	3	1	4
Sites	1645	3142	4787	235	310	474	465	470	512	488	565	562	706

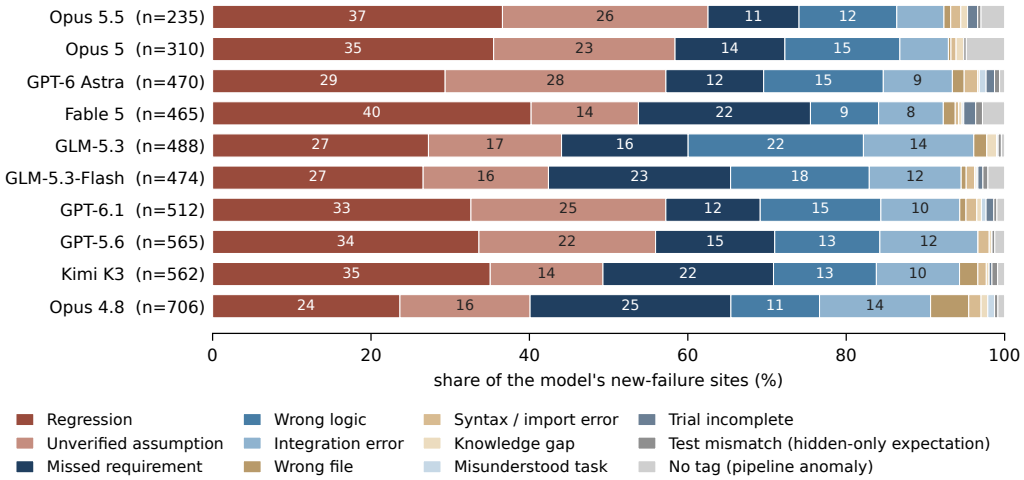


Figure 8: **Breaking what already worked is the most common new failure for nine of ten models and nearly half of all first failures.** Final root-cause label (static GPT-5.6 judge, Claude Fable 5.1 sandbox judge for the test-side and withheld tags, published merge rule; LLM labels, indicative) of the 4,787 requests at which a model's own hidden tests failed or a protected test that passed before the request failed after it: first attempt of the ten models on the 280 chains evaluated to request 20 without a stopping rule (leaderboard order; bar labels in per cent). Regressions are 31% of all new failures (48% of the chains' first failures but 23% of the 3,142 later ones); unverified assumptions (19%; 22% of later failures) and missed requirements (18%; 21%) follow.

commit" (its own request 8), concludes that there was "no earlier decorator contract to preserve", and redesigns them; two tests from request 8 then fail.

F.6.2 Missed requirements

One failure in seven leaves part of the specification unimplemented: the tests of the new behavior fail and nothing else does (98% of these sites fail only F2P nodes). The class is rarer at a chain's first failure (11%) than later (16%), and 220 of its 674 sites sit on a repository that was already carrying an earlier failure. It ranges from 8% of GPT-6.1's failures to 21% of Opus 4.8's, and its share is flat along the chain (13%, 14% and 15% by band).

F.6.3 Wrong logic and integration errors

A further 18% of failures implement the request but wrongly: the new code computes the wrong result (wrong logic, 11%) or is correct but never reached, wired into a literal but not into the constructor path the tests exercise, or into a sibling of the function the request names (integration error,

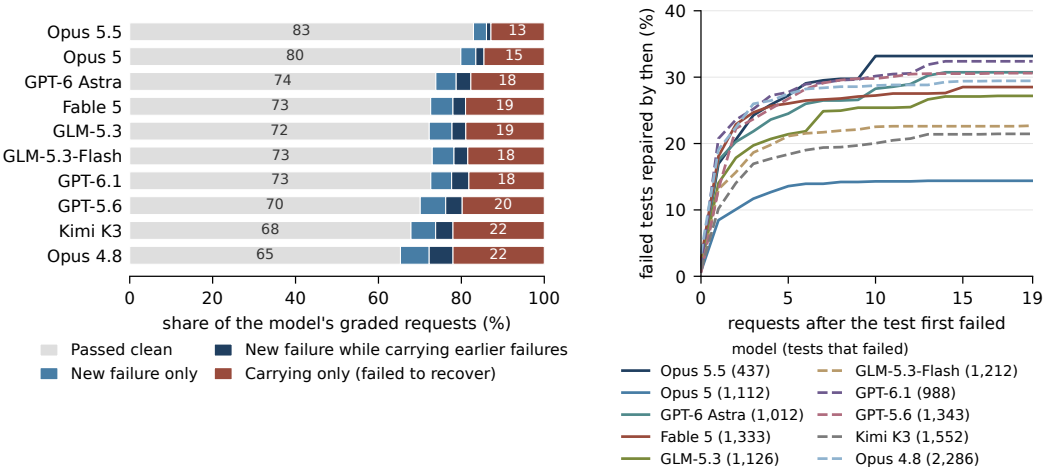


Figure 9: **Failures are seldom repaired: only 14–33% of the tests that fail during a chain ever pass again, and models carry an earlier failure at 14–28% of their requests.** Left: outcome of each of the 55,545 graded requests (first attempt, ten models, no stopping rule): passed clean, a new failure on an otherwise clean tree, a new failure while earlier failures were still carried, or carrying only (a test from an earlier request failing before and after the request with no new failure; bar labels = carrying-only share in per cent). Right: share of the 12,401 tests that failed during a chain (misses, regressions and silent breaks; tests failing only under a later step's test overlay excluded) repaired within d requests of their first failure; the repairs that happen come within 1–2 requests.

7%). Both are F2P-only failures in nearly every case. Integration errors are rarest for Opus 5.5 (4%) and most common for GLM-5.3 (10%) and Opus 4.8 (11%).

F.6.4 Unverified submission

The self-verification census (55,487 submitted requests of the ten models' first attempts on the 280 chains) asks what the agent did before it submitted. The agents test before almost every submission (97% of requests invoke a test runner; 90–100% per model), and a green local test run barely lowers the chance of a miss: submissions preceded by a passing run fail their request's hidden contract 8.3% of the time against 8.6% for all submissions (3.8–12.2% against 4.2–12.6% per model), and 59% of the misses follow a clean local run (47–77% per model; 96% follow some test run). The agents test what they wrote, not what was asked; a passing run of their own tests is what makes them believe a request complete.

F.7 Additional leaderboard details

Complete-chain rates. Table 25 gives the first attempt's pass@1, strict pass@1 and regression rate on the 280 chains with repository-clustered 95% bootstrap intervals, the values that Table 2 averages over the eight attempts, and the number of chains that pass@1 credits although a request was missed or a passing test broken along the way and repaired later. The converse, chains that delivered every request and never broke a test but fail pass@1, is rarer (0 to 5 per model): a test that never passed in the verifier environment is neither protected nor repairable and fails at request 20. pass@8 counts the chains that at least one of eight independent attempts completes under the same criterion, an attempt that broke a passing test being judged on its continuation to request 20.

Chains without a final state. An attempt counts as a failure in pass@1 and strict pass@1 whether or not it has a graded repository after request 20; Table 26 lists the first attempt's. Three kinds of chain have none (Table 26). A *verifier error* on a request leaves the chain without a verdict from that request on. A *budget exit* ends the chain when the two-hour tool-time budget or the model's context window is exhausted before request 20; the chain is graded for the request in which it happens and stops. A *failed run* is one whose model requests stopped returning (three Claude

Table 25: **Complete-chain rates on the 280 chains (140 repositories), first attempt.** Counts are chains; brackets are repository-clustered 95% bootstrap intervals in percent. Healed: chains counted by pass@1 but not by strict pass@1. pass@8: chains completed by at least one of the eight attempts. A chain without a request-20 state counts as a failure.

Model	pass@1 (attempt 1)	[95% CI]	strict pass@1	[95% CI]	regression rate	[95% CI]	healed	pass@8	[95% CI]
Claude Opus 5.5	187 (66.8)	[60.4, 73.0]	156 (55.7)	[49.1, 62.2]	59 (21.1)	[16.1, 26.4]	36	235 (83.9)	[79.2, 88.4]
Claude Opus 5	170 (60.7)	[54.3, 67.0]	143 (51.1)	[44.6, 57.4]	77 (27.5)	[22.0, 33.2]	30	232 (82.9)	[77.5, 88.0]
Claude Fable 5	145 (51.8)	[45.4, 58.0]	102 (36.4)	[29.8, 43.0]	118 (42.1)	[35.7, 48.9]	44	202 (72.1)	[66.3, 77.6]
GLM-5.3-Flash	143 (51.1)	[44.3, 57.8]	108 (38.6)	[31.8, 45.4]	87 (31.1)	[24.9, 37.4]	36	217 (77.5)	[71.6, 83.3]
GPT-6 Astra	143 (51.1)	[44.3, 57.8]	101 (36.1)	[29.6, 42.6]	98 (35.0)	[28.8, 41.5]	42	179 (63.9)	[57.2, 70.5]
GPT-6.1-sol	137 (48.9)	[42.3, 55.4]	102 (36.4)	[29.8, 43.2]	104 (37.1)	[30.9, 43.7]	36	170 (60.7)	[54.1, 67.2]
GLM-5.3	130 (46.4)	[39.7, 53.2]	107 (38.2)	[32.1, 44.4]	89 (31.8)	[25.6, 38.0]	28	213 (76.1)	[70.0, 82.1]
GPT-5.6-sol	128 (45.7)	[38.9, 52.5]	91 (32.5)	[26.0, 38.9]	121 (43.2)	[36.6, 50.0]	40	168 (60.0)	[53.4, 66.4]
Kimi K3	127 (45.4)	[38.4, 52.3]	92 (32.9)	[25.8, 40.1]	124 (44.3)	[37.5, 51.3]	36	191 (68.2)	[61.7, 74.4]
Claude Opus 4.8	107 (38.2)	[31.6, 44.9]	77 (27.5)	[21.4, 34.0]	115 (41.1)	[34.2, 48.3]	32	168 (60.0)	[53.0, 66.8]

Fable 5 chains, closed after a deterministic request-timeout loop). Counting all of these chains as failures is conservative. In the regression rate they count as not regressed unless a protected test had already broken, and in the request completion rate they contribute the requests they completed. Excluding them instead gives the last column of Table 26.

Table 26: **Chains without a request-20 state** on the 280 chains, by cause, and pass@1 over the chains that have one. Verifier errors and budget exits include those of the no-stop continuation of a chain.

Model	Verifier error	Budget exit	Failed run	pass@1 with a final state
Claude Opus 5.5	2	0	0	187/278 (67.3)
Claude Opus 5	5	0	0	170/275 (61.8)
Claude Fable 5	1	0	3	145/276 (52.5)
GLM-5.3-Flash	2	5	0	143/273 (52.4)
GPT-6 Astra	2	0	0	143/278 (51.4)
GPT-6.1-sol	2	0	0	137/278 (49.3)
GLM-5.3	3	3	0	130/274 (47.4)
GPT-5.6-sol	2	0	0	128/278 (46.0)
Kimi K3	3	3	0	127/274 (46.4)
Claude Opus 4.8	3	0	0	107/277 (38.6)

Failures along the chain. Table 27 follows the chains request by request. A chain is *clean through request* k if every request up to k passed its own tests on its own step and no passing test broke; clean through 20 is strict pass@1. The *first-failure hazard* of a quarter is the share of requests in that quarter at which a chain still clean picked up its first failure, a silent miss or a broken protected test, over the requests such chains reached in the quarter. For every model the hazard of the last quarter is at least that of the first, so the fall of the clean share over the chain is not front-loaded: it is the compounding of a roughly constant per-request rate. The miss and the regression are both common as a first failure: of the chains with one, 49 to 62% first missed a request, 31 to 45% first broke a passing test, and 4 to 8% did both at the same request. At request 20 the models pass that request's own tests in 84-94% of the chains with a final state, yet 29-55% of those chains carry a failing test from an earlier request or from the original repository.

Retries and the constant-rate fit. Every chain gets eight independent attempts; pass@8 counts the chains at least one of whose attempts passes under the pass@1 criterion. Eight attempts complete 12 to 30 points more chains than the first attempt alone (Table 25): 60 to 84% of the chains, against 38 to 67% on the first attempt and 39 to 66% per attempt on average (Table 2), and 46 to 78% with every request delivered on its own step and nothing broken in at least one attempt. The chains that no model completes in any of its eight attempts number 19 (44 are completed by no model on the first attempt). With h a model's pooled first-failure hazard, $(1-h)^{20}$ predicts a clean share at request 20 of 56.8, 53.0, 37.6, 40.1, 39.0, 38.6, 39.5, 34.3, 33.5 and 28.7% against the first attempt's strict pass@1 of 55.7, 51.1, 36.4, 38.6, 36.1, 36.4, 38.2, 32.5, 32.9 and 27.5%, so a constant per-request rate overestimates every model by 0.6 to 2.9 points: the strict record is well described by a flat hazard that compounds over twenty requests. pass@1 is not: it depends on what happens after a failure, which Appendix F.6 measures request by request.

Table 27: **Clean share by request and first-failure hazard by quarter on the 280 chains.** Clean through k: share of chains with no silent miss and no broken protected test at any request up to k (a chain without a verdict at some request $\leq k$ counts as not clean). Hazard: first failures at requests 1–5, 6–10, 11–15 and 16–20 over the requests reached by chains that were still clean. Request completion: share of all graded requests that passed their tests on their own step with no protected test broken.

	Clean through request k (%)				First-failure hazard by quarter (%)				Request
Model	k=5	k=10	k=15	k=20	1–5	6–10	11–15	16–20	completion (%)
Claude Opus 5.5	90.7	75.4	66.4	55.7	1.9	3.5	2.5	3.5	84.0
Claude Opus 5	88.2	73.9	62.1	51.1	2.5	3.2	3.4	3.8	80.8
Claude Fable 5	80.7	60.4	46.1	36.4	4.1	5.4	5.2	4.6	73.2
GLM-5.3-Flash	83.6	63.9	48.9	38.6	3.5	5.1	5.2	4.6	73.9
GPT-6 Astra	84.3	67.1	51.1	36.1	3.3	4.4	5.4	6.7	75.2
GPT-6.1-sol	83.6	64.6	49.6	36.4	3.5	4.9	5.2	5.9	74.2
GLM-5.3	82.5	63.2	49.3	38.2	3.8	5.1	4.7	4.9	73.0
GPT-5.6-sol	82.1	60.0	44.6	32.5	3.8	6.1	5.8	6.2	71.8
Kimi K3	76.8	57.5	43.2	32.9	5.1	5.5	5.4	5.4	68.8
Claude Opus 4.8	73.6	52.5	40.7	27.5	5.8	6.4	5.0	7.4	66.2

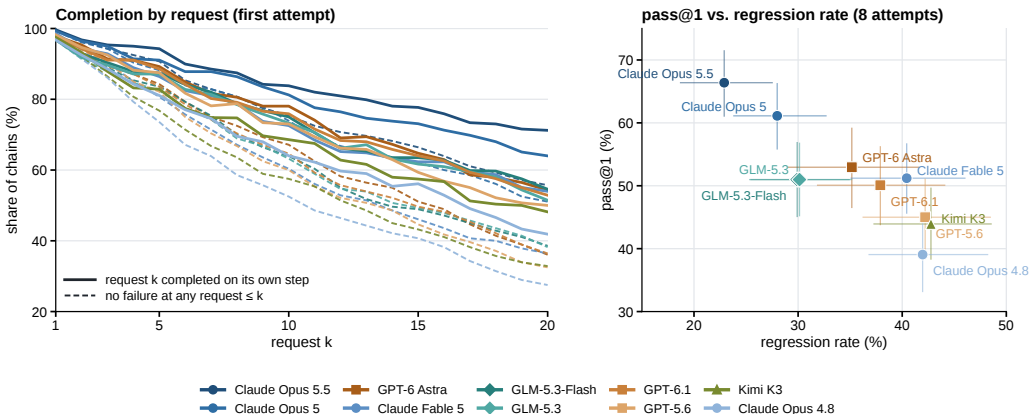


Figure 10: **Completion erodes request by request, and the models that break the fewest passing tests complete the most chains.** Left: for each model, the share of chains whose request k passed its own tests on its own step with no protected test broken (solid) and the share with no failure at any request up to k (dashed), first attempt, 280 chains. Right: pass@1 against the regression rate, both estimated from the eight attempts of each chain, with repository-clustered 95% intervals.

Shared outcomes. On the first attempt, 44, 21, 18, 23, 22, 23, 20, 22, 15, 26 and 46 chains are completed by 0, 1, . . . , 10 models; if models failed independently at their own rates, 0.2 chains would be completed by none and 0.3 by all ten. When two models both fail somewhere in a chain, their first failures fall on the same request in 44.6% of cases (2,624 of 5,885 pairs), against 5.4% if failure positions were independent. Any two models agree on pass or fail for 63–90% of chains (chance: 46–54%). The 44 chains that no model completes on the first attempt come from 35 repositories; on them the models complete a median of 7 to 10 of the 20 requests, and 19 of them resist every model's eight attempts.

Clustered intervals. Chains from one repository share code, tests and history, so every interval in Table 25 resamples the 140 repositories with replacement (10,000 draws), recomputes the statistic as a ratio of sums over the resampled chains (passing chains over chains) and reports the 2.5th and 97.5th percentiles; they are 1 to 3 points wider than the per-chain Wilson intervals of the same counts (Table 31 for pass@1). The 140 repositories contribute one to five chains each (34, 81, 18, 5 and 2 repositories respectively).

G Reproducibility and Evidence Scope

What the paper documents. Appendix D gives the construction pipeline, the specification prompt, grader rules, and every filtering round; Appendix F gives models, settings, and budgets; the `handoff/` directory of the release carries the scoring document, the per-model results with their intervals and the per-chain, per-request records behind Table 2; Table 28 gives the population of each study.

Replication. A replication needs the task images by digest, the grader, the specifications, and the per-attempt verdicts, together with the model and harness versions of Appendix F and, for the leaderboard, the scoring document and the per-request records of the release's `handoff/` directory, which define every verdict.

Statistical scope. Baseline intervals are repository-clustered bootstrap 95% intervals for the eight-attempt estimates and Wilson 95% intervals per chain for the single-attempt values, over all 280 chains of an arm; attempts without a request-20 record count as failures. 2,000 draws. Pairwise completion intervals use 20,000 draws. These counts belong to different analyses, not inconsistent repeats of one procedure. The prefix, review, and repair tables use the intervals and tests of each study. We infer no paired uncertainty from marginal rates.

H How well do coding benchmarks separate models?

This appendix gives the pairwise tests behind the leaderboard (Appendix F.2) and compares the resolving power of BOBA-Bench with that of the other repository-evolution leaderboards: that no two of the top five models on SWE-Milestone or SlopCodeBench can be told apart from the counts those papers report, that SWE-Chain reports no dispersion at all, and that on BOBA-Bench 26 of the 45 pairs among ten models are separated, the leader from every other model at $p \leq 0.036$. The script (`separation_analysis.py`) and its output accompany the paper.

Data. For each benchmark we take the per-model figures from the paper's main results table and the unit the figure is a fraction of (Table 29). Where the table reports a percentage rather than a count we recover the count as $\mathrm{round}(\text{rate} \times n)$; for SlopCodeBench the denominator is fixed at 196 by construction, and for SWE-Milestone the rates are per-evolution-range averages, so the implied count is approximate to within one milestone. SWE-Milestone's primary metric, Score, is a mean of recall$\times$precision and not a fraction of trials, so we use its Resolve Rate (a milestone is fully resolved or not). SWE-Chain's resolving rate is a mean over 12 chains of test-level micro-averages from a single run, with no per-chain values or dispersion published; it cannot be tested, and we show binomial bounds for orientation only. For BOBA-Bench we use attempt-1 outcomes on the 280-chain set of Section 3; a chain without a request-20 record (a verifier error, a budget or context exit before request 20, or a run whose model requests stopped returning; 2 to 7 chains per model) counts as a failure, as in the published rates.

Table 28: **Populations by study.** The unit and eligible subset depend on the experiment.

Study	Population and scope
Baseline	280 chains, 140 repositories; ten models; pass@1 estimated over eight attempts per chain (2,240 per model), single-attempt values alongside; pass@8 over the same attempts.
Whole-chain repair	280 chains, 140 repositories; five models, continuous-conversation first attempts; under the handoff protocol, ten models paired with their Autonomous first attempts (Appendix K).
Task-size and dependency statistics	280 chains, 140 repositories.
Verified-prefix comparison	Retained boundaries from the 140 repositories; eleven models, model-specific retention.
Failed-state continuation	Eligible silent misses from the continuous-conversation first attempts, with model-specific missing outcomes; gold-state control on 120 / 130 / 256 of those states (Fable 5 / GPT-6 Astra / Kimi K3), two attempts each.
Blind implementation review	109 wrong implementations per author from the Autonomous first attempts, each paired with a correct one: 2,180 items per reviewer, ten reviewers, 21,800 reviews.
Repair dose	Eligible misses on the 280 chains; GPT-5.6 and Kimi K3 in the continuous conversation; ten models under the handoff protocol (Appendix K).
Inference cost	Continuous-conversation first attempts on 280 chains, five models; usage available for 273 / 280 / 280 / 279 / 266 attempts (Fable / Astra / Kimi / GPT-5.6 / GLM). Standardized API estimates with a shared 98% cache-read assumption (Appendix F.4).
Verifier compute	Timed checks for Fable, GPT-5.6, Kimi and GLM, priced with an assumed CPU-hour rate; not measured supervision overhead.
False-alarm study	100 stratified chains; sealed run with 100 uncensored attempts per level for both models.
Harness comparison	58 chains, 52 repositories (a 60-chain draw made before the last filtering round, minus the two chains of a repository that round excluded); four models under three other harnesses, paired with their mini-swe-agent first attempts.
Prior-exposure probes	Hunk samples from the 140 repositories; all ten models.
Failure anatomy and self-checks	Failure anatomy: every graded request of the ten models' first attempts (55,545 requests, 4,787 failures); self-verification: every submitted request of the same attempts (Appendix F.6.4); self-checks: 60 silent misses and 60 accepted submissions per model from the continuous-conversation first attempts (Appendix I).

Table 29: Inputs. Unit: what one trial is. Count: how the per-model count was obtained.

Benchmark	Metric (source)	Unit	n	Count
SWE-Milestone (Deng et al., 2026)	Resolve Rate (Table 2), 15 agent–model configurations	milestone	98	round(rate$\times n$), ± 1
SlopCodeBench (Orlanski et al., 2026)	strict and isolated solve (Table 1; best run per model)	checkpoint	196	round(rate$\times n$)
SWE-Chain (Lam et al., 2026)	Build+Fix resolving (Table 3), single run	chain	12	not a count
BOBA-BENCH	pass@1, attempt 1	chain	280	exact

Tests. Intervals are Wilson 95% score intervals. For the other benchmarks the papers publish aggregates, not per-task outcomes, so two models can only be compared with an unpaired test: we use the two-sided Fisher exact test on the two counts at $\alpha = 0.05$. For BOBA-BENCH every model ran the same 280 chains, so we use the exact two-sided McNemar test on the discordant chains—the test Liu et al. (2026b) apply to adjacent SWE-bench Verified entries—and, because the 280 chains come from 140 repositories (one to five chains each), a repository-clustered paired bootstrap of the difference in pass@1 (repositories resampled with replacement, all their chains entering, 20,000 resamples, seed 20260921). We also report the unpaired Fisher test on our own counts so that the comparison with the other benchmarks uses the same test. No multiplicity correction is applied on either side; a correction would only reduce the number of separated pairs, and of the claims below it would remove the leader's separation from the runner-up (a Holm correction over the 45 pairs keeps 17 of the 26 separated pairs).

Other benchmarks. Table 30 gives, for each benchmark, how many of all model pairs, of adjacent pairs in the ranking, and of pairs among the top five are separated. On none of them is any adjacent pair or any pair among the top five separated. Separation exists only between a leader and clearly weaker models: on SWE-Milestone Gemini 3 Pro at 13.4% is separable only from MiniMax M2.5 at 1.3%; on SlopCodeBench GPT 5.5 at 14.8% is first separable from Sonnet 4.6 at 7.1% (strict) or Kimi K2.6 at 18.9% from 28.1% (isolated). On SWE-bench Verified, Liu et al. (2026b) find that exact McNemar tests separate none of the 29 adjacent pairs among the top thirty entries, whose scores span 8.8 points.

Table 30: Pairs of models separated at $\alpha = 0.05$ (two-sided Fisher exact test on reported counts). Spread: best to worst reported rate.

Benchmark	Metric	Models	Spread (%)	All pairs	Adjacent	Top five	Leader vs. runner-up (Wilson 95%)
SWE-Milestone	Resolve Rate	15	13.4–1.3	9/105	0/14	0/10	13/98 [7.9, 21.4] vs. 12/98 [7.1, 20.2]
SlopCodeBench	strict solve	15	14.8–2.0	27/105	0/14	0/10	29/196 [10.5, 20.4] vs. 22/196 [7.5, 16.4]
SlopCodeBench	isolated solve	15	28.1–4.1	46/105	0/14	0/10	55/196 [22.2, 34.7] vs. 51/196 [20.4, 32.6]
SWE-Chain	resolving, Build+Fix	9	60.8–20.2	not testable: single run, means over 12 chains, no dispersion published; a binomial bound with 12 units separates 0/36			

BOBA-BENCH. Table 31 gives the ten models' pass@1 with Wilson intervals and Table 32 every pairwise comparison. Twenty-six of the 45 pairs are separated by the McNemar test and 26 by the clustered bootstrap (25 in common: Opus 5.5 vs. Opus 5 is separated by the McNemar test alone, its bootstrap interval touching zero, and Fable 5 vs. GPT-5.6 by the bootstrap alone, $p = 0.064$), 20 by the unpaired Fisher test; three of the nine adjacent pairs are separated, and seven of the ten pairs among the top five. Opus 5.5 is separated from every other model ($p \leq 0.036$: the runner-up Opus 5 at $p = 0.036$, the other eight at $p \leq 2.1 \times 10^{-5}$; $p \leq 0.048$ if chains without a request-20 record are excluded instead of counted as failures). Below the two Opus models, seven models from 51.8% to 45.4% form a group within which only two of the 21 pairs are separated (GLM-5.3-Flash vs. Kimi K3, $p = 0.044$; GPT-6 Astra vs. GPT-5.6, $p = 0.049$), and Opus 4.8 is separated from all nine others; the ranking therefore has four tiers, Opus 5.5, then Opus 5 (the two separated by the McNemar test only), then the seven, then Opus 4.8. The number of models differs from the other leaderboards (10 against 9–15), so the comparable statements are about adjacent pairs, pairs among the top five, and the leader against the field, not the raw count of separated pairs.

Table 31: BOBA-BENCH, 280 chains, attempt 1: pass@1 with Wilson 95% intervals.

Model	pass@1	Wilson 95%
Opus 5.5	187/280 (66.8%)	[61.1, 72.0]
Opus 5	170/280 (60.7%)	[54.9, 66.3]
Fable 5	145/280 (51.8%)	[45.9, 57.6]
GLM-5.3-Flash	143/280 (51.1%)	[45.2, 56.9]
GPT-6 Astra	143/280 (51.1%)	[45.2, 56.9]
GPT-6.1	137/280 (48.9%)	[43.1, 54.8]
GLM-5.3	130/280 (46.4%)	[40.7, 52.3]
GPT-5.6	128/280 (45.7%)	[40.0, 51.6]
Kimi K3	127/280 (45.4%)	[39.6, 51.2]
Opus 4.8	107/280 (38.2%)	[32.7, 44.0]

Table 32: BOBA-BENCH pairwise comparisons on the same 280 chains. Discordant: chains passed only by the first / only by the second model. McNemar: exact two-sided p on the discordant chains. Fisher: unpaired two-sided p on the two counts. Bootstrap: repository-clustered 95% interval of the difference in percentage points.

		Diff. (pp)	Discordant	McNemar p	Fisher p	Bootstrap 95%
Opus 5.5	Opus 5	$+6.1$	38 / 21	0.036	0.16	$[+0.0, +11.9]$
Opus 5.5	Fable 5	$+15.0$	69 / 27	2.1×10^{-5}	4.1×10^{-4}	$[+8.1, +21.8]$
Opus 5.5	GLM-5.3-Flash	$+15.7$	68 / 24	4.9×10^{-6}	2.1×10^{-4}	$[+8.8, +22.6]$
Opus 5.5	GPT-6 Astra	$+15.7$	59 / 15	2.6×10^{-7}	2.1×10^{-4}	$[+9.8, +21.8]$
Opus 5.5	GPT-6.1	$+17.9$	66 / 16	2.3×10^{-8}	2.6×10^{-5}	$[+11.7, +24.1]$
Opus 5.5	GLM-5.3	$+20.4$	75 / 18	1.9×10^{-9}	1.7×10^{-6}	$[+13.2, +27.4]$
Opus 5.5	GPT-5.6	$+21.1$	75 / 16	2.7×10^{-10}	7.1×10^{-7}	$[+14.0, +28.0]$
Opus 5.5	Kimi K3	$+21.4$	78 / 18	4.4×10^{-10}	4.6×10^{-7}	$[+14.5, +28.4]$
Opus 5.5	Opus 4.8	$+28.6$	89 / 9	1.1×10^{-17}	1.7×10^{-11}	$[+22.0, +35.1]$
Opus 5	Fable 5	$+8.9$	51 / 26	5.9×10^{-3}	0.041	$[+2.9, +14.9]$
Opus 5	GLM-5.3-Flash	$+9.6$	53 / 26	3.2×10^{-3}	0.027	$[+3.1, +16.4]$
Opus 5	GPT-6 Astra	$+9.6$	53 / 26	3.2×10^{-3}	0.027	$[+3.2, +16.1]$
Opus 5	GPT-6.1	$+11.8$	58 / 25	3.8×10^{-4}	6.5×10^{-3}	$[+5.6, +18.2]$
Opus 5	GLM-5.3	$+14.3$	62 / 22	1.5×10^{-5}	9.3×10^{-4}	$[+7.5, +21.0]$
Opus 5	GPT-5.6	$+15.0$	58 / 16	9.7×10^{-7}	5.0×10^{-4}	$[+8.7, +21.3]$
Opus 5	Kimi K3	$+15.4$	64 / 21	3.3×10^{-6}	3.7×10^{-4}	$[+8.8, +21.9]$
Opus 5	Opus 4.8	$+22.5$	70 / 7	3.5×10^{-14}	1.4×10^{-7}	$[+16.8, +28.3]$
Fable 5	GLM-5.3-Flash	$+0.7$	46 / 44	0.92	0.93	$[-6.2, +7.7]$
Fable 5	GPT-6 Astra	$+0.7$	45 / 43	0.92	0.93	$[-5.5, +7.0]$
Fable 5	GPT-6.1	$+2.9$	46 / 38	0.45	0.55	$[-2.9, +8.7]$
Fable 5	GLM-5.3	$+5.4$	59 / 44	0.17	0.24	$[-1.8, +12.4]$
Fable 5	GPT-5.6	$+6.1$	46 / 29	0.064	0.18	$[+0.4, +11.8]$
Fable 5	Kimi K3	$+6.4$	48 / 30	0.054	0.15	$[+0.0, +13.0]$
Fable 5	Opus 4.8	$+13.6$	55 / 17	8.1×10^{-6}	1.6×10^{-3}	$[+8.0, +19.2]$
GLM-5.3-Flash	GPT-6 Astra	$+0.0$	41 / 41	1.0	1.0	$[-6.9, +6.9]$
GLM-5.3-Flash	GPT-6.1	$+2.1$	42 / 36	0.57	0.67	$[-4.3, +8.4]$
GLM-5.3-Flash	GLM-5.3	$+4.6$	42 / 29	0.15	0.31	$[-1.5, +10.6]$
GLM-5.3-Flash	GPT-5.6	$+5.4$	46 / 31	0.11	0.24	$[-1.1, +11.6]$
GLM-5.3-Flash	Kimi K3	$+5.7$	36 / 20	0.044	0.20	$[+0.4, +10.9]$
GLM-5.3-Flash	Opus 4.8	$+12.9$	55 / 19	3.4×10^{-5}	2.9×10^{-3}	$[+6.8, +19.0]$
GPT-6 Astra	GPT-6.1	$+2.1$	17 / 11	0.34	0.67	$[-1.7, +5.9]$
GPT-6 Astra	GLM-5.3	$+4.6$	52 / 39	0.21	0.31	$[-2.7, +12.0]$
GPT-6 Astra	GPT-5.6	$+5.4$	33 / 18	0.049	0.24	$[+0.4, +10.4]$
GPT-6 Astra	Kimi K3	$+5.7$	47 / 31	0.089	0.20	$[-1.1, +12.4]$
GPT-6 Astra	Opus 4.8	$+12.9$	53 / 17	1.9×10^{-5}	2.9×10^{-3}	$[+7.1, +18.8]$
GPT-6.1	GLM-5.3	$+2.5$	51 / 44	0.54	0.61	$[-4.9, +9.7]$
GPT-6.1	GPT-5.6	$+3.2$	29 / 20	0.25	0.50	$[-1.4, +7.8]$
GPT-6.1	Kimi K3	$+3.6$	39 / 29	0.27	0.45	$[-2.7, +9.7]$
GPT-6.1	Opus 4.8	$+10.7$	51 / 21	5.3×10^{-4}	0.013	$[+5.0, +16.4]$
GLM-5.3	GPT-5.6	$+0.7$	46 / 44	0.92	0.93	$[-6.6, +8.0]$
GLM-5.3	Kimi K3	$+1.1$	41 / 38	0.82	0.87	$[-5.3, +7.5]$
GLM-5.3	Opus 4.8	$+8.2$	53 / 30	0.015	0.060	$[+2.2, +14.3]$
GPT-5.6	Kimi K3	$+0.4$	30 / 29	1.0	1.0	$[-5.6, +6.2]$
GPT-5.6	Opus 4.8	$+7.5$	40 / 19	8.6×10^{-3}	0.087	$[+2.1, +13.1]$
Kimi K3	Opus 4.8	$+7.1$	40 / 20	0.013	0.10	$[+1.7, +12.6]$

What the comparison does and does not establish. Three caveats bound the claim. First, the other benchmarks are tested unpaired because per-task outcomes are not published; with those outcomes a paired test could separate more pairs than we find, which is why Section 2 says "from their reported counts". Second, treating tasks, milestones or checkpoints as independent trials ignores

their clustering within repositories or problems (SWE-Milestone's 98 milestones come from seven repositories, SlopCodeBench's 196 checkpoints from 36 problems); the true intervals are therefore wider than those in Table 30, which makes the finding of no separation conservative in that respect. Third, the leaderboards evaluate different model sets under different scaffolds, so the comparison is about the resolving power of each benchmark on the models its authors ran, not about any particular model.

I TRIAL RECORDS: FAILURE ANATOMY, SELF-CHECKS, AND TRACED REGRESSIONS

This appendix gives the long form of three results of Section 4.2 ("How agents miss"). It reads the per-request grader records and the agent conversations of the first attempts on the 280 chains; the failure anatomy covers every graded request of the handoff first attempts, and the traced cases are first failures, the first request of a chain whose own tests fail or that breaks a protected test. Unless stated otherwise, intervals are repository-clustered bootstrap 95% intervals.

I.1 FAILURE ANATOMY

Method. The grader of request k runs every test of the request's universe, its own hidden tests ($F2P_k$), the pass-to-pass list of state k and the accumulated tests of earlier requests (Appendix D.9), once on the tree the agent received and once on the tree it submitted, and records each node as passed, failed, uncollectable or missing (a deleted or renamed test keeps its frozen id and counts as missing). A request has a *new failure* if one of its own tests does not pass afterwards (a silent miss) or if a protected test, one that passed before the request, does not pass afterwards (a regression); it is *carrying* a failure if a test that was already failing when it was issued is still failing afterwards. Every graded request of every chain, 55,545 across the ten models, therefore has one of four outcomes: passed clean, new failure only, carrying only, or both. A new failure is *own only* if only the request's own tests fail, *earlier only* if only protected tests fail, and a *regression* if at least one protected test fails. Every class comes from grader output; no model judges it. Sixty of the 4,787 new failures have a failing node that passes when re-run in the sealed image (flaky or order-dependent tests); they are kept as graded. Tests that fail on the received tree only because the request's own test overlay rewrote or newly imports them are grader artefacts and are excluded from the carried counts; so are original repository tests that never pass in the verifier environment. Every classified failure followed the agent's own submission of the request as complete. The *first failure* of a chain is its first request with a new failure (1,645 of the 2,800 model $\times$ chain slots; 1,079 chains passed all 20 requests, 31 ended on a verifier or budget error, or did not finish, before any failure, and 45 chains' first graded failure was only a carried earlier test): it is the one failure that happens on a state in which every earlier request passed, so nothing it breaks was inherited. Appendix F.6 labels the same 4,787 sites with a judge model and reaches a regression share of 44.0% of first failures against 44.4% here, and 27.8% of all failures against 29.3%, by a different route: the judge places some broken protected tests on the benchmark side and tags 78 failures of a request's own tests as regressions of behavior the new test exercises.

Where and how far back. The regression share is 37.8% [29.8, 46.3] of first failures at requests 1–5, 49.6% at requests 6–10, 51.5% at requests 11–15, and 38.3% at requests 16–20. The rise from requests 1–5 to requests 6–15 is 12.6 points [2.6, 22.8], but the share falls again at the end of the chain, so we do not claim a monotone trend. Of the 731 first-failure regressions, 624 (85.4% [79.3, 90.8]) break a test that an earlier request of the chain added or changed; 103 break only tests of the original repository. The nearest such request lies a median of three requests back (interquartile range 2–6), and 35.3% [28.7, 42.0] of these regressions break a request five or more back. Every broken test had passed on the tree the agent received: the grader protects only such tests, and at a first failure every earlier request had passed. In 38.5% [33.4, 44.1] of first failures, and in 24.1% [20.3, 28.0] of all failures, every own test passes and only earlier behavior breaks, so a grader of the new request's tests alone would accept the submission. The contract checks only tested behavior. Breakage of untested behavior is not measured, so the true regression share is likely higher.

After a failure. Nothing stops a chain, so what follows a failure is observed rather than inferred (Table 34). A new failure occurs at 8.6% of all graded requests: at 3.6–9.6% of the requests issued

Table 33: **The first failure of a first attempt, and the failures after it**, from grader output alone (280 chains, ten models, first attempts). Each first failure is classed by its failing tests: those of the new request (own) and those of behavior that already existed (protected tests of the original repository or of an earlier request). A regression fails at least one protected test. Shares are percentages of a model's first failures (n); the pooled row gives repository-clustered bootstrap 95% intervals. "Earlier request broken": first failures that break a test added or changed by an earlier request of the same chain. The last two columns give the new failures after the first one and their regression share.

	First failure						Later failures	
	n	Own tests only	Earlier tests only	Both	Regression	Earlier request broken	n	Regression
Opus 5.5	116	56.0	37.9	6.0	44.0	37.9	119	17.6
Opus 5	130	53.8	41.5	4.6	46.2	41.5	180	18.3
GLM-5.3-Flash	165	61.2	31.5	7.3	38.8	33.3	309	17.5
Fable 5	170	47.6	45.9	6.5	52.4	47.6	295	31.2
GPT-6 Astra	170	55.3	39.4	5.3	44.7	37.1	300	19.3
GPT-6.1	170	53.5	40.6	5.9	46.5	37.1	342	21.1
GLM-5.3	163	60.1	35.0	4.9	39.9	35.0	325	16.3
GPT-5.6	184	53.8	42.4	3.8	46.2	37.5	381	23.6
Kimi K3	180	52.2	40.0	7.8	47.8	40.0	382	25.9
Opus 4.8	197	61.4	32.0	6.6	38.6	33.5	509	19.6
Pooled	1645	55.6 [50.6, 60.8]	38.5 [33.2, 43.8]	5.9 [3.5, 8.5]	44.4 [39.4, 49.6]	37.9 [32.8, 43.1]	3142	21.4 [17.4, 26.0]

on a repository in which every earlier test passed and at 7.7–20.7% of those issued on a repository already carrying a failure; the second group sits later in the chain and in chains that were already failing, so the gap is descriptive, not a controlled contrast (Appendix F.3 gives the controlled one). The failures that follow a first failure are of a different kind: 44% of first failures are regressions against 21% of the 3,142 later ones, which are mostly misses of the request's own contract (79%). What breaks mostly stays broken. Of the 12,401 tests that fail at some request (5,919 as a miss, 5,639 as a regression, 843 broken while outside the universe of the intervening requests, so that the break is seen only later), 3,314 (27%; 14–33% per model) pass again at a later request, typically one or two requests later (median per model 1–2), and 9,283 are still failing at the chain's end. At 14–28% of all graded requests a model is carrying at least one earlier failure, and a model's chains spend 17–35% of their 20 requests in that state. Of the 1,697 chains (of 2,800) that break at some request, 360 (16–29% per model) are fully green again at request 20; these are the chains that pass@1 counts and strict pass@1 does not (Section 4.1).

Table 34: **Carried failures**: every graded request of the first attempts (280 chains per model) by its outcome, and what happens to the tests that fail. Clean: every test of the universe passes; New: a new failure on a repository with no earlier failure; Carrying: an earlier failure is still failing and nothing new fails; Both. "New failure rate" is the share of requests with a new failure among those issued on a clean repository and among those issued on a repository carrying a failure (descriptive; the two groups differ in position and chain). Recovered: tests that fail at some request and pass again at a later one (median delay in requests). Healed: chains broken at some request whose repository passes every test at request 20. Tests that fail on the received tree only because the request's test overlay rewrote them are excluded.

	Graded requests (%)					New failure rate (%)		Tests that fail			Chains
	n	Clean	New	Carrying	Both	clean repo.	carrying	n	Recovered (%)	Delay	Healed / broken
Opus 5.5	5574	82.9	3.1	12.9	1.1	3.6	7.7	437	33.2	2	36 / 123
Opus 5	5549	79.9	3.6	14.5	2.0	4.3	11.9	1112	14.4	1	30 / 134
GLM-5.3-Flash	5528	73.0	5.2	18.5	3.3	6.7	15.3	1212	22.7	1	36 / 168
Fable 5	5525	72.6	5.4	19.0	3.0	6.9	13.8	1333	28.5	1	44 / 174
GPT-6 Astra	5567	73.8	4.9	17.7	3.5	6.2	16.7	1012	30.7	1	42 / 177
GPT-6.1	5571	72.6	5.0	18.2	4.2	6.5	18.6	988	32.4	1	36 / 176
GLM-5.3	5536	72.3	5.5	18.9	3.3	7.1	14.9	1126	27.2	2	28 / 171
GPT-5.6	5581	70.1	6.1	19.8	4.0	8.1	16.7	1343	30.6	2	40 / 188
Kimi K3	5545	67.8	6.0	22.0	4.2	8.1	16.0	1552	21.5	2	36 / 186
Opus 4.8	5569	65.3	6.9	22.0	5.7	9.6	20.7	2286	29.4	1	32 / 200
All	55545	73.0	5.2	18.3	3.4	6.6	15.8	12401	26.7	–	360 / 1697

I.2 SELF-CHECKS BEFORE SUBMISSION

Sample. The sample comes from the first attempts of our earlier runs of the five models of Appendix F.3, in which one conversation ran through a chain and the chain ended at its first failed request; the fixed rules below read a request's commands and outputs and do not depend on that protocol. From first attempts with a finished record on the 280 chains, we drew, per model, 60 silent misses and 60 accepted submissions uniformly at random with a fixed seed: 600 submitted and graded requests from 525 attempts in 136 repositories. A miss is the failed request that ended its attempt, so the 300 misses come from 300 different attempts.

Rules. Fixed regular-expression rules read only the sampled request's commands and tool outputs; no model labels the reported rates. The agent *ran tests* if it called a test runner (pytest, unittest, tox, nox, or a `make` test target) or ran a Python assertion script that writes no repository file. It *ran the repository suite* if a runner call had no targets, or only repository paths that the agent had not written in that request. *Wrote test* counts writes of test files, scratch files included. The *last run* is the outcome of the last test command. It failed if a runner summary reports failures or errors, a test is listed as failed, or a script raises a traceback or exits with an error; it passed if a summary reports passes and no failures, or a script exits cleanly; otherwise it is unknown. The agent *claimed success* if one of its last three non-empty messages states success (for example, that all tests pass or that the work is verified or complete) and the final message has no hedge. *Green, then claimed* requires a test run, a passing last run, and a success claim.

Rule check. We checked the rules against labels that a language model assigned from the full transcripts, on requests not used to write the rules. On 20 such requests, the action rules agreed on 100 of 100 labels and the last-run outcome on 18 of 20. After one revision, the success-claim rule agreed on 18 of 20 further requests. An independent re-implementation of the rules gives 100% for running tests, 82.0% for the repository suite, and 56.3% for green, then claimed, against 99.7%, 85.0%, and 61.3% in Table 35.

Results. Because the sample is stratified on the outcome, we compare misses with accepted submissions through Mantel–Haenszel odds ratios across models. Neither running the repository suite (1.44 [0.90, 2.29]) nor a passing last run (1.33 [0.91, 1.98]) is associated with fewer silent misses. In half of the green-then-claimed misses (93 of 184), the passing last run was the repository suite; in the rest it was the agent's own script or a targeted run. In at least 43 of the 300 misses (14.3%, Wilson [10.8, 18.8]), the agent's own test output named a test that the grader later failed, and the agent's last mention of that test still reported a failure. This count is a lower bound: agents often print only summary lines, and the records keep at most 50 failing test identifiers per request. Appendix F.6.4 counts every submitted request of the ten models' handoff first attempts instead of a sample and finds that 59% of the misses (47–77% per model) follow a clean local run; that census applies its own test for a clean run and requires no success claim, so its shares are not directly comparable with Table 35.

Table 35: **Self-checks before submission**, first attempts. Each cell gives the percentage of sampled silent misses / accepted submissions (60 each per model) that show the behavior. "Green, then claimed": the last test run passed and the closing message claimed success. "Wrote test" includes scratch files. Claim rates depend on whether a model writes closing text, and the success-claim odds ratio is driven by Fable 5 (without it: 1.35, 0.71–2.70). Pooled intervals and odds ratios (OR, Mantel–Haenszel across models, silent miss vs. accepted) use a repository-clustered bootstrap.

Model	Ran tests	Repo suite	Wrote test	Saw failure	Last run green	Claimed success	Green, then claimed
Fable 5	100.0 / 98.3	73.3 / 58.3	26.7 / 5.0	56.7 / 41.7	71.7 / 68.3	65.0 / 26.7	46.7 / 20.0
GPT-6 Astra	100.0 / 100.0	91.7 / 90.0	6.7 / 0.0	90.0 / 78.3	78.3 / 65.0	98.3 / 96.7	76.7 / 61.7
Kimi K3	98.3 / 98.3	85.0 / 83.3	25.0 / 15.0	75.0 / 65.0	66.7 / 65.0	96.7 / 98.3	66.7 / 65.0
GPT-5.6	100.0 / 100.0	86.7 / 88.3	13.3 / 15.0	80.0 / 61.7	95.0 / 91.7	40.0 / 35.0	40.0 / 31.7
GLM-5.2	100.0 / 96.7	88.3 / 80.0	36.7 / 15.0	81.7 / 65.0	78.3 / 75.0	96.7 / 91.7	76.7 / 70.0
All (300 / 300)	99.7 / 98.7	85.0 / 80.0	21.7 / 10.0	76.7 / 62.3	78.0 / 73.0	79.3 / 69.7	61.3 / 49.7
95% CI, misses	98.9–100	80.4–89.3	16.7–26.6	71.2–81.9	73.1–82.8	74.1–84.1	55.7–66.9
OR (95% CI)	–	1.44 (0.90–2.29)	2.56 (1.57–4.62)	2.08 (1.41–3.07)	1.33 (0.91–1.98)	2.26 (1.40–3.71)	1.71 (1.21–2.42)

Table 36: **A regression that undoes accepted work** (`python-poetry/cleo`, GPT-5.6, first attempt; abridged, quotes verbatim). Request 9 is silent on a lone decorator, and the agent's conversation holds nothing of request 8. A grader of request 9's new test alone would accept this code. Chosen for clarity, not at random (Appendix I.3).

Request 8	Add `@app.command`, `@app.argument` and `@app.option`; a lone `@app.argument` or `@app.option` registers a command under the function's name. GPT-5.6's request-8 code does this, and the request passes.
Request 9	Let the three decorators stack on one function. In a fresh conversation, the agent reads the history, finds the decorators "only just added in the latest commit" and concludes: "There was no earlier decorator contract to preserve". It makes `argument` and `option` store definitions only, checks the stacked form ("all 41 application tests pass") and submits.
Hidden tests	Request 9's new test passes. Two request-8 tests fail: `application.has(...)` is `False` after a lone decorator. Opus 5.5, Opus 5, Fable 5, GLM-5.3 and GLM-5.3-Flash pass request 9 on their first attempts.

I.3 Two traced regressions

We chose these two first-attempt regressions for clarity, not at random. Both meet five conditions: the specification states the behavior; the agent tested its change and declared success; the failing hidden test shows what was broken; that test belongs to an earlier request; and another model or attempt passed the same request. Quotes are verbatim from the request's conversation, square brackets mark our edits, and code is abridged.

Case 1: `python-poetry/cleo`, GPT-5.6, request 9 (Table 36). Request 8 adds three decorators to `Application`. Each registers a command. Used alone, `@app.argument` and `@app.option` register the command under the decorated function's name. The agent's request-8 code does exactly that, each decorator building a `Command` and calling `self.add`, and the request passes. Request 9, in a new conversation that holds nothing of request 8, asks that `@command`, `@argument` and `@option` stack on one function, with `@command` applied last. It states: "The `argument` and `option` decorators contribute definition entries; the `command` decorator finalizes registration." It says nothing about a decorator used alone. The agent reads the three methods ("each currently creates and registers its own independent command") and, before editing, looks for "any prior compatibility expectations (such as standalone `argument`/`option` behavior), since preserving those where possible will reduce regressions". The history shows the methods added by the previous commit, its own request 8, and it concludes: "There was no earlier decorator contract to preserve—the methods were only just added in the latest commit—so I'll implement the specified finalize-on-`command` behavior directly." It rewrites `argument` and `option` so that they only store a definition on the function for `@command` to collect. Its checks use the stacked form; the repository's visible tests, which do not contain request 8's hidden tests, pass ("all 41 application tests pass, and the broader suite passes 152 tests when excluding the two pre-existing unrelated progress-rendering failures"). After 15 model calls it submits: "Implementation and validation are complete." The one new test of request 9 passes. Two of the 197 protected tests fail: `test_argument_decorator` and `test_option_decorator`, both from request 8, at `self.assertTrue(application.has('decorated_foo_code'))` with `AssertionError: False is not true`. A correct change keeps the request-8 behavior for a decorator used alone and adds the stacked behavior. Opus 5.5, Opus 5, Fable 5, GLM-5.3 and GLM-5.3-Flash pass request 9 on their first attempts; GPT-6 Astra, GPT-6.1, Kimi K3 and Opus 4.8 fail the same two tests, Opus 4.8 in all eight of its attempts and Opus 5.5 in none. Across all graded handoff attempts, 12 of 24 pass request 9. The two tests stay broken for the rest of GPT-5.6's chain: every later request is issued on a repository carrying them, request 16 adds a miss of its own, and the final state fails.

Case 2: `joshtemple/lkml`, Fable 5, request 4. Request 2 adds `serialize_dict`; when the dictionary has a `name` key, "The output begins with the value of `"name"`, followed by a space, then {". Request 3 extends it to nested dictionaries and repeats the form: "use `<name_value>` {, with no key or colon before the name value". Both pass. Request 4 asks only for a

`serialize_list` method and says nothing about `serialize_dict`. The agent, in a new conversation, adds the method, checks the two examples of the specification, and runs the repository's test suite. Two of the repository's own serializer tests fail: they predate the chain and still expect `dimension: {`, with a colon; the hidden test patch of request 2 rewrote them, but hidden patches are applied only in the verifier. The agent confirms with `git stash` that they also fail on the tree it received: "Pre-existing failures. These are in `serialize_dict` (missing colon), outside my segment but let me check if fixing it breaks anything else"; then: "The dict tests fail pre-existing because `serialize_dict` yields `name {` instead of `name: {`. Since the serializer isn't used elsewhere, I'll fix that too so all tests pass". It adds the colon, reports "All 87 tests pass" and submits after 15 model calls. Both new tests of request 4 pass. Two of the 89 protected tests fail: `test_serialize_dict_with_name` from request 2 and `test_serialize_nested_dict` from request 3, each expecting `<name> {` and receiving `<name>: {`. A correct change adds `serialize_list` and leaves `serialize_dict` alone. The nine other models pass request 4 on their first attempts, and Fable 5 passes it in five of its eight. Across all graded handoff attempts, 43 of 47 pass request 4. Here the chain heals: the two tests are still failing after requests 5 and 6 and pass again after request 7, and Fable 5's final state passes every test.

J Supporting Details for Related Work

This appendix retains quantitative and construction details omitted from Section 2. These details describe the evaluation setting and its controls; they do not establish that tasks across benchmarks have equal difficulty or that a larger benchmark necessarily distinguishes every model pair.

Counting requests consistently. We count one separately delivered specification document as one request, without splitting its internal requirements into additional tasks. SWE-Milestone contains 98 graded specifications and three additional, ungraded specifications that agents still implement (Deng et al., 2026). SWE-Chain contains 155 version-transition specifications, each bundling upgrade requirements (Lam et al., 2026). SlopCodeBench contains 196 checkpoint specifications across 36 problems (Orlanski et al., 2026). Including the ungraded SWE-Milestone requests gives 452 specifications in total. These counts were checked against SWE-Milestone dataset revision `3500478e`, SWE-Chain dataset revision `7080d652`, and the SlopCodeBench problems release `v1.0` (`4d38d300`). The corresponding source snapshots and counting scripts accompany our local research artifacts.

BOBA-Bench contains 280 chains of 20 requests across 140 repositories, yielding 5,600 request occurrences but 4,404 distinct specifications. The final window of a retained run overlaps the preceding window when needed to include its tail, producing 1,196 repeated occurrences at different chain positions. The distinct-request ratio is therefore $4{,}404/452 = 9.74$, not the occurrence ratio $5{,}600/452 = 12.39$. Repeated requests have different prefixes but are not independent observations for repository-level inference. These are counts of specification documents, not comparable amounts of implementation work.

Task-quality evidence. OpenAI's SWE-bench Verified audit establishes a known-defect floor above 16% of the full benchmark, based on an audit that targeted frequently failed tasks rather than a random sample (OpenAI, 2026e). Its SWE-bench Pro analysis reports defects in roughly 30% of the public split, with the exact proportion depending on the audit procedure (OpenAI, 2026d). These figures concern different auditing procedures and should not be interpreted as directly comparable prevalence estimates. Other studies identify solution leakage, weak tests, and opportunities for exploitation (Aleithan et al., 2024; Yu et al., 2025; Zhong et al., 2025).

SWE-Milestone generates requirements from reference patches, aligns them with tests, and uses expert review to check solvability and implementation leakage (Deng et al., 2026). BOBA specification generation receives test changes and development descriptions, but not the source patch. This removes direct access to the reference implementation at generation time, rather than proving that every resulting specification is complete or free of hints. Appendix D.4 describes generation and reference-code validation.

The isolated-success figure formerly quoted in Related Work concerns sampled requests, not every request in every chain. For Claude Fable 5, 1,089 of 1,120 scheduled attempts on one sampled

request per repository passed, or 97.23%, counting the 28 ungraded attempts as failures. Among graded attempts the fraction was 1,089/1,092. These denominators must not be confused with distinct-request coverage or the probability of at least one success across repeated attempts. The count comes from the filtered per-target isolated-attempt summary, not from the earlier candidate-certification campaign.

Model comparison and uncertainty. The manuscript's earlier Related Work combined illustrative binomial interval widths with claims about ranking models. Counts of repositories, chains, and checkpoints are not interchangeable statistical units. Within-repository dependence must be accounted for, and an aggregate rate cannot in general be multiplied by a nominal task count to reconstruct valid binomial outcomes. In particular, a macro-averaged milestone rate is not a binomial proportion over all milestones. The absence of published uncertainty does not demonstrate that models are indistinguishable. We therefore do not retain the earlier numerical resolution argument or the blanket non-separation claim in Section 2. The predecessor calculations in Appendix H require revision before they can support comparative significance claims. BOBA comparisons should use the observed paired outcomes and repository-clustered uncertainty, without treating a nonsignificant difference as evidence of equivalence.

K TRUST, BUT VERIFY: PROTOCOLS AND ESTIMATES

This appendix gives, for each arm of Section 5, what the agent was told, which runs were used, and the per-model estimates with intervals. The mechanism studies of Appendices K.1–K.5 ran in the continuous conversation: one conversation per chain, continued after the miss so that the verifier's message arrives as the observation to the agent's submit command. There, completion means that every request passes on its own step, and the baseline is the first attempt under that conversation, not the pass@1 of Table 2, which starts every request in a fresh conversation and grades the final state; the two are not interchangeable. Appendix K.6 delivers the same message under the benchmark's protocol and reports single-attempt pass@1, and Appendix K.7 compares harnesses under it. Intervals are Wilson 95% for rates, repository-clustered bootstrap 95% for contrasts (2,000 replicates in the mechanism studies, 10,000 in the handoff sections), unless stated.

Table 37 summarizes the section's estimates; the subsections below give counts, intervals and protocols.

	Reject (once, at the miss)				Gate (every request): completion (%)	
		Next request fails (%) on a state that is				
Model	Repair rate (%)	repaired	unrepaired	certified	No supervisor	Perfect supervisor
Fable 5	44.6	5.7	22.7	3.8	65.7	77.5
GPT-6 Astra	45.4	5.1	28.2	4.3	52.1	68.2
Kimi K3	29.4	3.4	25.0	9.6	48.2	56.1†
GPT-5.6	30.0	12.4	30.0	16.9	43.9	59.3
GLM-5.2	37.4	8.5	31.1	11.9	27.9	41.4

Noisy supervisor: completion (%), 100-chain subset per model

Model	No supervisor	p=0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5
GPT-6 Astra	49	64	64	58	61	53	53
GPT-5.6	44	60	58	54	46	49	47

Table 37: **What one bit from outside recovers: summary of Section 5** (280 chains; one conversation per chain, continued after the miss; completion = every request passes on its own step). *Reject*: the message once, at a silent miss, with one repair round; *Gate*: the same after every failed submission of a chain. The certified column is the reference of a state that passed its contract. The lower block reports the separate 100-chain *Noisy* study: p is the false-alarm rate and p=0 a perfect supervisor. †Lower bound (Appendix K).

K.1 ARMS AND MESSAGES

Starting points. The Reject, Name, Insist and fresh-conversation arms start from the silent misses of §4.2, taken from the continuous-conversation first attempts: every request $k \leq 19$ that a model

submitted as complete and that failed its hidden contract. The failing state is restored and the same conversation is continued. Counts per model are in Table 21.

The message. In every arm the supervisor's message is the same string, sent as the observation to the agent's submit command:

> *External verification of your previous submission failed: the repository does not satisfy the previous specification. Fix the repository so that the previous specification is satisfied, then submit again with the same command. Do not modify or create test files.*

The message is an oracle bit, and it bounds what any verifier can deliver: the checks an agent can run without hidden tests—its own test run or test suite, another model's review—fall far short of it (§4.2, Appendix C, §4.3). *Reject* (arm E) sends it once and allows one repair round before request $k{+}1$ is issued. *Name* (arm E1-names) appends the identifiers of the failing test nodes. *Insist* (arm E8) re-sends it after each failed repair, for up to eight rounds. *Fresh conversation* (arm F) restores the same broken state into a new conversation with no message, as a control for the effect of a restart. The next request is graded on its own tests (own-work failure), so a failure inherited from the open miss is not counted against it; the reference rate is the same model's own-work failure after a certified pass of request k, standardised to the arm's distribution of k (Table 21). The repaired and unrepaired subsets are selected on outcome; their rates describe, they do not estimate a causal effect.

Gate. The continuous-conversation first attempts are audited from the exact state at which they first failed (identity-checked restore): one repair round is allowed at that request and at every later request that fails, including the twentieth, and a chain counts as complete only if all twenty requests pass. Kimi K3's continuations that reached the 12-hour wall while still passing are counted as failures (18 of 145 audited chains), so its completion is a lower bound; every other model's audit is fully resolved on the 280 chains.

Noisy supervisor. Fresh chains are run from the first request on 100 chains of the 280-chain set (100 distinct repositories; 44 that GPT-5.6 completed on its first attempt, 30 that it failed by request 10 and 26 that it failed later), with GPT-5.6-sol and GPT-6 Astra in the continuous conversation, with the budgets of the Gate audit. After every submission the hidden grader runs: after a failure the message above is sent (a true alarm); after a pass it is sent with probability p (a false alarm), otherwise the usual acceptance. One uniform draw per (chain, request), fixed before launch, decides the false alarms at every level $p \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, so the levels are nested. Each alarm allows one repair round, after which the verdict is final: a false alarm that makes the agent break a correct submission ends the chain. The references are the same 100 chains without a supervisor (first attempt in the continuous conversation) and under the Gate audit above ($p{=}0$). A first run of this arm wrote its bookkeeping—the level, the chain's alarm schedule and the kind of each alarm—into a directory that the harness mounts into the agent's sandbox; GPT-5.6 read at least one of these files in 13–19 of 100 attempts per level and GPT-6 Astra in at most one. The arm was re-run with the bookkeeping sealed outside the sandbox and a per-attempt check that nothing of it was visible. Section 5.4 reports the sealed run; the first run is reported below as a sensitivity analysis.

K.2 Reject, one round

Table 21 gives the per-model estimates on the 280-chain set: starting points, repair rate with Wilson intervals, own-work failure at $k{+}1$ on repaired, unrepaired and certified states, and the paired contrast between the fresh-conversation and silent arms ($F - D$: Fable 5 $+4.1$ pp [0.7, 8.2] on 465 pairs; Kimi K3 $+0.4$ [-1.9, 2.7] on 773; GPT-5.6 -1.2 [-5.1, 2.5] on 939; GPT-6 Astra and GLM-5.2 have no arm F).

K.3 Name and Insist

Both variants were run for GPT-5.6 and Kimi K3 on the same starting points as their Reject arm and are reported on the 280 chains, recomputed from the continuation records; Table 38 gives the estimates. Contrasts are paired on starting points and clustered by repository. Insist is intention-to-treat: a continuation censored by the tool-time cap or the wall before a pass counts as unrepaired at every round, and the arms' denominators differ slightly for Kimi K3 for that reason.

Table 38: **Name and Insist** (280 chains). Repair rate of the starting-point miss after r rounds; increments per round; calls until the miss passes among repaired misses and calls spent among misses never repaired (Insist arm). Wilson intervals for rates, repository-clustered bootstrap for the paired contrast.

	GPT-5.6	Kimi K3
Starting points, Reject / Name / Insist (repositories)	939 / 939 / 939 (100)	834 / 827 / 821 (96)
Reject, repair rate after one round (%)	30.0 [27.2, 33.0]	29.5 [26.5, 32.7]
Name, repair rate after one round (%)	46.4 [43.3, 49.6]	46.8 [43.4, 50.2]
Name − Reject, paired (pp)	16.4 [11.0, 22.0]	17.1 [11.2, 22.8]
Insist, repair rate after round 1, 2, . . . , 8 (%)	30.0, 38.1, 44.3, 47.6, 49.8, 52.2, 53.0, 53.9	31.1, 42.3, 48.2, 52.6, 54.1, 56.0, 57.6, 58.7
Increment of rounds 2, . . . , 8 (pp)	8.1, 6.2, 3.3, 2.2, 2.3, 0.9, 0.9	11.2, 6.0, 4.4, 1.5, 2.0, 1.6, 1.1
Round 8, Wilson interval (%)	53.9 [50.7, 57.1]	58.7 [55.3, 62.0]
Calls until pass, repaired misses, median	16.5	19
Calls spent, misses never repaired, median (mean)	78 (87.8)	73 (75.5)

K.4 GATE

Table 39 gives the audit on the 280 chains: every failed first attempt is resumed from the exact state at its failure with the rejection and one repair round, and the chain continues under the same rule until it completes or a repair round fails.

Table 39: **Gate** (280 chains). Completion with no supervisor (first attempt in the continuous conversation) and with a perfect supervisor (one repair round at every failed request); Wilson intervals over 280 chains.

	Fable 5	GPT-6 Astra	Kimi K3	GPT-5.6	GLM-5.2
No supervisor: first attempts completing 20 (of 280)	184	146	135	123	78
Failed first attempts audited	96	134	145	157	202
Rescued by verification and repair	33	45	22	43	38
Perfect supervisor: completion (%)	77.5 [72.3, 82.0]	68.2 [62.5, 73.4]	56.1 [50.2, 61.8]	59.3 [53.4, 64.9]	41.4 [35.8, 47.3]
Gain (pp; relative)	11.8 (18%)	16.1 (31%)	7.9 (16%)	15.4 (35%)	13.6 (49%)
Unresolved continuations (counted as failures)	0	0	18	0	0
Upper bound if all unresolved had completed (%)	—	—	62.5	—	—

K.5 NOISY SUPERVISOR

Table 40 gives the sealed run (its bands are drawn in Figure 7): ten arms of 100 chains, 1,000 attempts, none censored, every model call at the benchmark's reasoning setting, and no harness file visible in any sandbox. Completion intervals are clustered by chain (each chain is its own repository in this sample). We also report, per model, the outcome of every phantom round—unchanged resubmit (identical tree), changed and still passing, or broken—the calls it required, its share of all model calls, and the repair rate of true alarms, which is comparable in protocol but not in population to the Reject arm (here the misses arise inside fresh attempts). The call shares are pooled within each arm, not averaged across chains. We report calls rather than dollar shares because the cache and pricing uncertainties of Appendix F.4 also affect the monetary attribution of phantom rounds.

Pooled over the five levels, GPT-5.6's 2,143 phantom rounds ended in an unchanged resubmit 1.4% of the time [0.6, 2.4], in a change that still passed 97.2%, and in a broken submission 1.4% [0.8, 2.1]; GPT-6 Astra's 2,295 in 0.7% [0.0, 1.8], 97.6% and 1.7% [0.9, 2.7]. A phantom round cost a median of 8 (GPT-5.6) and 6 (GPT-6 Astra) model calls and about 6 and 3 s of tool execution. True alarms were repaired in 28.7% [22.4, 34.8] and 38.1% [27.9, 48.4] of cases, against 30.0% and 45.4% in the Reject arm's population.

Sensitivity: the first run. The first run, in which the alarm schedule was readable, gives completion within 7 points of the sealed run at every level for both models (GPT-5.6: 55, 61, 47, 42, 44%;

Table 40: **Noisy supervisor** (sealed run; 100 chains per level, 0 censored). Completion against the false-alarm rate p, its paired difference from the perfect supervisor on the same chains, and the fate of the phantom rounds. Brackets: repository-clustered bootstrap 95% intervals (2,000 replicates).

	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5
GPT-5.6-sol — no supervisor 44 [35, 54]; perfect supervisor 60 [50, 70]					
Completion (%)	58 [49, 67]	54 [44, 64]	46 [37, 56]	49 [39, 59]	47 [37, 57]
Difference from perfect supervisor, paired (pp)	−2 [−9, 5]	−6 [−13, 1]	−14 [−23, −5]	−11 [−19, −2]	−13 [−21, −5]
Chains ended by a false alarm (%)	3	2	8	7	10
Phantom rounds (n); broken (%); unchanged (%)	147; 2.0; 2.0	298; 0.7; 1.7	416; 1.9; 0.2	581; 1.2; 1.2	701; 1.4; 2.0
Calls per phantom round, median	8	8	7.5	8	8
Phantom rounds' share of all model calls (%)	10	19	23	28	34
True alarms repaired (%)	33 (19/58)	34 (23/67)	27 (17/63)	24 (14/58)	25 (14/57)
GPT-6 Astra — no supervisor 49 [40, 59]; perfect supervisor 64 [54, 73]					
Completion (%)	64 [55, 73]	58 [48, 68]	61 [51, 71]	53 [44, 63]	53 [42, 62]
Difference from perfect supervisor, paired (pp)	0 [−7, 7]	−6 [−13, 1]	−3 [−10, 4]	−11 [−19, −3]	−11 [−19, −2]
Chains ended by a false alarm (%)	6	6	7	8	12
Phantom rounds (n); broken (%); unchanged (%)	157; 3.8; 0.0	311; 1.9; 0.3	463; 1.5; 0.9	593; 1.3; 0.8	771; 1.6; 0.6
Calls per phantom round, median	6	6	6	5.5	6
Phantom rounds' share of all model calls (%)	9	16	21	27	31
True alarms repaired (%)	47 (27/57)	36 (20/56)	40 (21/53)	33 (19/58)	35 (19/54)

GPT-6 Astra: 63, 60, 57, 55, 57%) and the same broken rate per phantom round (1.5% for both). Where it differs is in how GPT-5.6 answered a false alarm: 11.3% [7.8, 15.2] of its 2,145 phantom rounds were unchanged resubmits, against 1.4% once sealed; restricted to the attempts whose model never read a bookkeeping file, the first run's rate is 1.2%, so the unchanged resubmits were the exposed attempts. GPT-6 Astra, which read the files in at most one attempt per level, shows no difference (0.8% against 0.7%). One first-run cell, GPT-5.6's true-alarm repair rate of 10% at p=0.5 (5 of 50), did not recur in the sealed run (25%, 14 of 57).

K.6 Verification under the handoff protocol

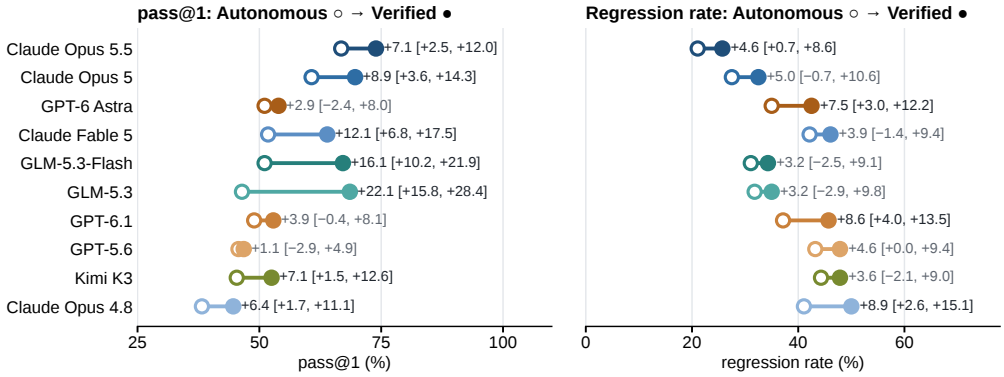
The arms above continue the conversation in which the miss was made. Under the benchmark's protocol (§3.3) that conversation is gone when the verdict arrives, so the bit is delivered as follows: whenever request $k \leq 19$ leaves a test of requests $1..k$ failing, the agent receives, before request $k{+}1$, one repair round in a fresh conversation whose task is the specification of request k followed by the message of Appendix K.1; request $k{+}1$ then starts in its own fresh conversation. The repair is graded together with request $k{+}1$, so strict pass@1 never credits it; a failing original test of the repository does not trigger the message; nothing stops the chain, and a failed request 20 gets no repair. Harness, budgets and settings are those of the AUTONOMOUS run, and every contrast is paired with its first attempt on the same 280 chains (Table 41): the Verified arm has one attempt, so these are single-attempt comparisons, not the eight-attempt estimate of Table 2, and the Autonomous columns below are attempt-1 values. Eight of the 2,800 Verified chains were still running when the table was built and count as failures (three for Fable 5, two each for GPT-5.6 and GLM-5.3-Flash, one for Kimi K3); no number can move by more than 1.1 points.

The bit repairs misses: the share of new failing tests that pass after the next request rises from 8–30% to 28–60%, and pass@1 rises by 1.1 to 22.1 points, with seven of ten intervals excluding zero (Figure 11). It also breaks things: the regression rate rises for every model, by 3.2 to 8.9 points (four intervals exclude zero), because the repair round starts without the conversation in which the work was written and has to rediscover it before it can mend it. Strict pass@1, which grades each request on its own step, cannot see a repair and moves by -2.5 to $+2.5$ points; the repair rounds add 47 to 294 calls per chain (12 to 98 per round).

Name and Insist under handoff. Both variants start from every repair slot of the Verified arm—a request $k \leq 19$ after which fixable tests were failing—restored to the exact repository state after k with the arm's budget counters, and send the message in a fresh conversation. *Name* appends the failing tests' identifiers; *Insist* re-sends the bare message after each failed round, up to four rounds (rounds 5–8 of Appendix K.3 added under 2 points each), its first round being Reject once in isolation, the paired control for Name on identical slots (Table 42).

Table 41: **Verified under the handoff protocol** (ten models, 280 chains each, one attempt per arm; Wilson 95% intervals per arm, repository-clustered bootstrap 95% intervals for the paired differences Verified − Autonomous; A = Autonomous (attempt 1), V = Verified). *Tests repaired* is the share of tests that fail for the first time after a request and pass after the next one: after the repair round in the Verified arm, by the model's own later work in the Autonomous arm. Calls per chain are medians and include the repair rounds.

	pass@1 (%)		Regression rate (%)		Strict (%)	Tests repaired (%)	Calls
Model	A / V	Δ (pp)	A / V	Δ (pp)	A / V	A / V	A / V
Opus 5.5	66.8 / 73.9	+7.1 [2.5, 12.0]	21.1 / 25.7	+4.6 [0.7, 8.6]	55.7 / 57.9	30 / 60	217 / 234
Opus 5	60.7 / 69.6	+8.9 [3.6, 14.3]	27.5 / 32.5	+5.0 [−0.7, 10.6]	51.1 / 50.0	26 / 43	401 / 486
GPT-6 Astra	51.1 / 53.9	+2.9 [−2.4, 8.0]	35.0 / 42.5	+7.5 [3.0, 12.2]	36.1 / 36.4	12 / 39	182 / 209
Fable 5	51.8 / 63.9	+12.1 [6.8, 17.5]	42.1 / 46.1	+3.9 [−1.4, 9.4]	36.4 / 38.9	19 / 38	293 / 350
GLM-5.3-Flash	51.1 / 67.1	+16.1 [10.2, 21.9]	31.1 / 34.3	+3.2 [−2.5, 9.1]	38.6 / 40.7	16 / 49	739 / 902
GLM-5.3	46.4 / 68.6	+22.1 [15.8, 28.4]	31.8 / 35.0	+3.2 [−2.9, 9.8]	38.2 / 40.7	16 / 49	560 / 684
GPT-6.1 Sol	48.9 / 52.9	+3.9 [−0.4, 8.1]	37.1 / 45.7	+8.6 [4.0, 13.5]	36.4 / 35.0	17 / 37	190 / 222
GPT-5.6 Sol	45.7 / 46.8	+1.1 [−2.9, 4.9]	43.2 / 47.9	+4.6 [0.0, 9.4]	32.5 / 30.0	12 / 31	299 / 355
Kimi K3	45.4 / 52.5	+7.1 [1.5, 12.6]	44.3 / 47.9	+3.6 [−2.1, 9.0]	32.9 / 35.4	8 / 28	389 / 497
Opus 4.8	38.2 / 44.6	+6.4 [1.7, 11.1]	41.1 / 50.0	+8.9 [2.6, 15.1]	27.5 / 27.9	22 / 36	367 / 496



Labels: paired difference Verified − Autonomous (points) with repository-clustered 95% interval; dark = excludes zero, grey = covers zero.
One attempt per arm, 280 chains per model.

Figure 11: **One "verification failed" message raises pass@1 for every model, and raises the regression rate too.** Single-attempt pass@1 (left) and regression rate (right) of the AUTONOMOUS run (hollow) and of the Verified arm (filled) for ten models on the same 280 chains; labels give the paired difference with its repository-clustered 95% interval, dark where the interval excludes zero (seven of ten for pass@1, four of ten for the regression rate).

For Claude Opus 5.5 and GPT-6.1 Sol, naming the failing tests repairs 44.5% and 37.1% of the misses in one round against 16.8% and 12.6% for the bare message, which four rounds of insisting (28.8% and 14.4%) do not reach; the named round breaks a protected test more often than the bare one (12.0% and 12.6% against 7.2% and 4.9% of rounds), because it changes more.

Noisy watcher under handoff. Fresh chains on the 100-chain subset of Appendix K.1, at $p \in \{0.1, 0.5\}$ with the same alarm schedule, under the handoff protocol: a failing request gets the repair round of the Verified arm (a true alarm), a passing one gets it with probability p (a phantom alarm), one round per alarm, nothing stops the chain, and the references are the AUTONOMOUS run and the Verified arm (p=0) on the same chains (Table 43). Because a phantom round is a fresh conversation that is told to fix work that is not broken, its cost shows in what it breaks rather than in chains it ends.

Where the cells are final, a phantom round breaks a passing test in 3.5–11.4% of cases at p=0.1 and in 4.0–4.5% at p=0.5, against 1.4–1.7% inside a continuous conversation (Table 40); the regression rate of the noisy chains exceeds the Verified arm's by up to 16 points (Claude Opus 5.5 at p=0.5, [6, 26]). On pass@1 one false alarm in ten is within the interval of a perfect supervisor for every final

Table 42: **Name and Insist under the handoff protocol.** Repair rate of the starting-point miss after one round with the failing tests named (Name) and with the bare message (Reject once = Insist round 1), their paired difference on identical slots (repository-clustered bootstrap 95%), the bare message after four rounds, the Insist rounds needed to reach Name's one round, and the share of rounds that broke a protected test. Wilson 95% intervals in the source; '–': slots still running when the table was built.

Model	Slots (Name / Insist)	Name, 1 round (%)	Reject once, 1 round (%)	Name − Reject once, paired (pp)	Insist, 4 rounds (%)	Insist rounds to match Name	Round broke a test (%), Name / Reject once
Claude Opus 5.5	517 / 517	44.5	16.8	+27.7 [18.5, 37.3]	28.8	> 4	12.0 / 7.2
Claude Opus 5	632 / 632	–	–	–	–	–	–
GPT-6 Astra	1090 / 1090	35.6	–	–	–	–	12.2 / –
Claude Fable 5	875 / 875	–	–	–	–	–	–
GLM-5.3-Flash	762 / 762	–	–	–	–	–	–
GLM-5.3	808 / 808	27.2	–	–	–	–	9.8 / –
GPT-6.1 Sol	1155 / 1155	37.1	12.6	+24.5 [18.6, 30.9]	14.4	> 4	12.6 / 4.9
GPT-5.6 Sol	1304 / 1304	28.2	–	–	–	–	12.9 / –
Kimi K3	1212 / 1212	32.5	–	–	–	–	12.4 / –
Claude Opus 4.8	1430 / 1430	–	–	–	–	–	–

Table 43: **Noisy watcher under the handoff protocol** (100 chains per cell, one attempt per arm; paired differences with repository-clustered bootstrap 95% intervals). Single-attempt pass@1 of the noisy arm next to the AUTONOMOUS run (attempt 1) and the Verified arm (p=0) on the same chains; regression rate of the three; the fate of the phantom rounds (share that broke a passing test) and their share of all model calls. '–': chains still running when the table was built.

Model	p	pass@1 (%) noisy / Auton. / p=0	Δ vs Autonomous (pp)	Δ vs p=0 (pp)	Regression (%) noisy / Auton. / p=0	Phantom rounds (n; broke, %)	Phantom share of calls (%)
Claude Opus 5.5	0.1	71 / 69 / 76	+2 [−6, 10]	−5 [−13, 2]	41 / 20 / 28	169; 9.5	11
	0.5	67 / 69 / 76	−2 [−11, 7]	−9 [−18, −1]	44 / 20 / 28	769; 4.5	33
Claude Opus 5	0.1	–	–	–	–	–	–
	0.5	–	–	–	–	–	–
GPT-6 Astra	0.1	55 / 52 / 53	+3 [−5, 11]	+2 [−6, 10]	48 / 41 / 47	143; 3.5	6
	0.5	51 / 52 / 53	−1 [−9, 7]	−2 [−11, 7]	52 / 41 / 47	719; 4.0	24
Claude Fable 5	0.1	–	–	–	–	–	–
	0.5	–	–	–	–	–	–
GLM-5.3-Flash	0.1	–	–	–	–	–	–
	0.5	–	–	–	–	–	–
GLM-5.3	0.1	–	–	–	–	–	–
	0.5	–	–	–	–	–	–
GPT-6.1 Sol	0.1	57 / 51 / 53	+6 [−2, 14]	+4 [−5, 13]	46 / 41 / 49	149; 5.4	6
	0.5	49 / 51 / 53	−2 [−10, 6]	−4 [−11, 3]	56 / 41 / 49	716; 4.2	24
GPT-5.6 Sol	0.1	41 / 49 / 45	−8 [−16, 0]	−4 [−12, 4]	59 / 47 / 55	142; 7.0	6
	0.5	–	–	–	–	–	–
Kimi K3	0.1	49 / 48 / 56	+1 [−10, 12]	−7 [−17, 4]	51 / 46 / 46	140; 11.4	7
	0.5	–	–	–	–	–	–
Claude Opus 4.8	0.1	–	–	–	–	–	–
	0.5	–	–	–	–	–	–

model (−7 to +4 points), and at p=0.5 Claude Opus 5.5 gives back the whole gain (−9 [−18, −1] against p=0) and sits where the AUTONOMOUS run sits (−2 [−11, 7]).

K.7 SCAFFOLD SENSITIVITY

Table 44 gives the harness comparison used in Section 5.3 (and referenced in Section 4.1): on a stratified 58-chain subset (52 repositories; a 60-chain draw made before the last filtering round, minus the two chains of a repository that round excluded), four models run under Codex CLI 0.146.1, the OpenHands SDK 1.46.0 and pi 0.84.2, each installed inside the task container and invoked afresh for every request on the persisting repository, with the hidden grader, the no-stop rule and the model settings of the AUTONOMOUS run; every arm is paired with the same model's mini-swe-agent first attempt on the same chains. No harness raises pass@1 for any model: the paired difference runs from 0.0 (GPT-5.6, OpenHands SDK) to −29.3 points (GLM-5.3-Flash, Codex CLI), seven of the twelve intervals cover zero and five lie below it (Figure 12). Between 67 and 90% of the chains get the same verdict under a harness as under mini-swe-agent, and when both fail, 139 of 264 pairs

fail first at the same request. GLM-5.3-Flash is the exception, and it is a property of the model: under every harness it breaks protected tests far more often than under mini-swe-agent (regression rate 36–62% against 26%). pi has no per-command timeout; four chains stalled in one shell call for more than two hours and were closed under the benchmark's tool-time rule, and nine chains ended in a verifier error; both count as failures, as in the AUTONOMOUS run. The cumulative tool-time budget cannot be enforced inside the harnesses; a per-call limit stands in for it, and the OpenHands SDK, the one harness that reports its tool time, stays far below the budget.

Table 44: **Scaffold sensitivity** (58 chains, 52 repositories; handoff protocol; one attempt per arm). Single-attempt pass@1 of each model under three harnesses next to its mini-swe-agent run (attempt 1) on the same chains; Wilson 95% intervals per arm, repository-clustered bootstrap 95% intervals for the paired difference; strict pass@1 and regression rate as chain counts; chains with the same pass@1 verdict under the harness and under mini-swe-agent.

Model	Harness	pass@1 (of 58)	Paired Δ (pp)	Strict	Regressed	Same verdict
Claude Opus 5.5	mini-swe-agent	39 [54.4, 77.9]	—	29	15	—
	Codex CLI	31 [40.8, 65.7]	−13.8 [−26.7, −1.7]	27	16	44
	OpenHands SDK	34 [45.8, 70.4]	−8.6 [−21.3, 3.4]	30	17	47
	pi	35 [47.5, 71.9]	−6.9 [−17.5, 3.6]	29	12	48
GPT-5.6-sol	mini-swe-agent	32 [42.5, 67.3]	—	20	27	—
	Codex CLI	28 [35.9, 60.8]	−6.9 [−15.5, 0.0]	21	25	52
	OpenHands SDK	32 [42.5, 67.3]	0.0 [−9.3, 10.0]	21	23	52
	pi	23 [28.1, 52.5]	−15.5 [−27.3, −4.7]	13	27	45
Kimi K3	mini-swe-agent	30 [39.2, 64.1]	—	19	26	—
	Codex CLI	27 [34.3, 59.2]	−5.2 [−17.9, 7.8]	20	27	43
	OpenHands SDK	28 [35.9, 60.8]	−3.4 [−15.3, 8.1]	20	29	46
	pi	22 [26.6, 50.8]	−13.8 [−26.8, 0.0]	16	26	42
GLM-5.3-Flash	mini-swe-agent	32 [42.5, 67.3]	—	24	15	—
	Codex CLI	15 [16.3, 38.4]	−29.3 [−42.1, −16.9]	11	36	39
	OpenHands SDK	22 [26.6, 50.8]	−17.2 [−28.6, −7.0]	15	27	46
	pi	24 [29.6, 54.2]	−13.8 [−24.6, −4.5]	20	21	48

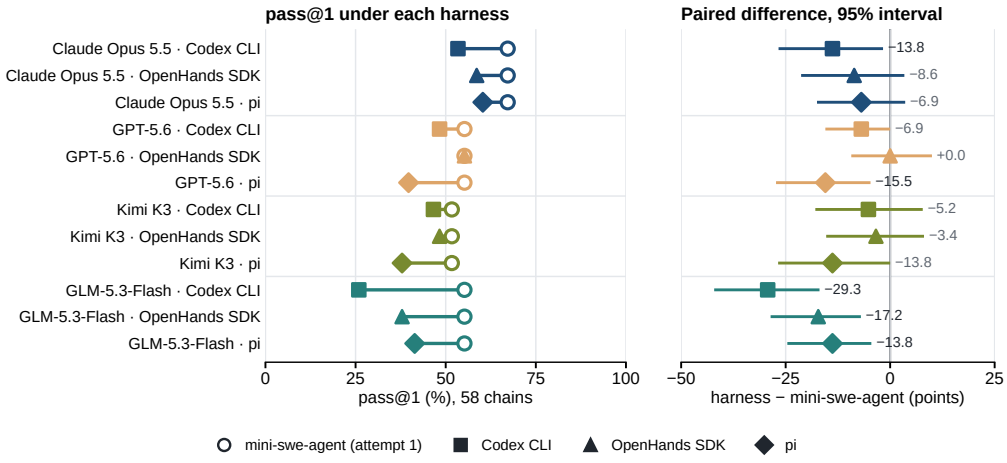


Figure 12: **A different harness does not help: none raises pass@1 for any model.** Single-attempt pass@1 of four models under Codex CLI, the OpenHands SDK and pi next to their mini-swe-agent first attempt on the same 58 chains (left), and the paired difference harness − mini-swe-agent with its repository-clustered 95% interval (right; dark labels where the interval excludes zero): five of the twelve intervals lie entirely below zero and none above it.